



# Chapter 10

## CMOS 工艺流程

### Process of CMOS

李秀妍

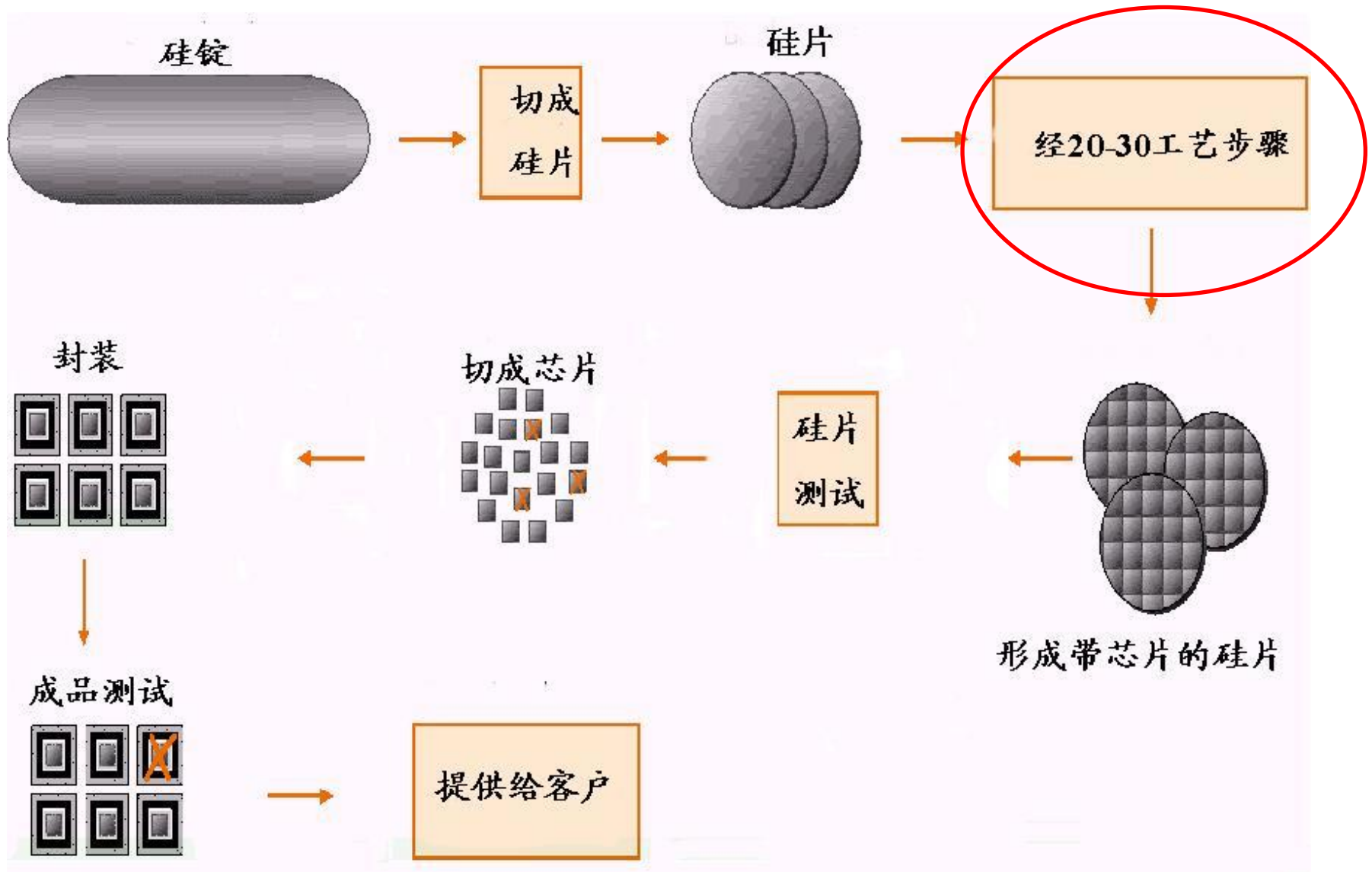
[xiuyanli@sjtu.edu.cn](mailto:xiuyanli@sjtu.edu.cn)

2025年12月22日



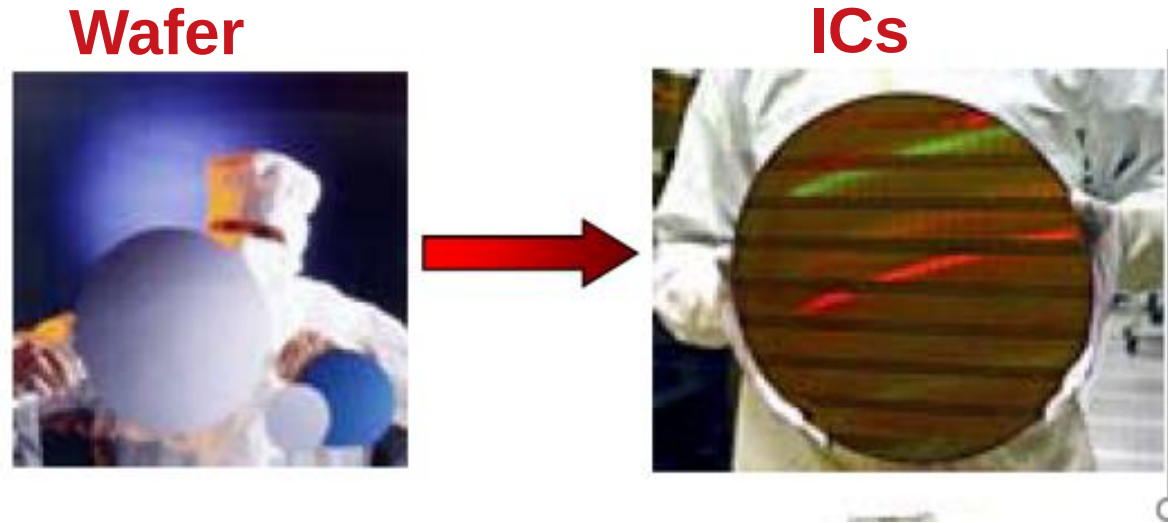
- ✎ **10. 1 Basics of CMOS process**
- ✎ **10. 2 Single-well process**
- ✎ **10. 3 Twin-well process**
- ✎ **10. 4 FinFET process**
- ✎ **10. 5 3D NAND process**
- ✎ **10. 6 Summary and homework**

# 10.1 IC fabrication process



# 10.1 IC fabrication process



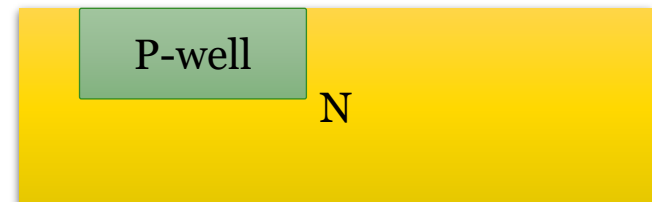
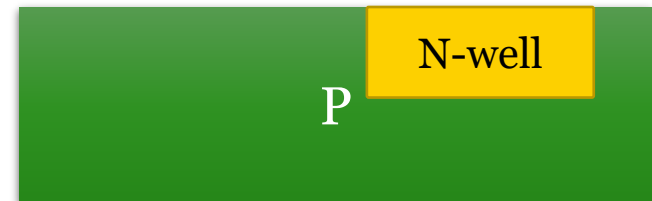


- Chose wafer size according to equipment
- Chose the doping type and resistivity according to specific process
- Chose crystalline orientation  
(100) has smallest  $D_{it}$  and highest mobility;  
(111) has highest  $D_{it}$  and largest stress.

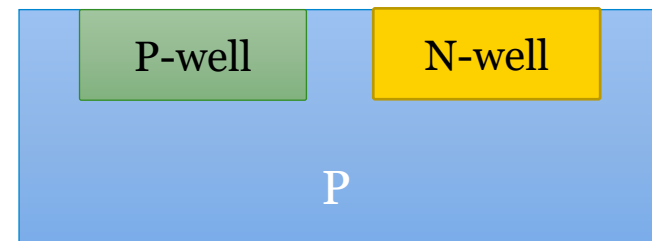
Two kinds of devices (CMOS) can be fabricated on one kind of substrate through formation of well.

➤ Three types of well

-N-well } Single-well  
-P-well }  
-Twin-well

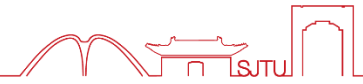


➤ Critical parameters of well include junction depths and well resistivity





# 10.1 22nm Processing Modules

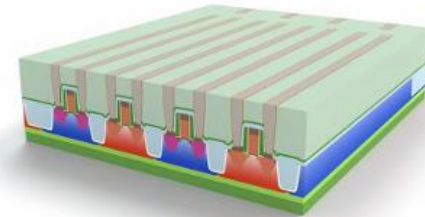


*The STI Module*



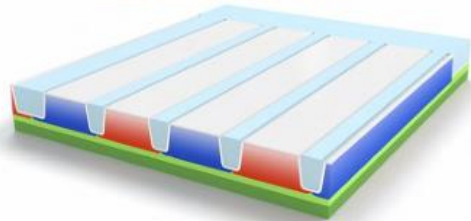
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*The Contact Module*



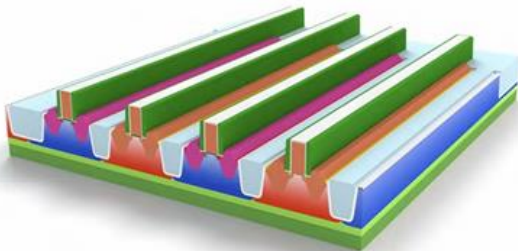
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*The Well Module*



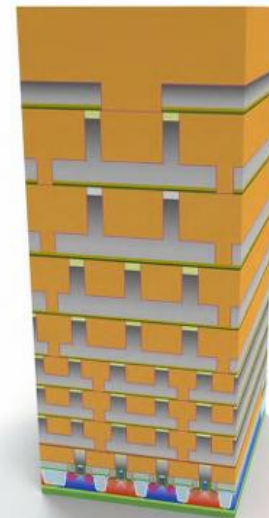
© 2016 Threshold Systems Inc.

*The Device Module*



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*Metallization*

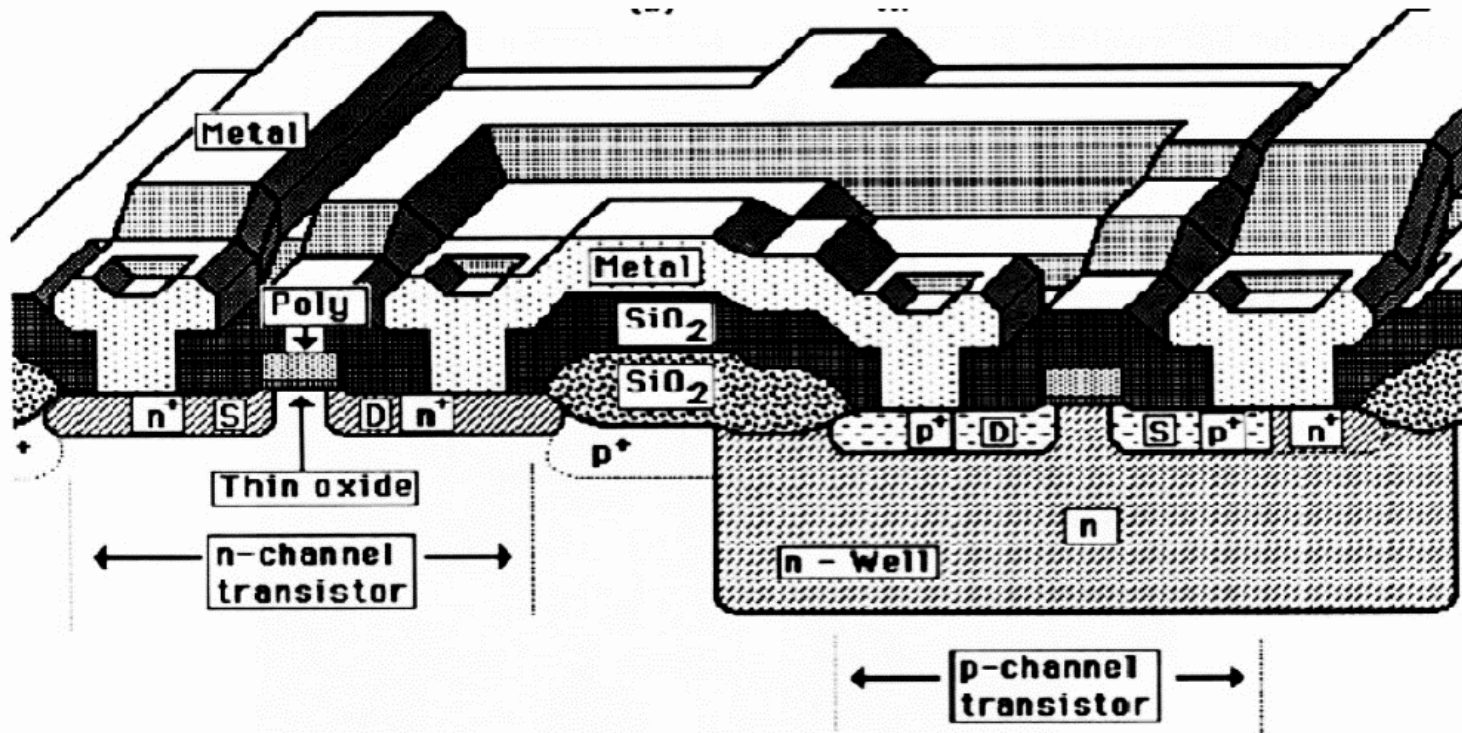


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- ✎ **10. 1 Basics of CMOS process**
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## 10.2.0 N-well CMOS structure



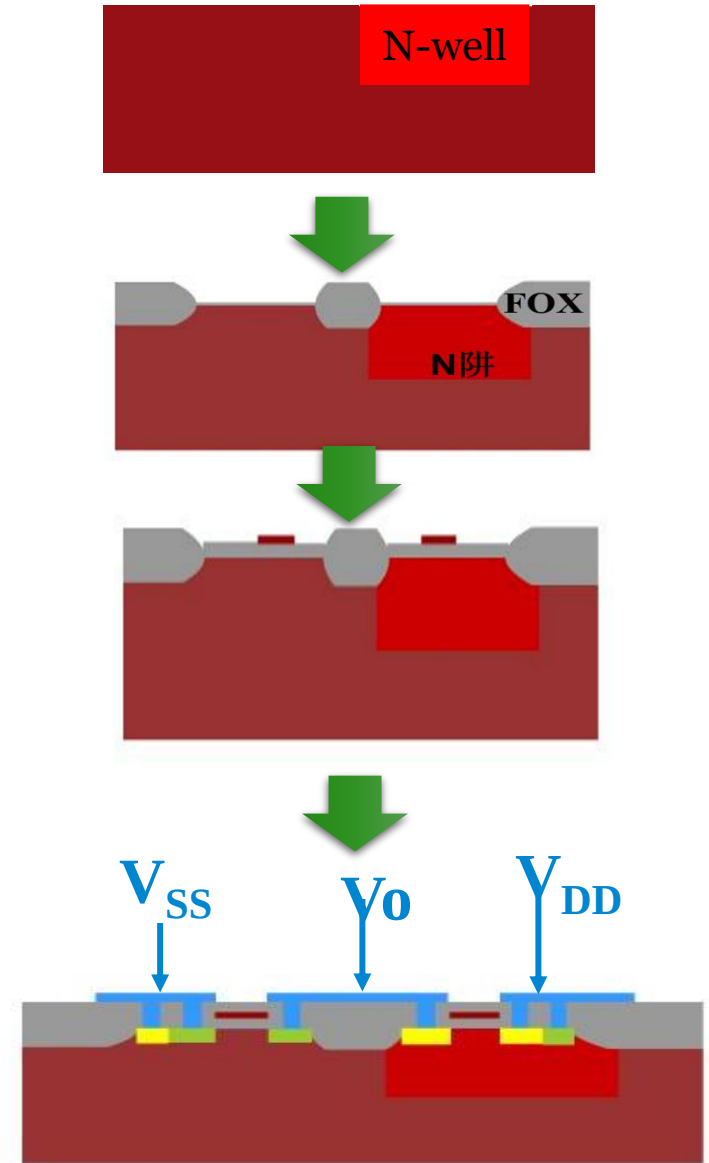
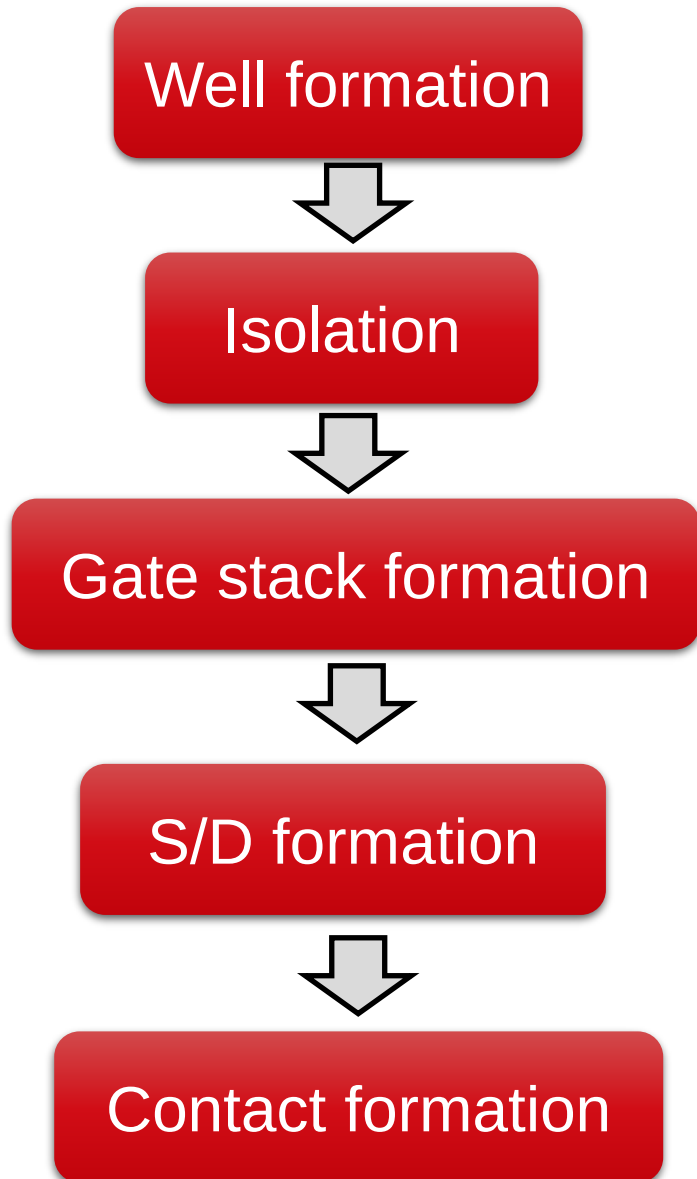
N-well  
CMOS  
structure

How many  
masks?  
Process flow?

# 10.2.0 N-well CMOS process flow



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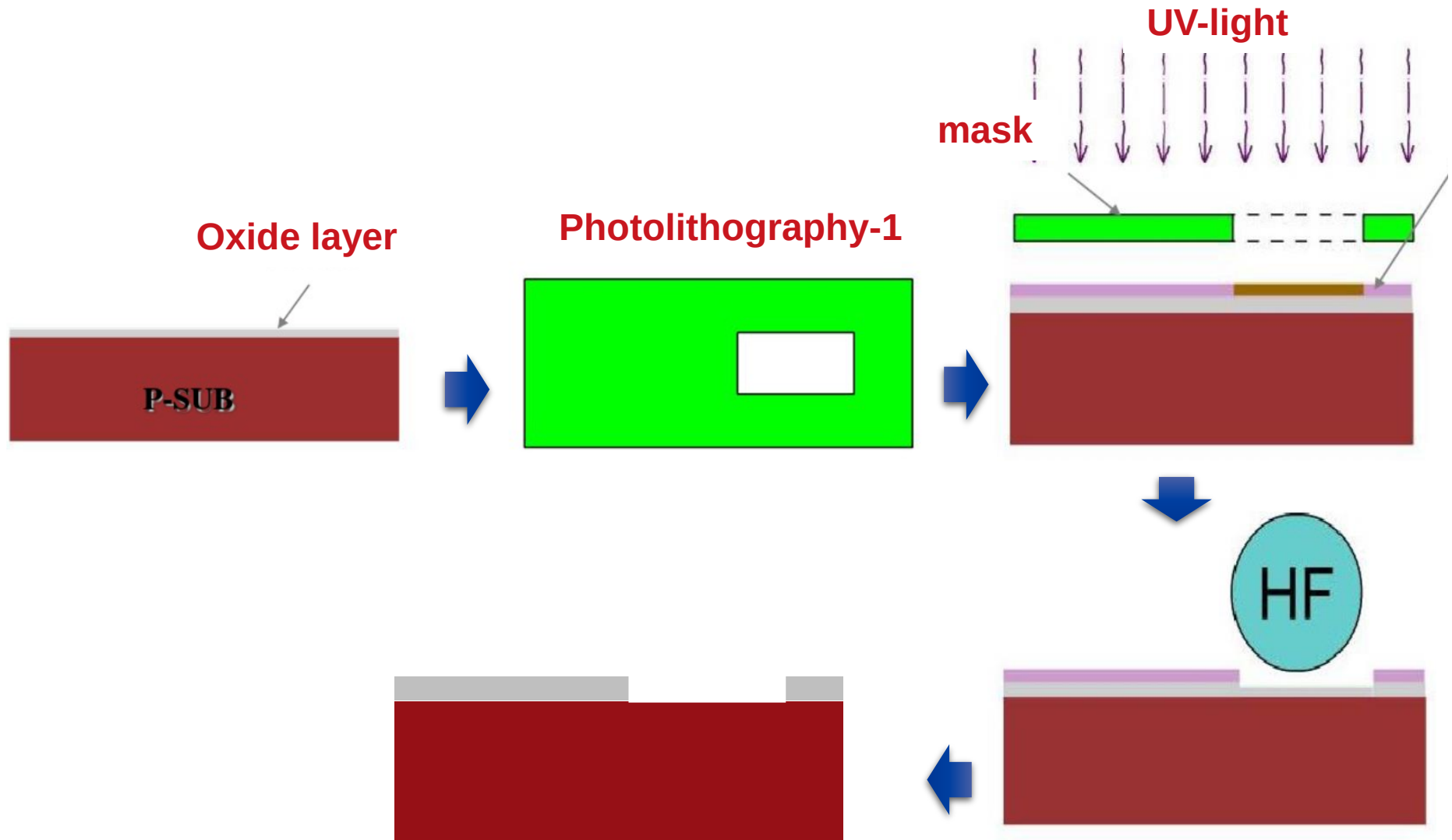
### Starting

P sub

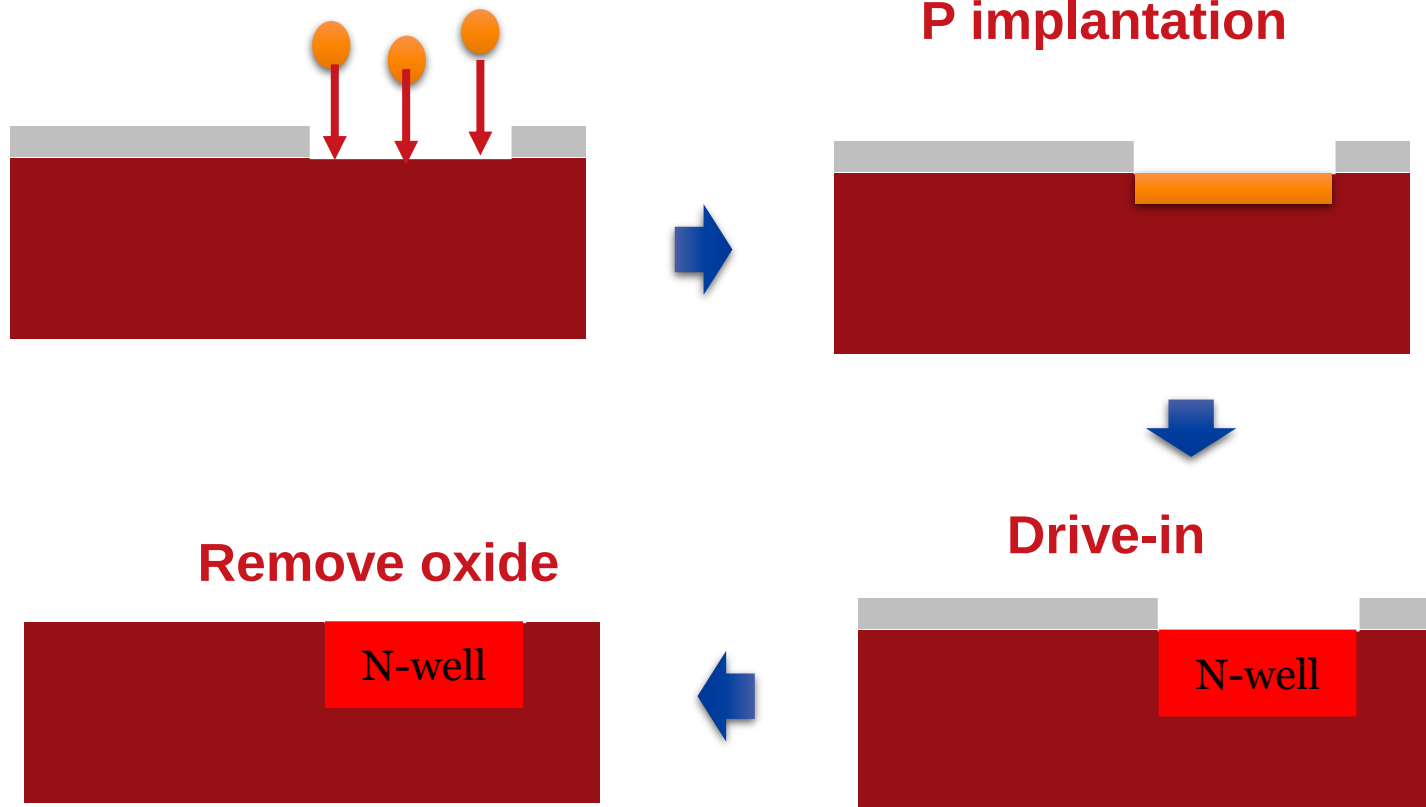
### Objective

N-well

### (1) Pattern mask opening for N-well implantation



### (2) Shallow implantation and drive-in

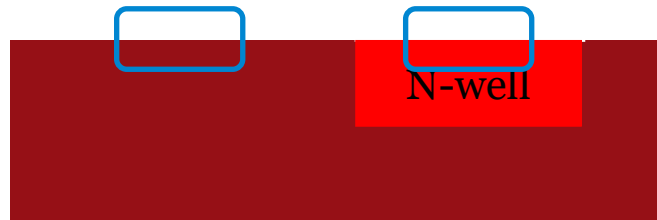


## 10.2.1 N-well formation

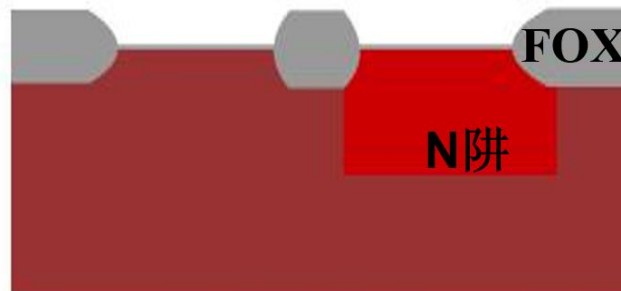


|      |      |                     |      |      |                  |
|------|------|---------------------|------|------|------------------|
| 1000 | 编批打标 | P<100>15- 25 ohm.cm | 1120 | 湿法去胶 | PR-L/R           |
| 1010 | 擦片   | 40#                 | 1140 | 涂胶   | 14#              |
| 1020 | 清洗   | B-CLEAN             | 1150 | P阱曝光 | <b>XXXX.PT</b>   |
| 1030 | 1次氧化 | G0224 ( 1000℃ )     | 1160 | 显影   | 1#               |
| 1040 | 测产品片 | 1500-1700-1900Å     | 1170 | 显检   |                  |
| 1050 | 涂胶   | 14#                 | 1180 | 注硼   | B/100Kev/7.5E12  |
| 1060 | N阱曝光 | <b>XXXX.TB</b>      | 1190 | 湿法去胶 | PR-L/R           |
| 1070 | 显影   | 1#                  | 1210 | 清洗   | B-CLEAN          |
| 1080 | 显检   |                     | 1220 | 推阱   | G0315 (1150℃)    |
| 1090 | 注磷   | P/180Kev/1.25E13    | 1230 | 测产品片 | 2280-2500-2720Å  |
| 1100 | 湿法腐蚀 | WE-N61(2.5MIN浸润)    | 1240 | 全剥   | WE-N62 (5.5 MIN) |

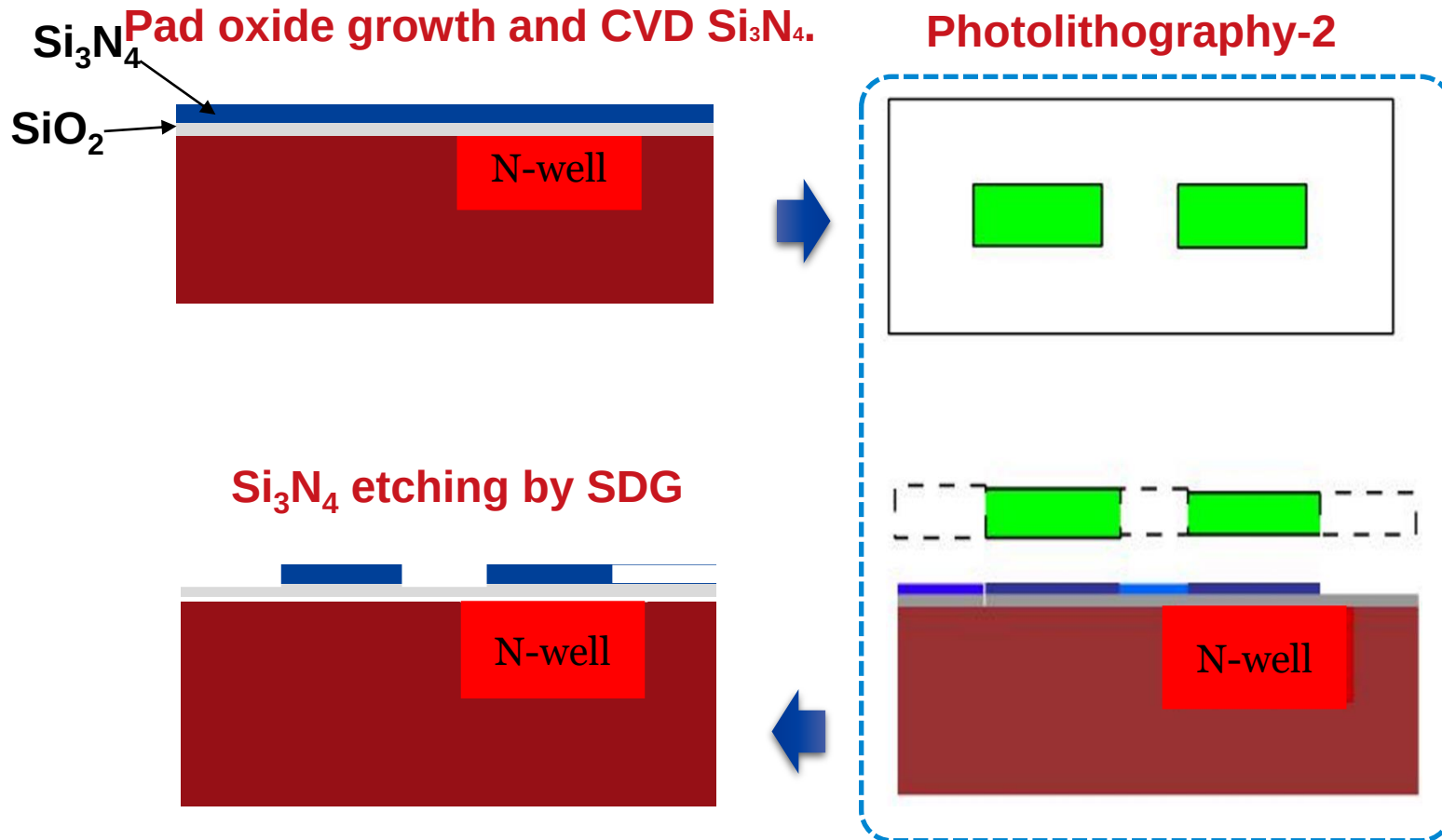
**Starting point**



**Objective**

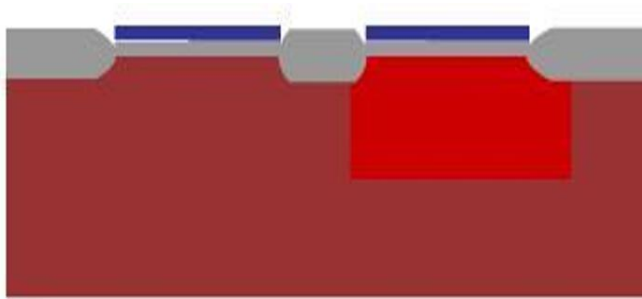


### (1) Pattern field oxide regions

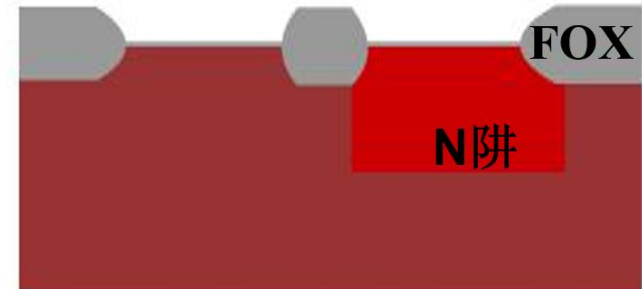


### (2) Field oxide formation

LOCOS

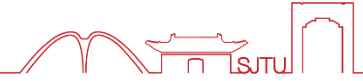


Etching  $\text{Si}_3\text{N}_4$ .



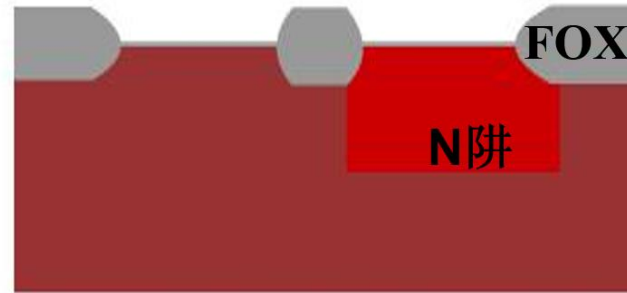
|      |         |                 |      |       |                 |
|------|---------|-----------------|------|-------|-----------------|
| 1250 | 清洗      | B-CLEAN         |      |       |                 |
| 1260 | 二次氧化    | G0222(900℃ )    |      |       | <b>NF</b>       |
| 1270 | 测产品片    | 150-200-250Å    | 1430 | 涂胶    | 14#             |
| 1280 | LPSiN淀积 | G0121           | 1440 | NF曝光  | <b>XXXX.NF</b>  |
| 1290 | 测产品片    | 1010-1175-1340Å | 1450 | 显影    | 1#              |
| 1300 | 涂胶      | 20#             | 1460 | 显检    |                 |
| 1310 | T0曝光    | <b>XXXX.TO</b>  | 1470 | 注硼    | B/25Kev/5E13    |
| 1320 | 显影      | 10#             | 1480 | APT注B | B+/180Kev/3E12  |
| 1330 | 显检      |                 | 1490 | 湿法去胶  | PR-L/R          |
| 1340 | 套刻测量    |                 | 1500 | 检查    |                 |
| 1350 | 测条宽     | 1.0-1.20-1.40um | 1510 | 清洗    | S/P 5*1         |
| 1360 | SDG腐蚀   | GASAD 1175      | 1520 | 场氧化   | G0631(975℃)     |
| 1370 | 测产品片    | 40--180Å        | 1530 | 测产品片  | 4800-5200-5600Å |
| 1380 | 干法去胶    | BC              | 1540 | 湿法腐蚀  | WE12-L/R        |
| 1390 | 湿法去胶    | PR-L/R          | 1560 | 漂洗    | WE22            |

## 10.2.3 Gate stack formation

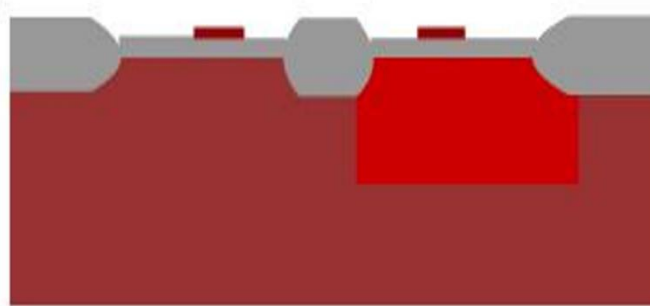


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**Starting**

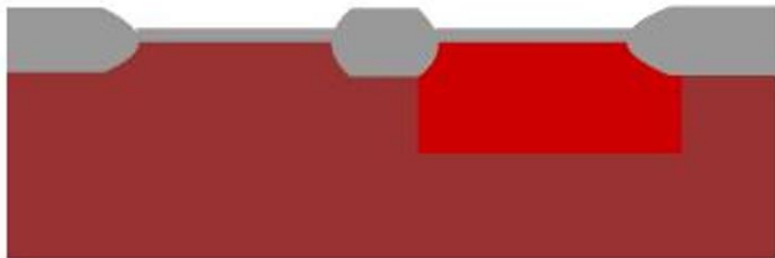


**Objective**

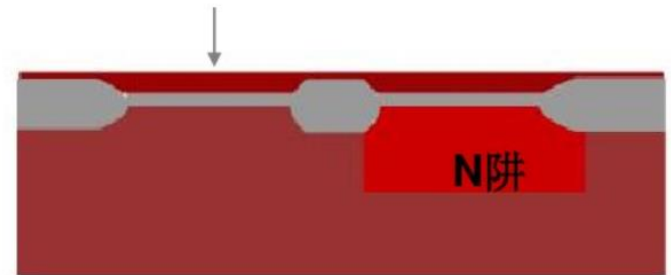


### (1) Oxide regrowth and CVD poly-Si deposition

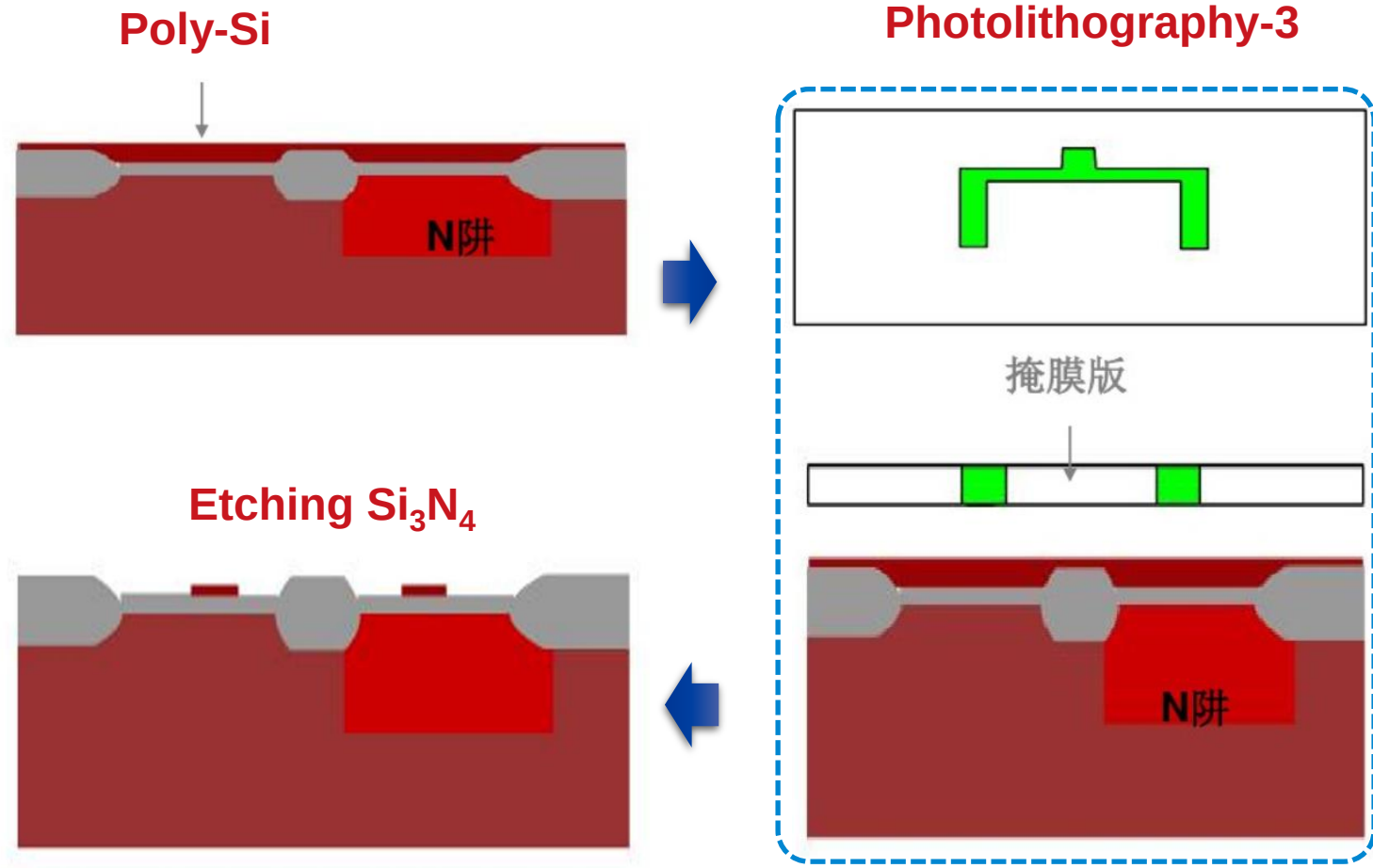
Regrowth oxide thermally



Poly-Si deposition by CVD



### (2) Pattern of gate stack

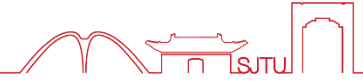


## 10.2.3 Gate stack formation

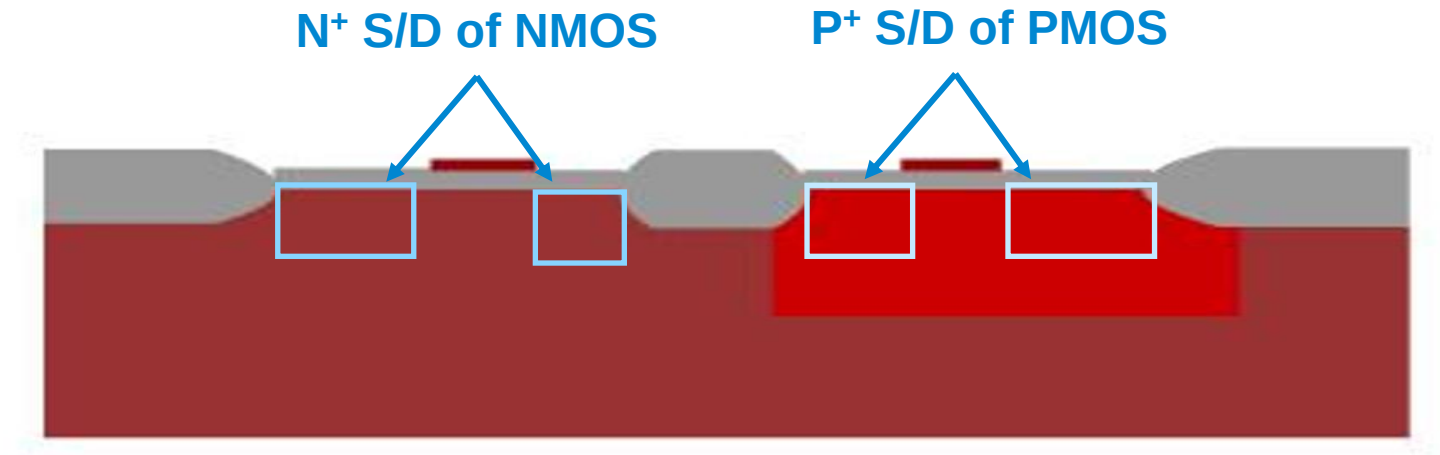


|      |        |                   |      |        |              |
|------|--------|-------------------|------|--------|--------------|
| 1570 | 清洗     | B-CLEAN           | 1740 | 清洗     | POLY-BCLN    |
| 1580 | 牺牲氧化   | G0237 (850℃)      | 1750 | 涂胶     | 20#          |
| 1590 | 测产品片   | 385-425-465Å      | 1760 | Poly曝光 | XXXX.GT      |
| 1600 | 漂洗     | WE-N27            | 1770 | 显影     | 10#          |
| 1610 | 清洗     | B-CLEAN           | 1780 | 显检     |              |
| 1620 | 预栅氧    | G0322             | 1790 | 测条宽    | 0.6+/-0.06um |
| 1630 | 测产品片   | 127-150-173 Å     | 1800 | Poly腐蚀 | TEL-0.6      |
| 1640 | 注BF2   | BF2+/50Kev/3.0E12 | 1810 | 测残氧    | 80-150A      |
| 1650 | 漂洗     | WE22              | 1820 | 漂洗     | WE31         |
| 1660 | 清洗     | B-CLEAN           | 1830 | 干法去胶   | BC           |
| 1670 | 栅氧     | TUNN125A          | 1840 | 湿法去胶   | PR-L/R       |
| 1680 | 测产品片   | 100-125-150Å      | 1850 | 漂洗     | WE32         |
| 1690 | LP-多晶  | G0331             | 1860 | 检查     |              |
| 1700 | 测产品片   | 3780-4200-4620Å   | 1880 | 测条宽    | 0.6+/-0.06um |
| 1710 | TENC监测 |                   | 1890 | 清洗     | S/P 5*1      |
| 1720 | 磷掺杂    | G0144 (900℃)      | 1900 | 多晶氧化   | G0421 (850℃) |
| 1730 | 测控制片   | 15-20-25Ω/sq(多晶)  | 1910 | 测控制片   | 80-100-120Å  |

## 10.2.4 S/D formation

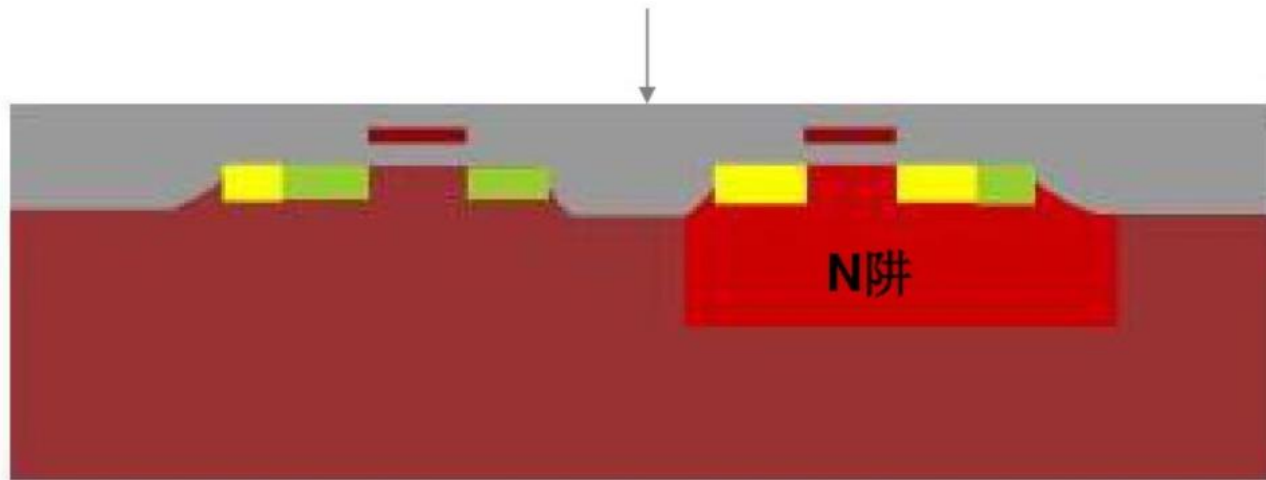


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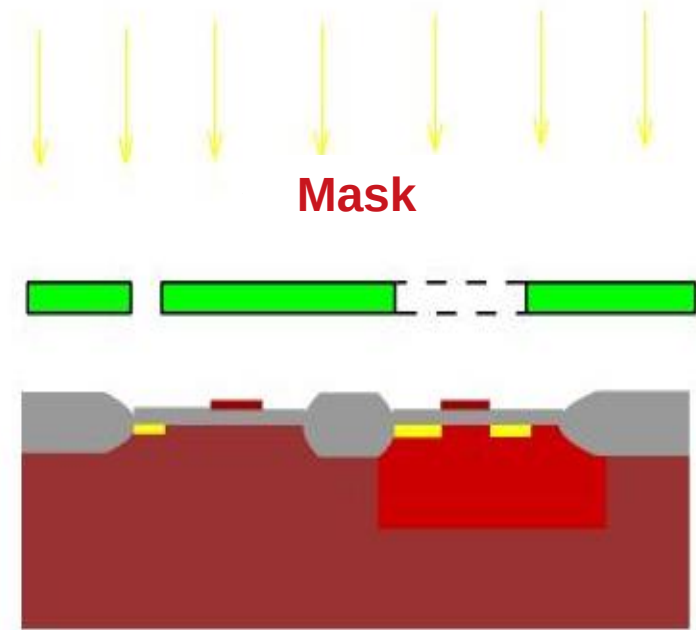
**Objective**

LT SiO<sub>2</sub> or PSG



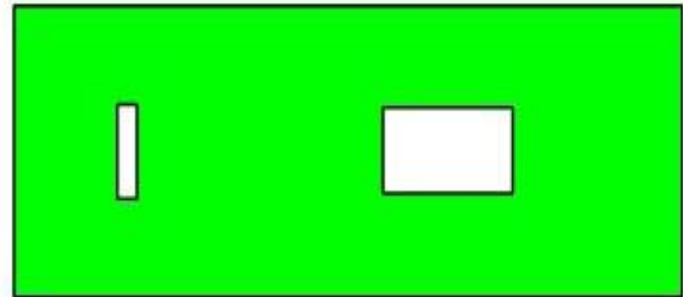
### (1) S/D implantation of PMOS

**B+**

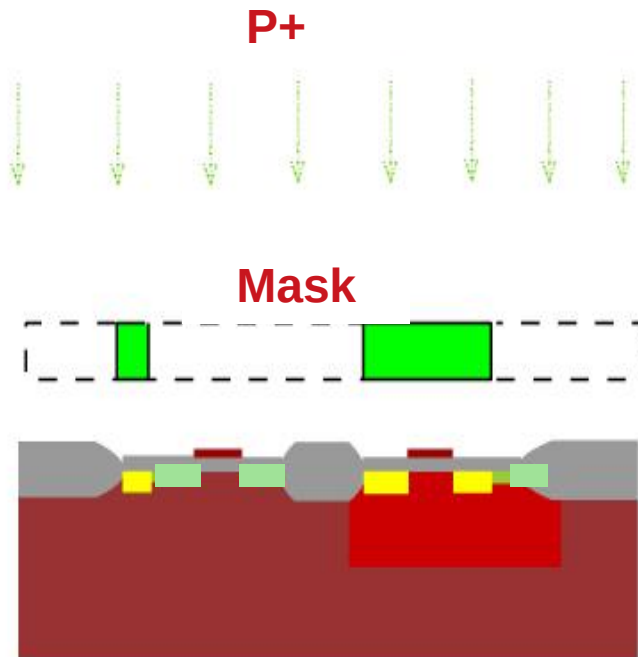


**Mask**

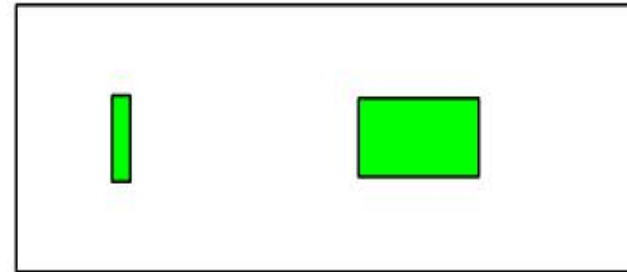
**Photolithography-4**



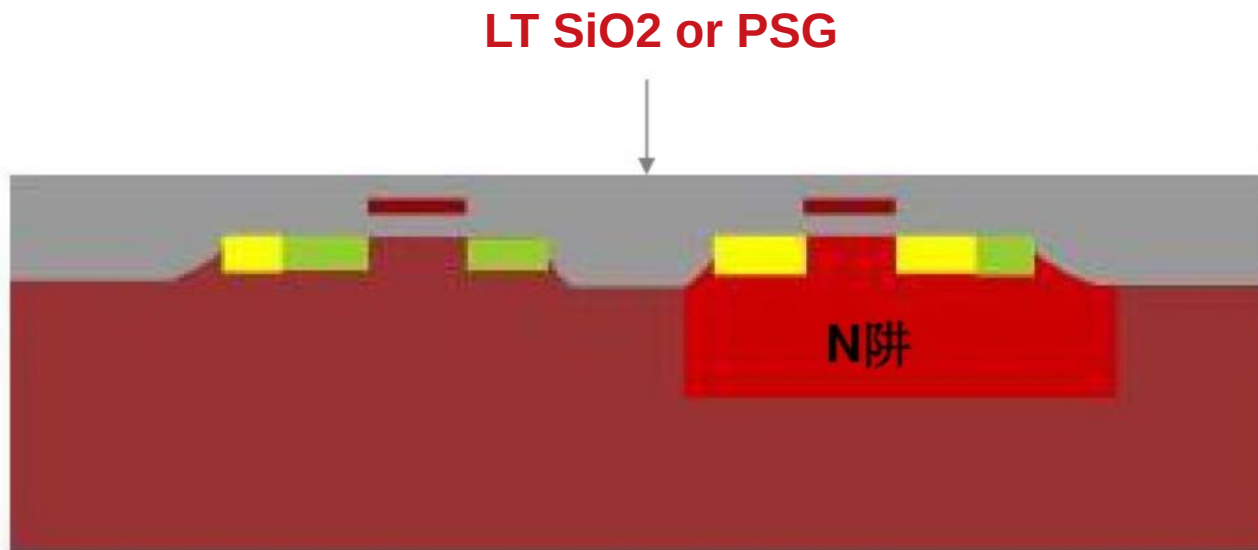
### (2) S/D implantation of NMOS



### Photolithography-5



### (3) CVD SiO<sub>2</sub> (Low--temperature oxide)



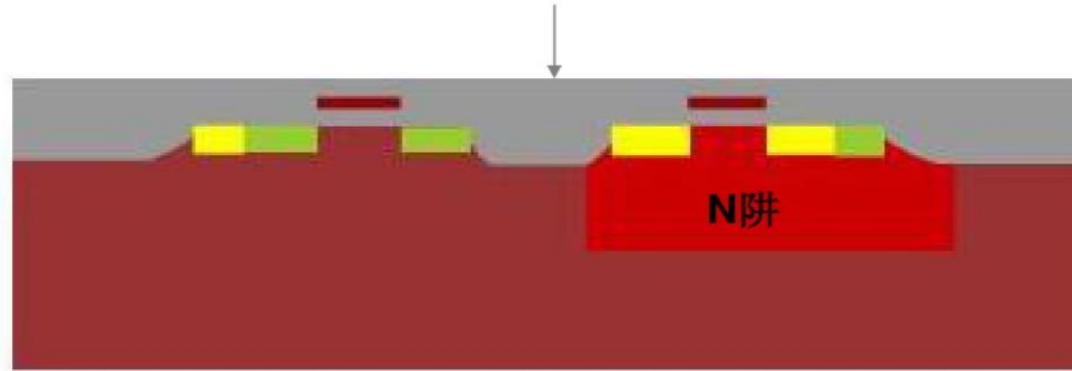
# 10.2.4 S/D formation



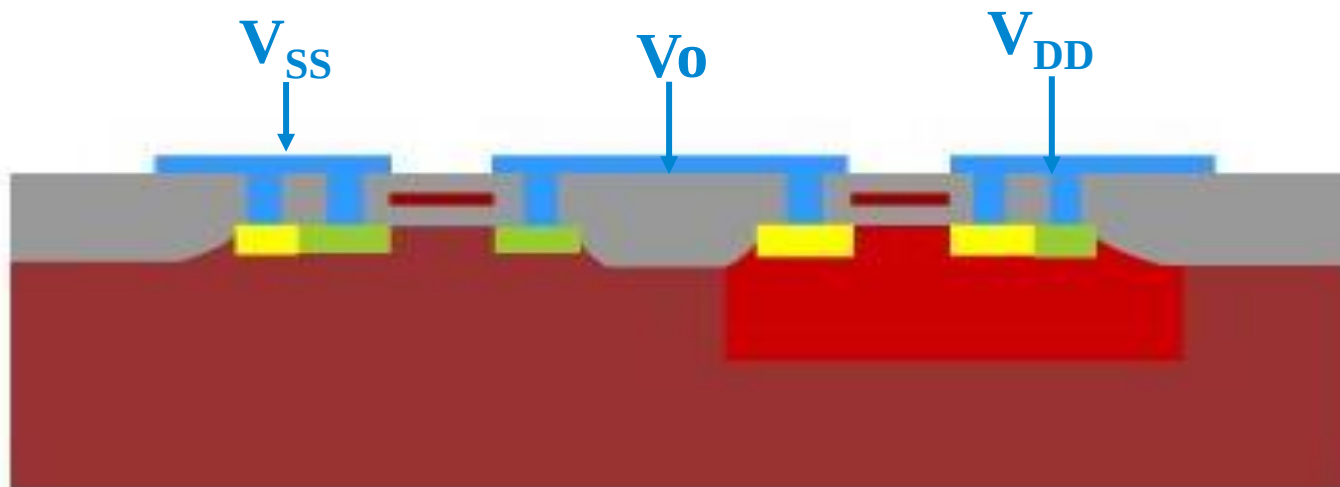
|      |        |                  |      |       |                  |
|------|--------|------------------|------|-------|------------------|
|      |        | <b>NLDD</b>      | 2140 | 清洗    | S/P 5*1          |
| 1920 | 涂胶     | 14#              | 2150 | 氧化    | G0422 (800°C)    |
| 1930 | LDD曝光  | <b>XXXX.SN</b>   | 2160 | 测控制片  | 25-50-75Å        |
| 1940 | 显影     | 1#               |      |       | <b>N-SD</b>      |
| 1950 | 显检     |                  | 2170 | 涂胶    | 14#              |
| 1960 | 注磷     | P/50Kev/3.0E13   | 2180 | NSD曝光 | <b>XXXX.SN</b>   |
| 1970 | 湿法去胶   | PR-L/R           | 2190 | 显影    | 1#               |
| 1980 | 检查     |                  | 2200 | 显检    |                  |
|      |        | <b>PLDD</b>      | 2210 | 注砷    | As/50Kev/3.0E15  |
| 1990 | 涂胶     | 14#              | 2220 | 干法去胶  | BC               |
| 2000 | LDD曝光  | <b>XXXX.SP</b>   | 2230 | 湿法去胶  | PR-L/R           |
| 2010 | 显影     | 1#               | 2240 | 漂洗    | WE99             |
| 2020 | 显检     |                  | 2250 | 检查    |                  |
| 2030 | 注BF2   | BF2/50Kev/1.6E13 |      |       | <b>P-SD</b>      |
|      |        | <b>0 度角注入!</b>   | 2260 | 涂胶    | 14#              |
| 2040 | 注磷     | P/110Kev/4.0E12  | 2270 | PSD曝光 | <b>XXXX.SP</b>   |
| 2050 | 湿法去胶   | PR-L/R           | 2280 | 显影    | 1#               |
| 2060 | 检查     |                  | 2290 | 显检    |                  |
| 2070 | 清洗     | S/P 5*1          | 2300 | 注BF2  | BF2/50KeV/3.0E15 |
| 2080 | 测FOX   | 3600±300A        | 2310 | 干法去胶  | BC               |
| 2090 | LPTEOS | G0131            | 2320 | 湿法去胶  | PR-L/R           |
| 2100 | 测控制片   | 3600-4000-4400Å  | 2330 | 漂洗    | WE99             |
| 2110 | TENC监测 |                  | 2340 | 检查    |                  |
| 2120 | SPAC腐蚀 | TEOS4000         | 2350 | 清洗    | S/P 5*1          |
| 2130 | 测FOX   | 3200±400A        | 2360 | 退火    | G0344            |

### Starting

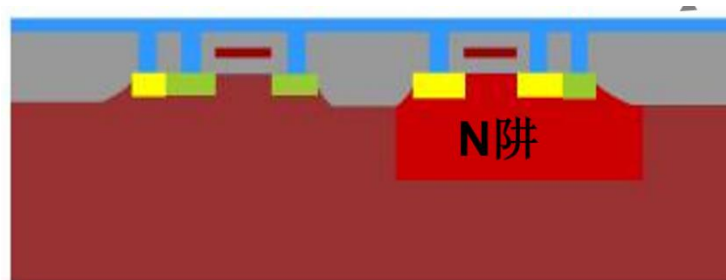
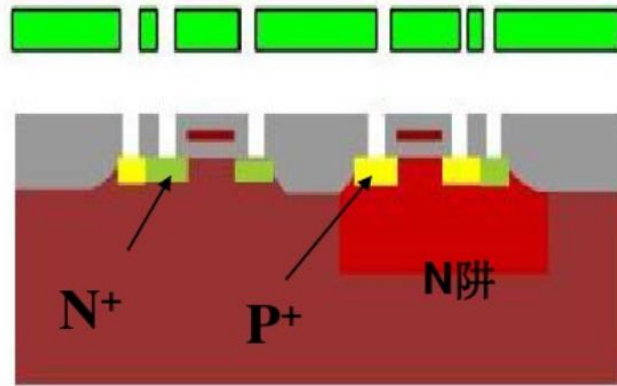
LT SiO<sub>2</sub> or PSG



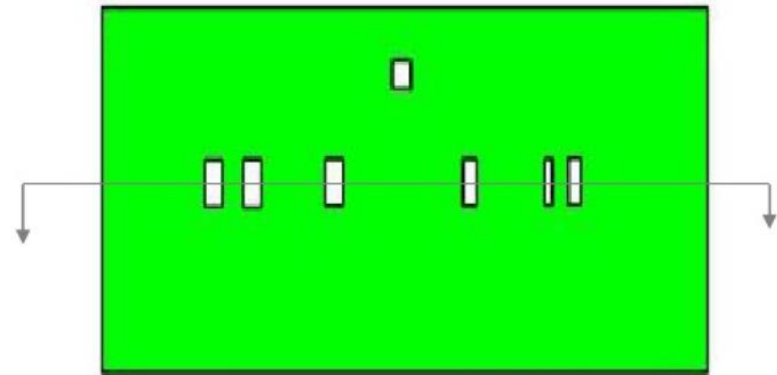
### Objective



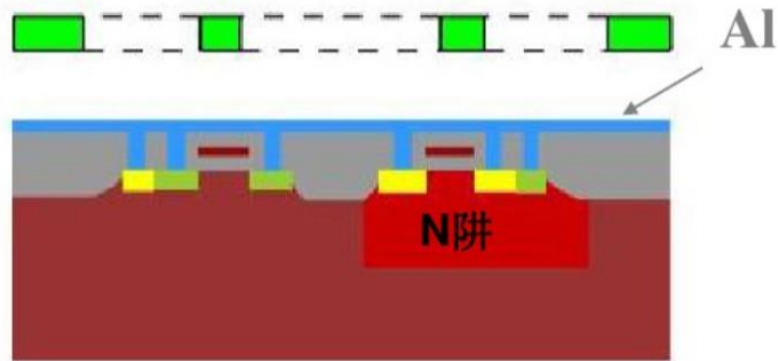
## (1) Space etch and metal deposition



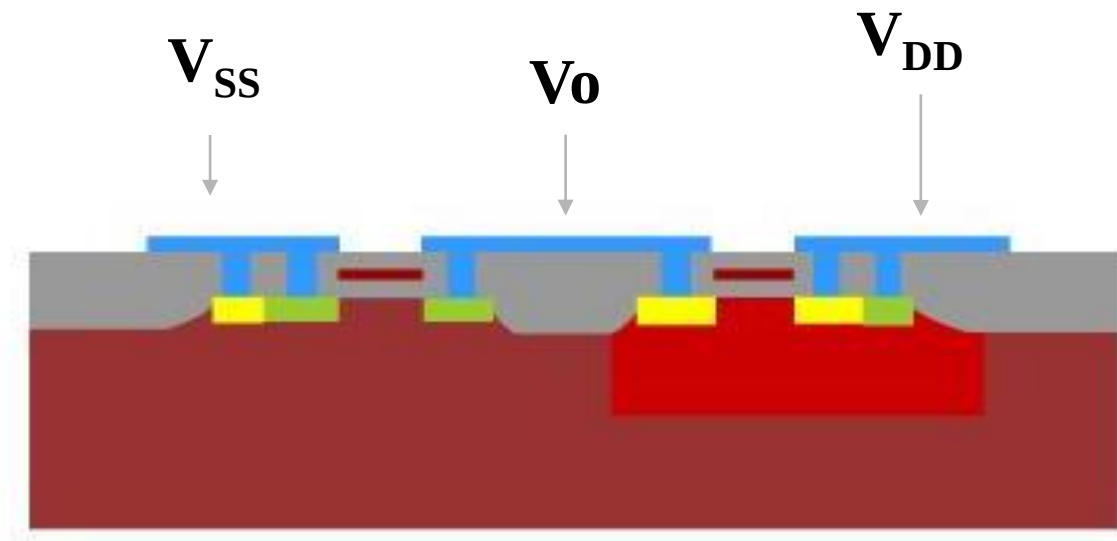
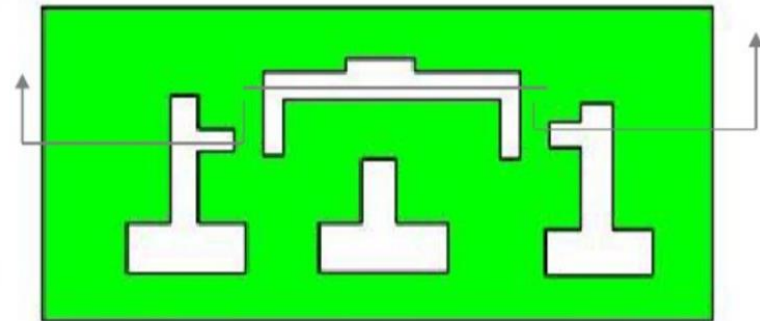
### Photolithography-6



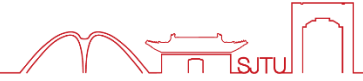
## (2) Metal etch and contact formation



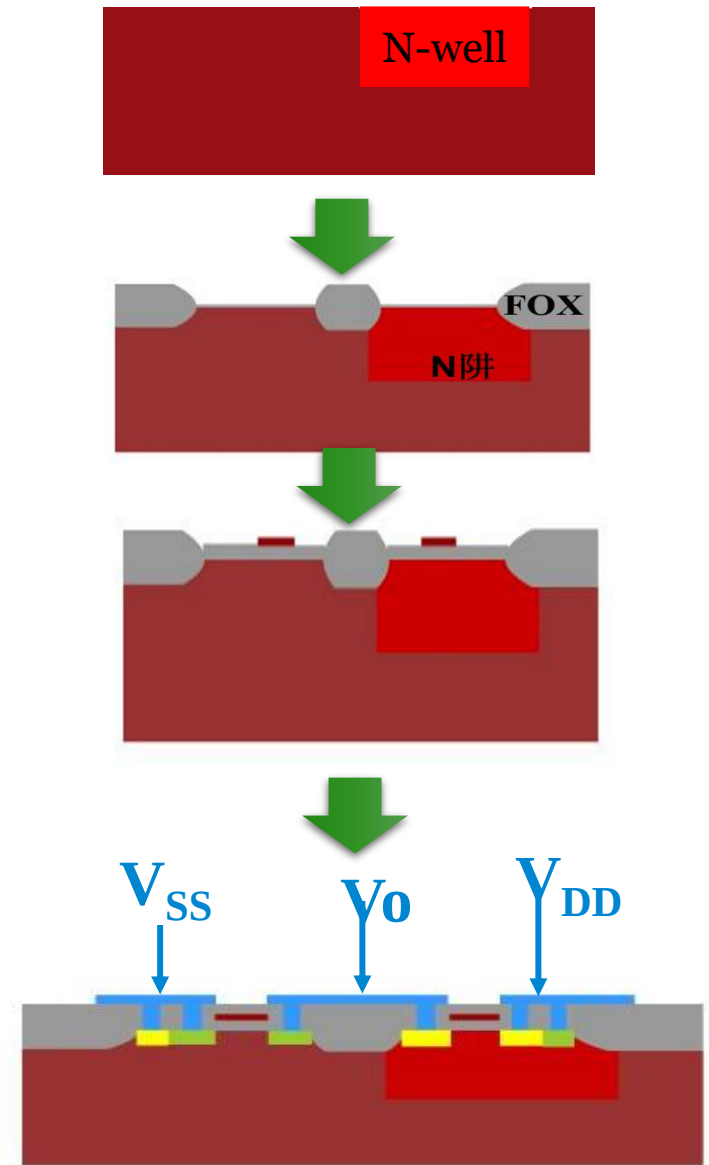
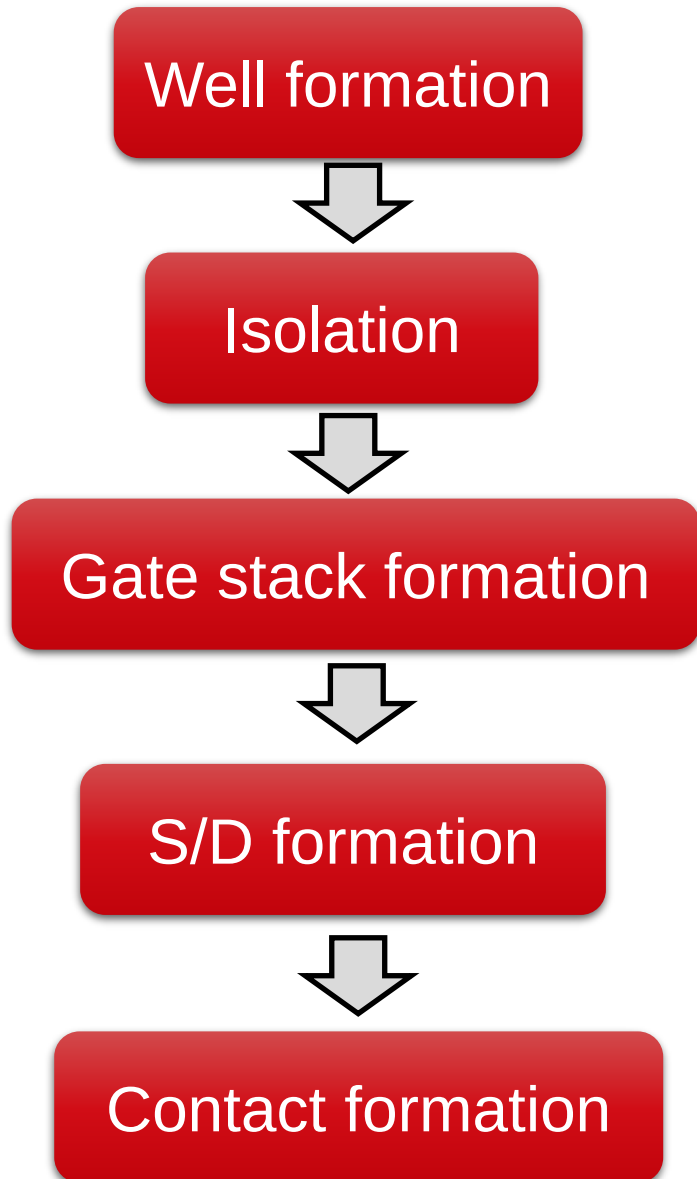
Photolithography-7



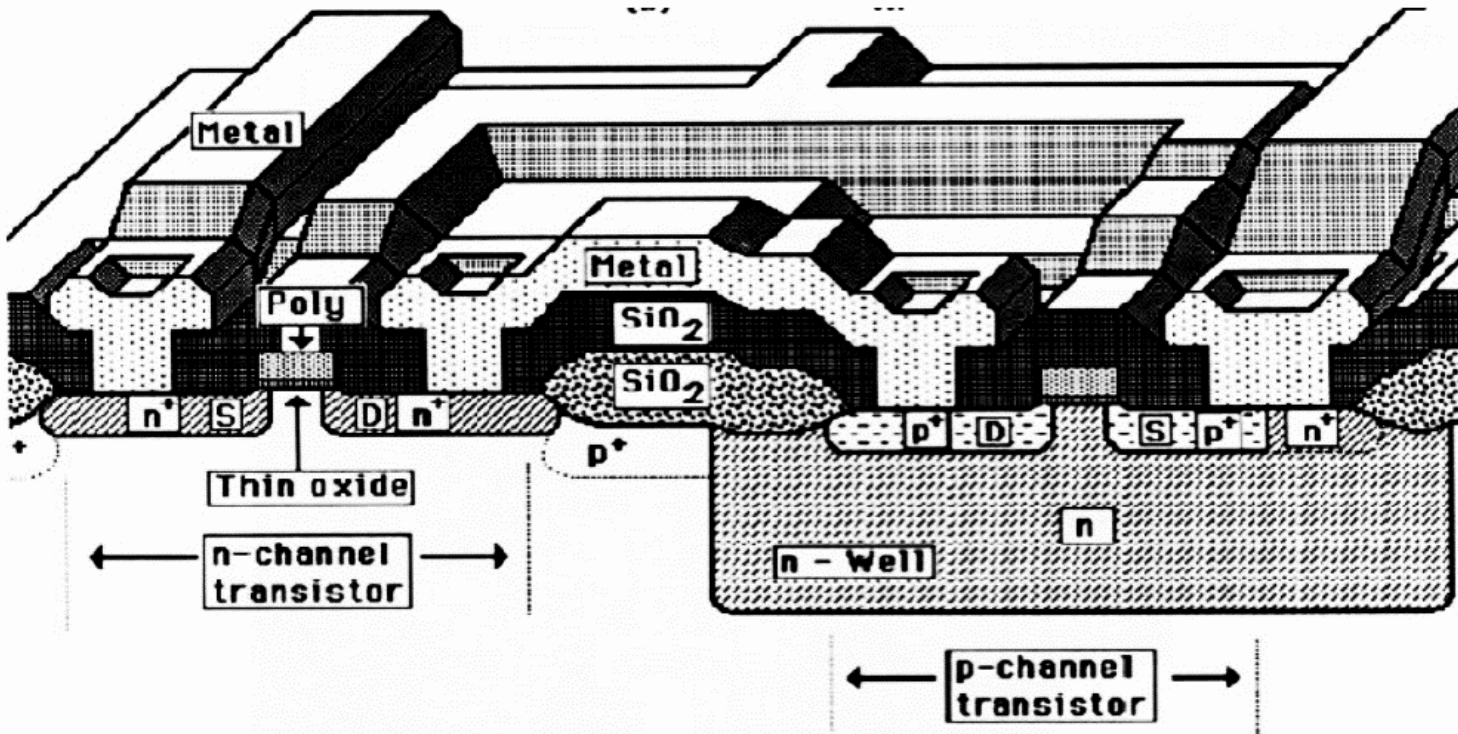
## 10.2 N-well CMOS process flow



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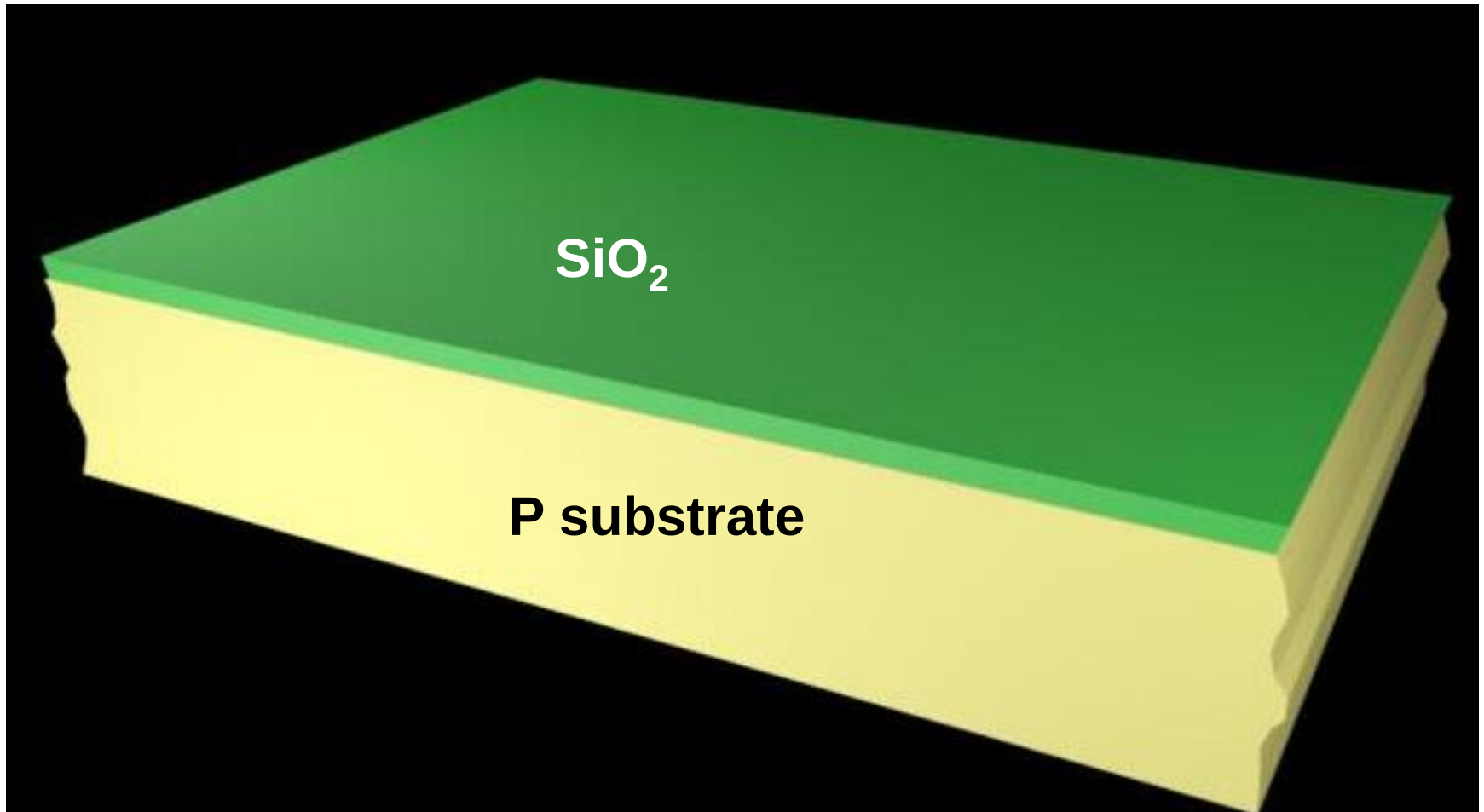
## 10.2 Structure of N-well CMOS



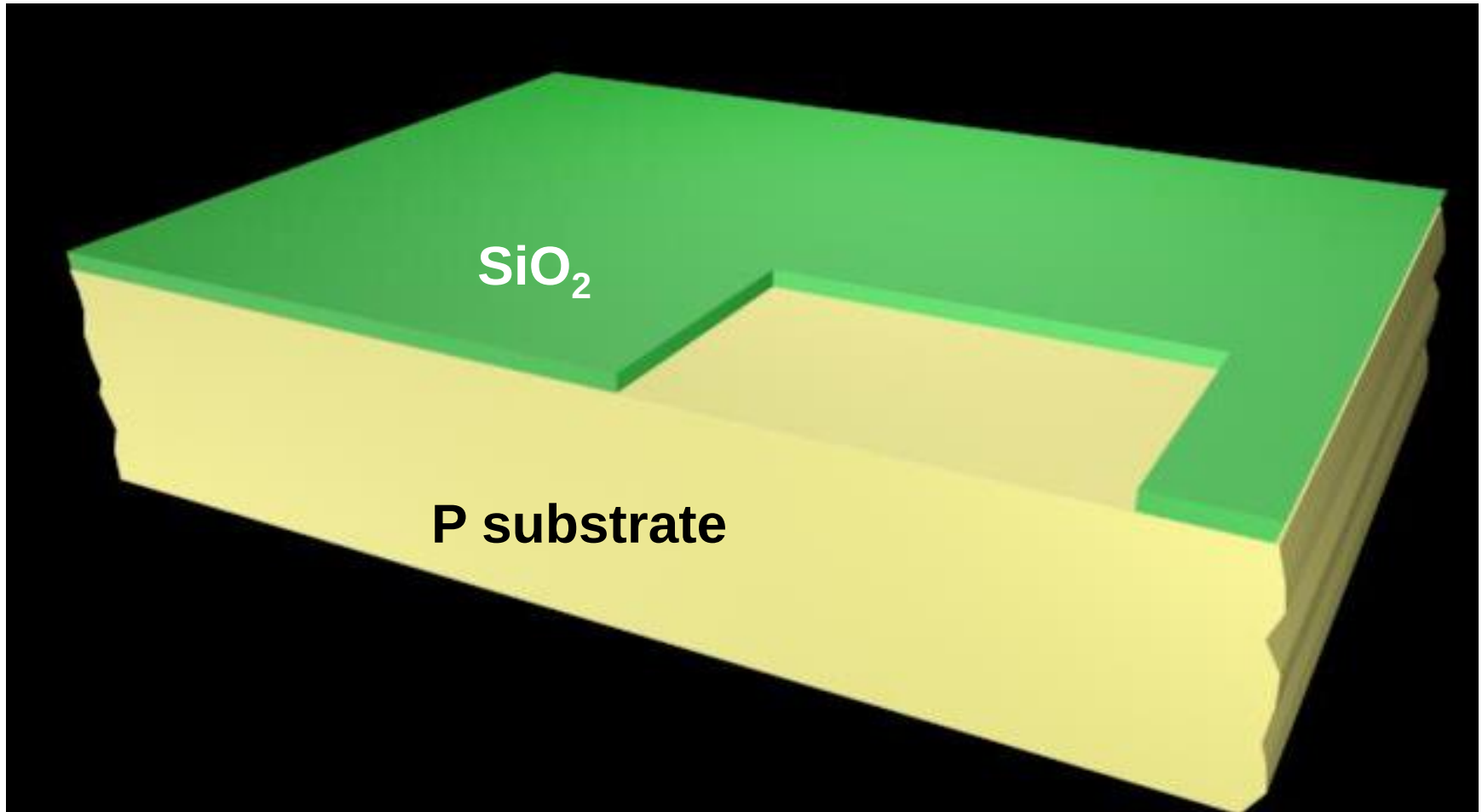
N-well  
CMOS  
structure

How many  
masks?  
Process flow?

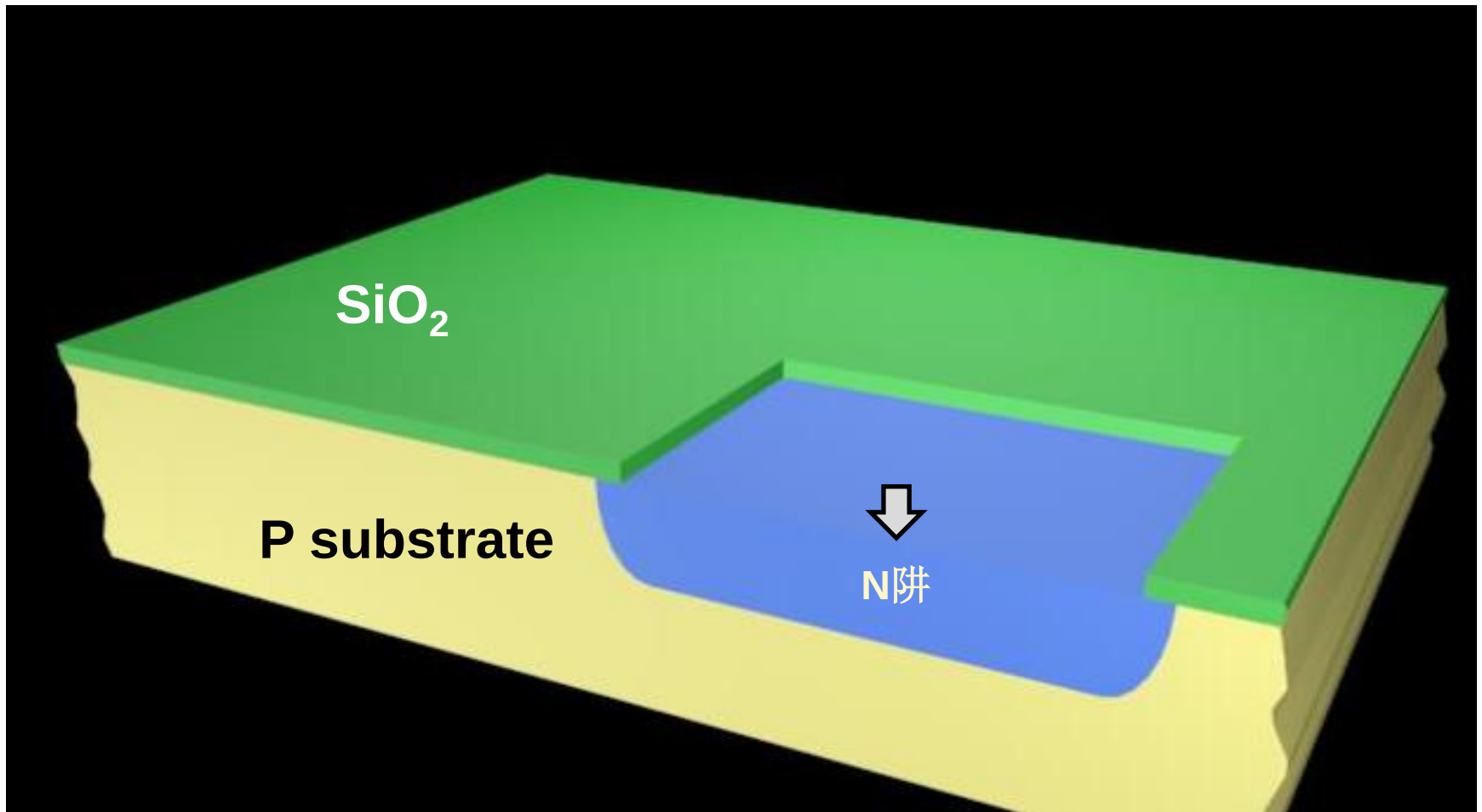
### Initial oxidation



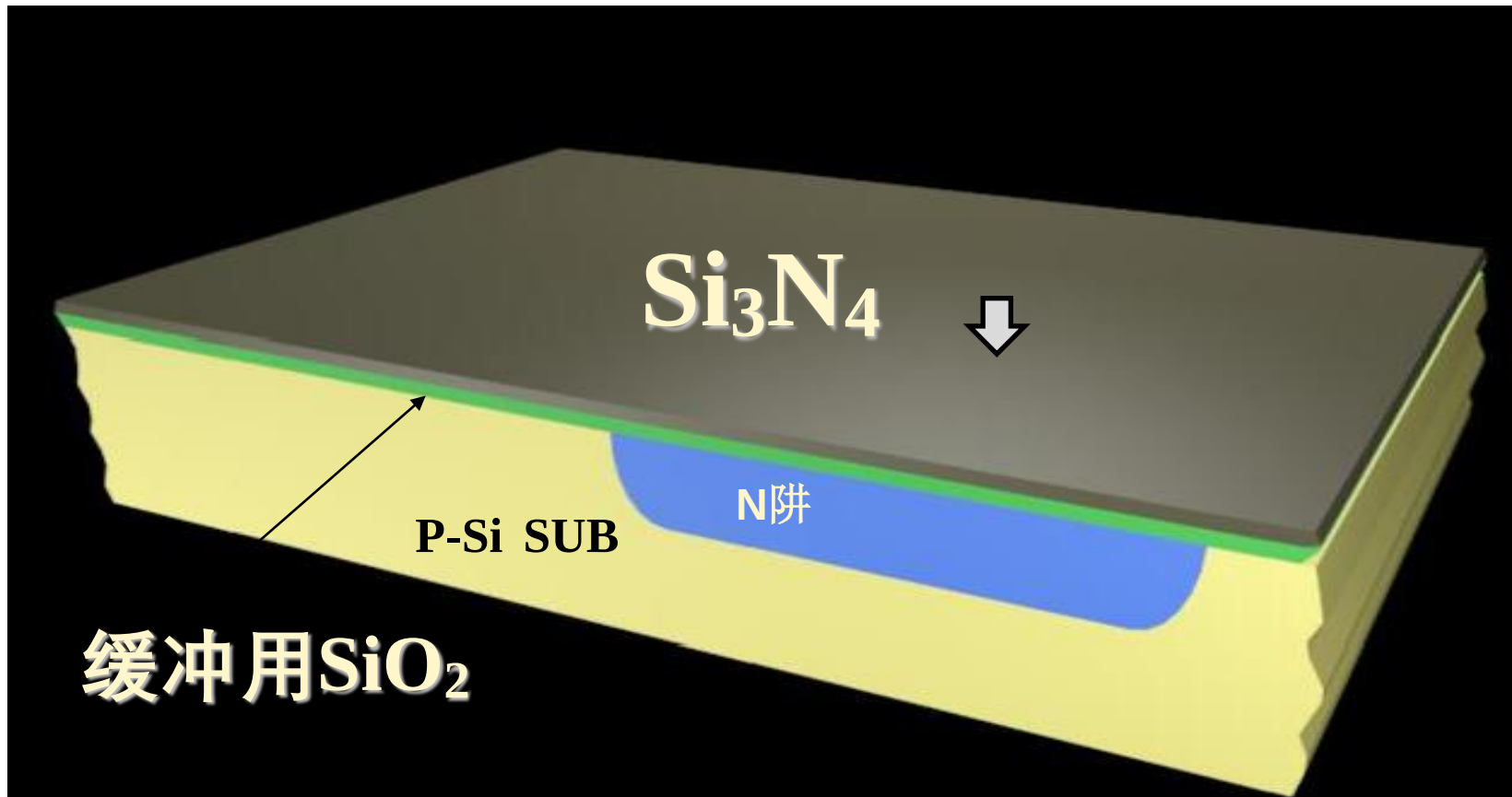
### Pattern mask opening for N-well implantation



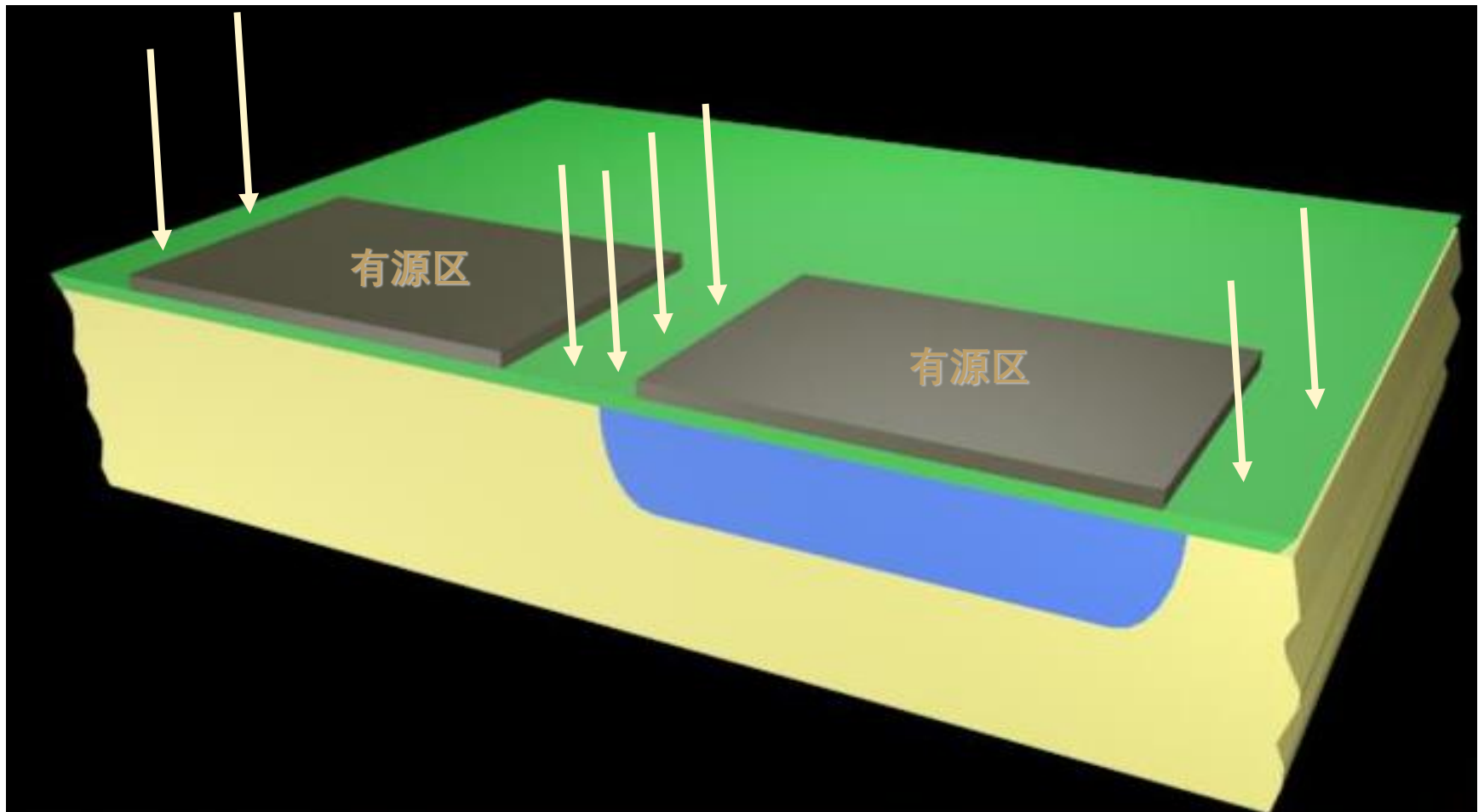
### shallow implantation and drive-in



### $\text{SiO}_2$ growth and $\text{Si}_3\text{N}_4$ deposition

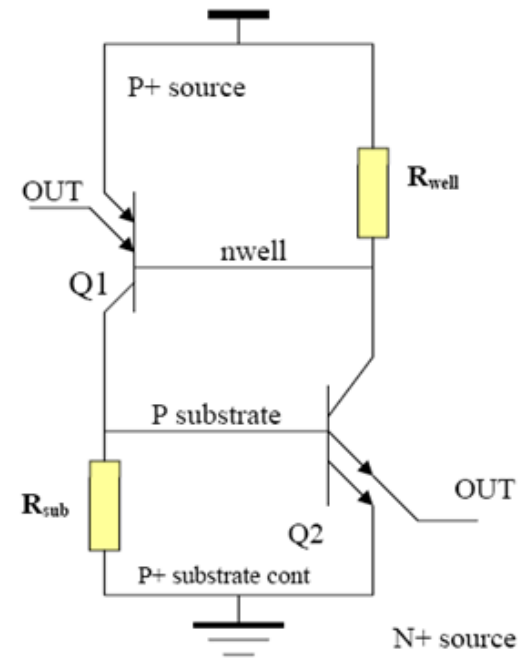


### Field area implantation

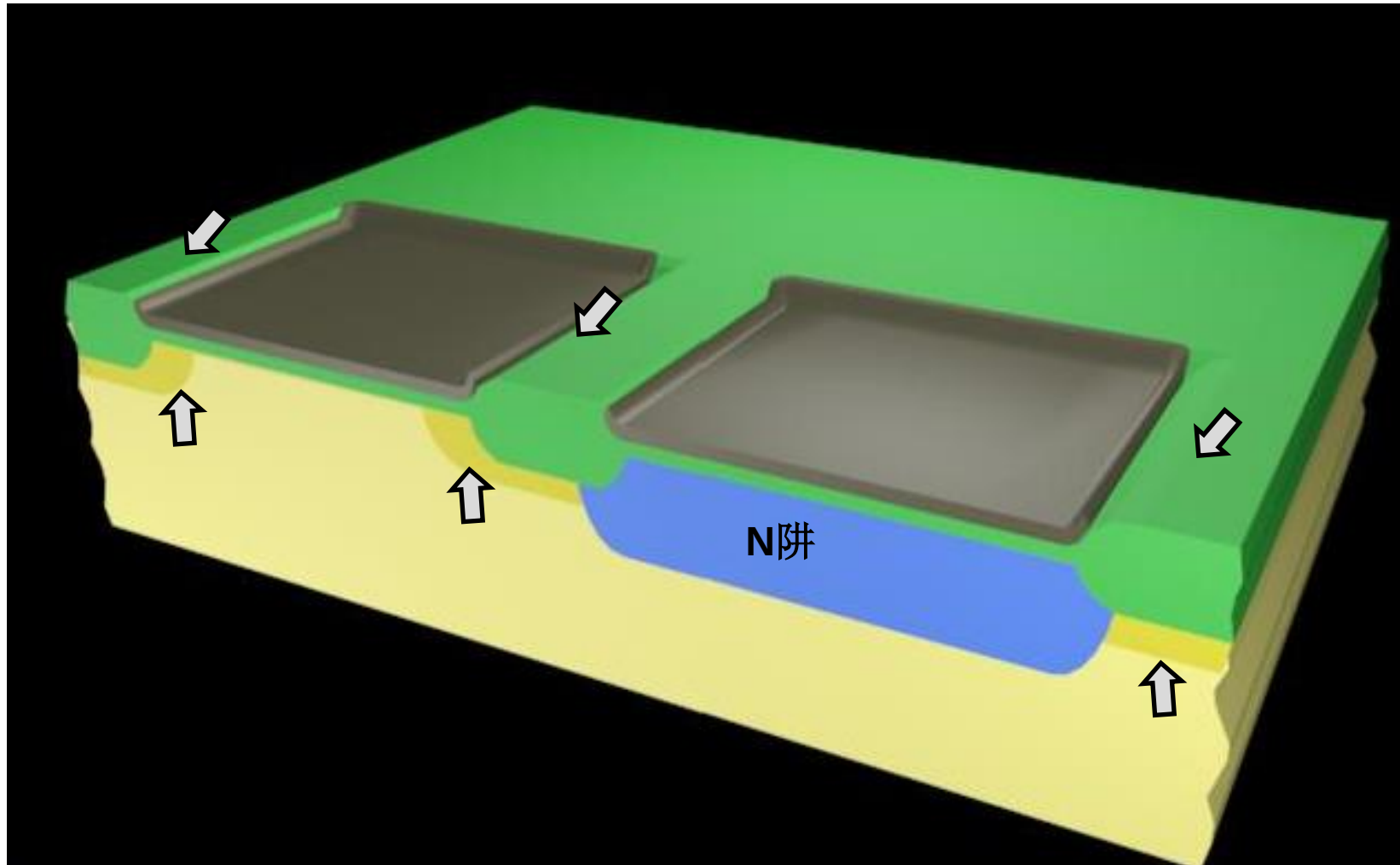




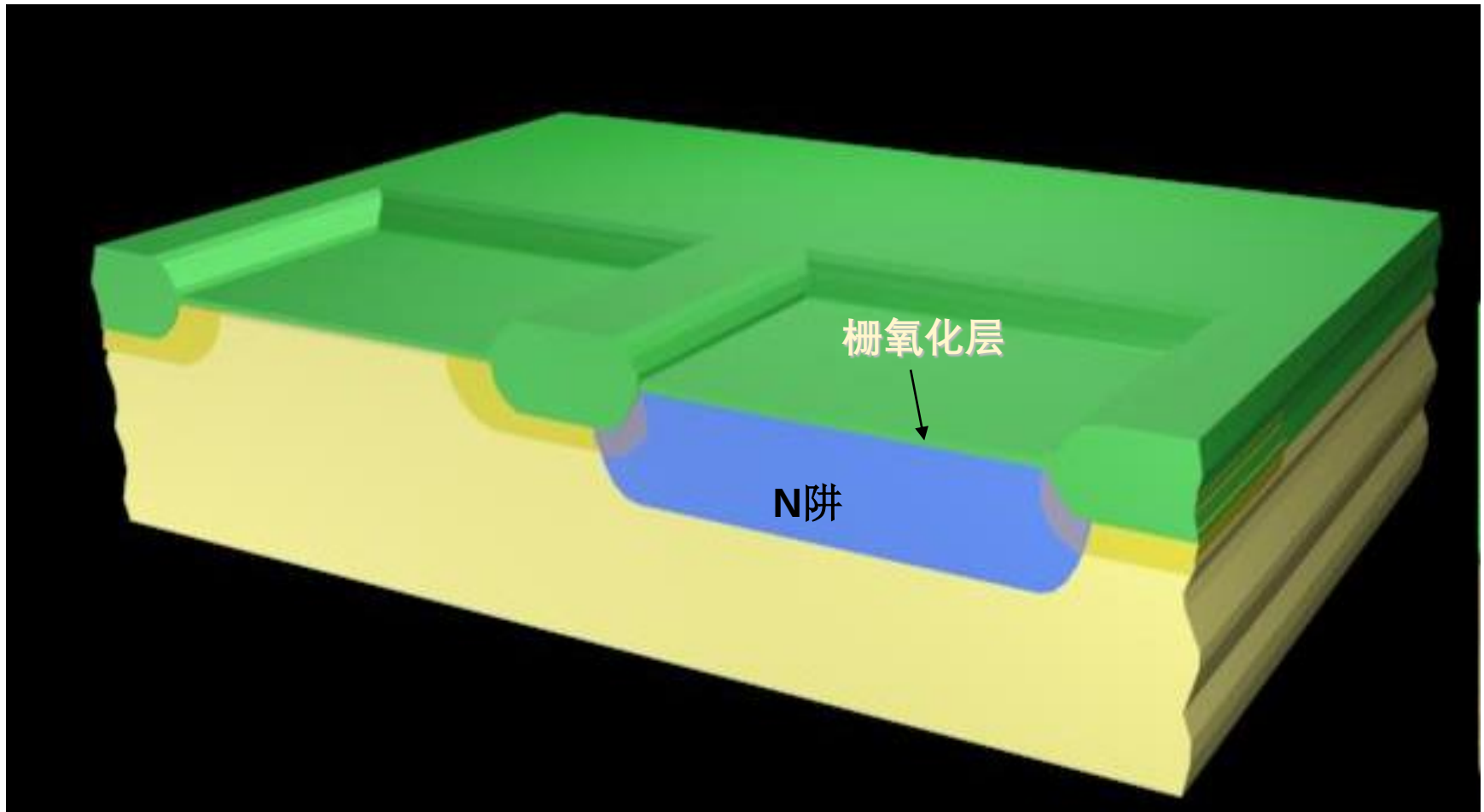
### 寄生BJT形成SCR的电路模型



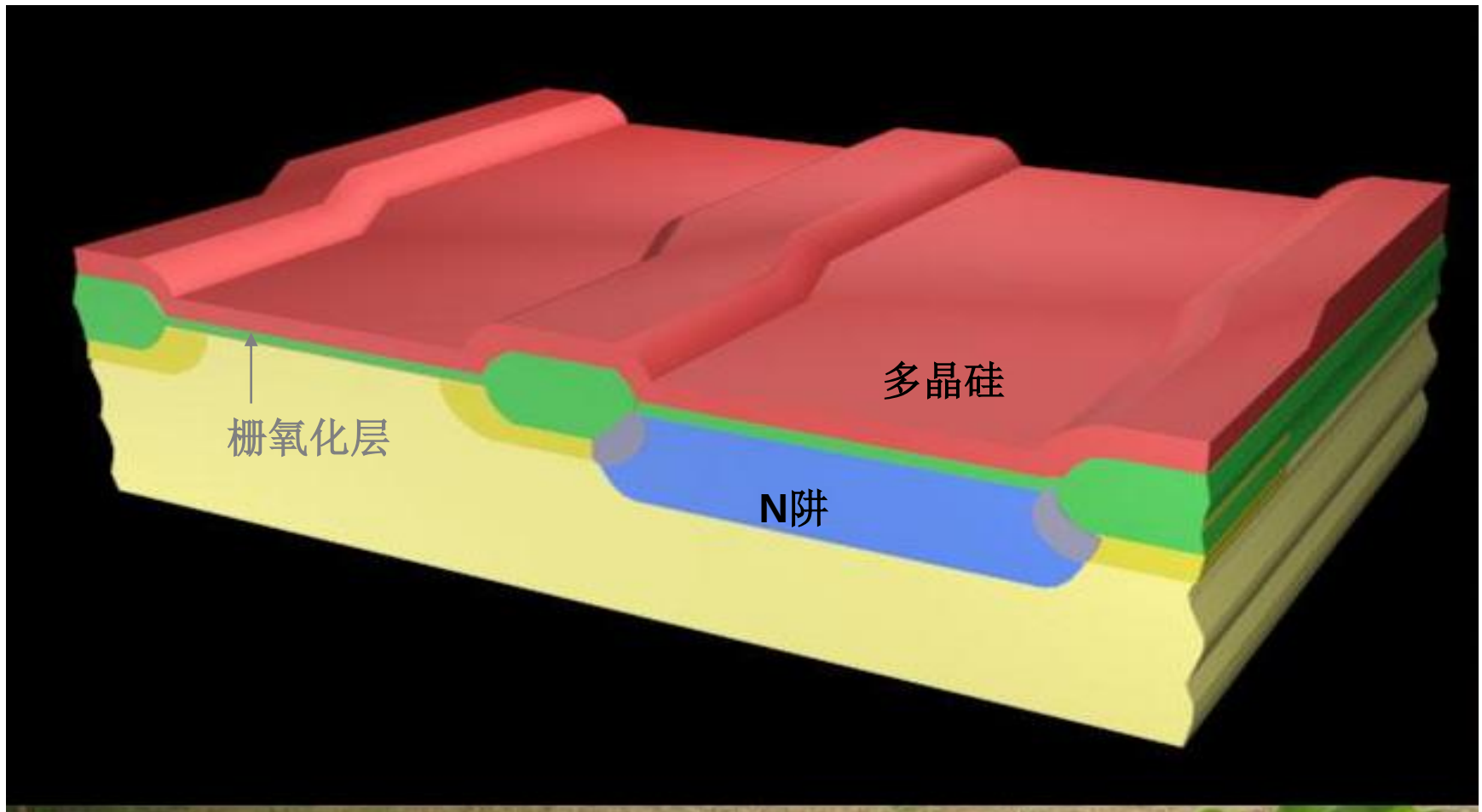
### Isolation: LOCOS



### Gate oxide growth

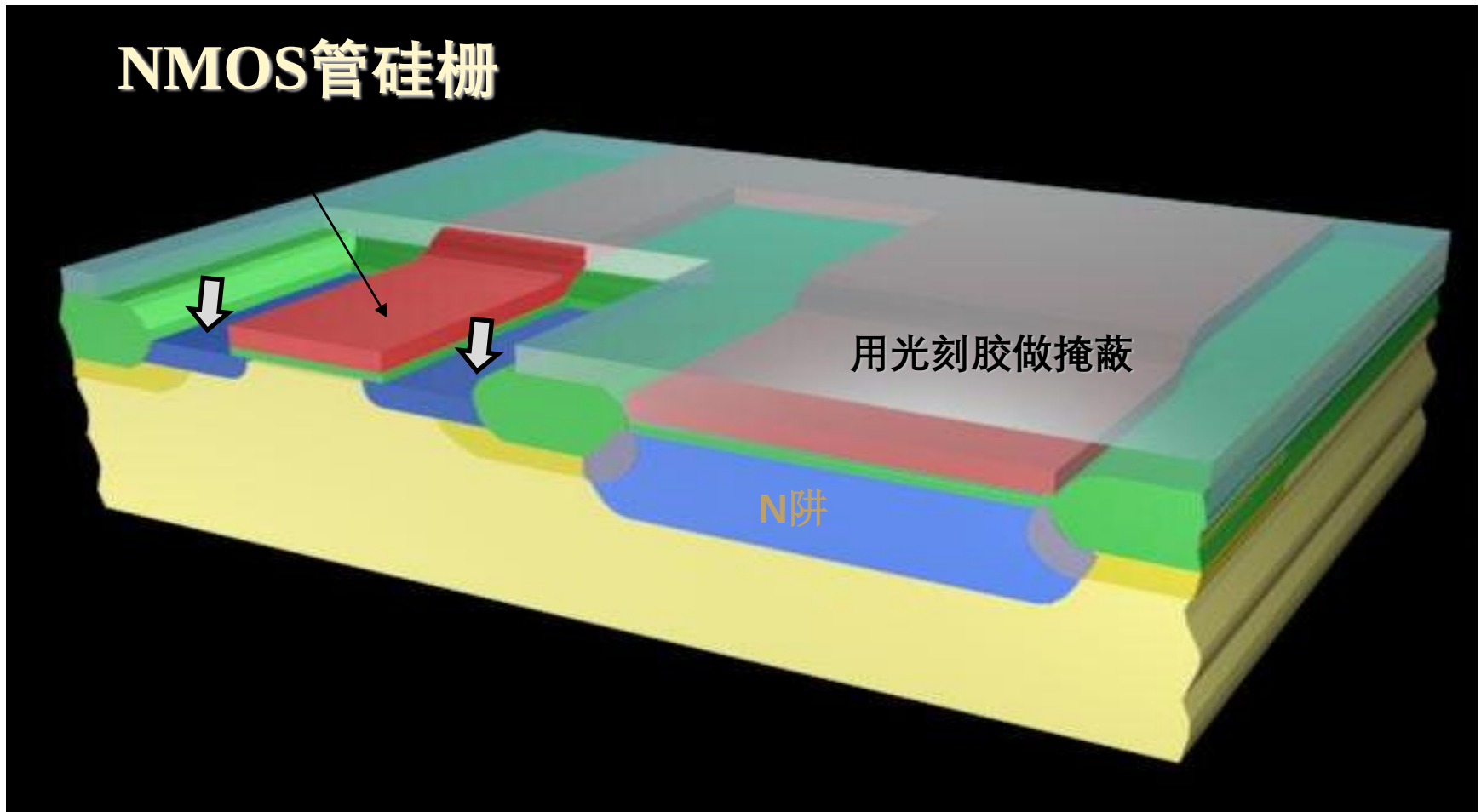


### Poly-Si deposition

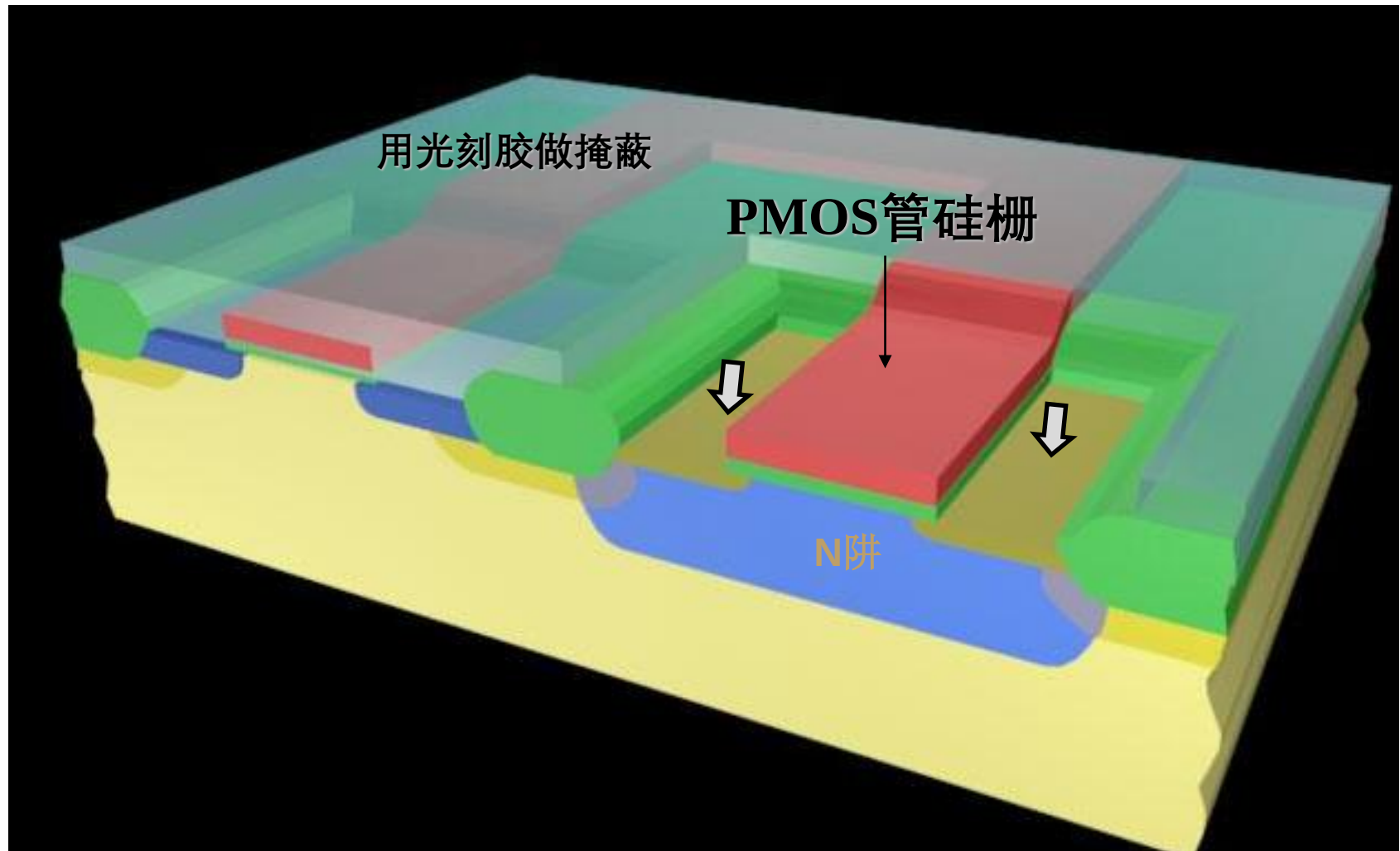


### S/D formation of NMOS

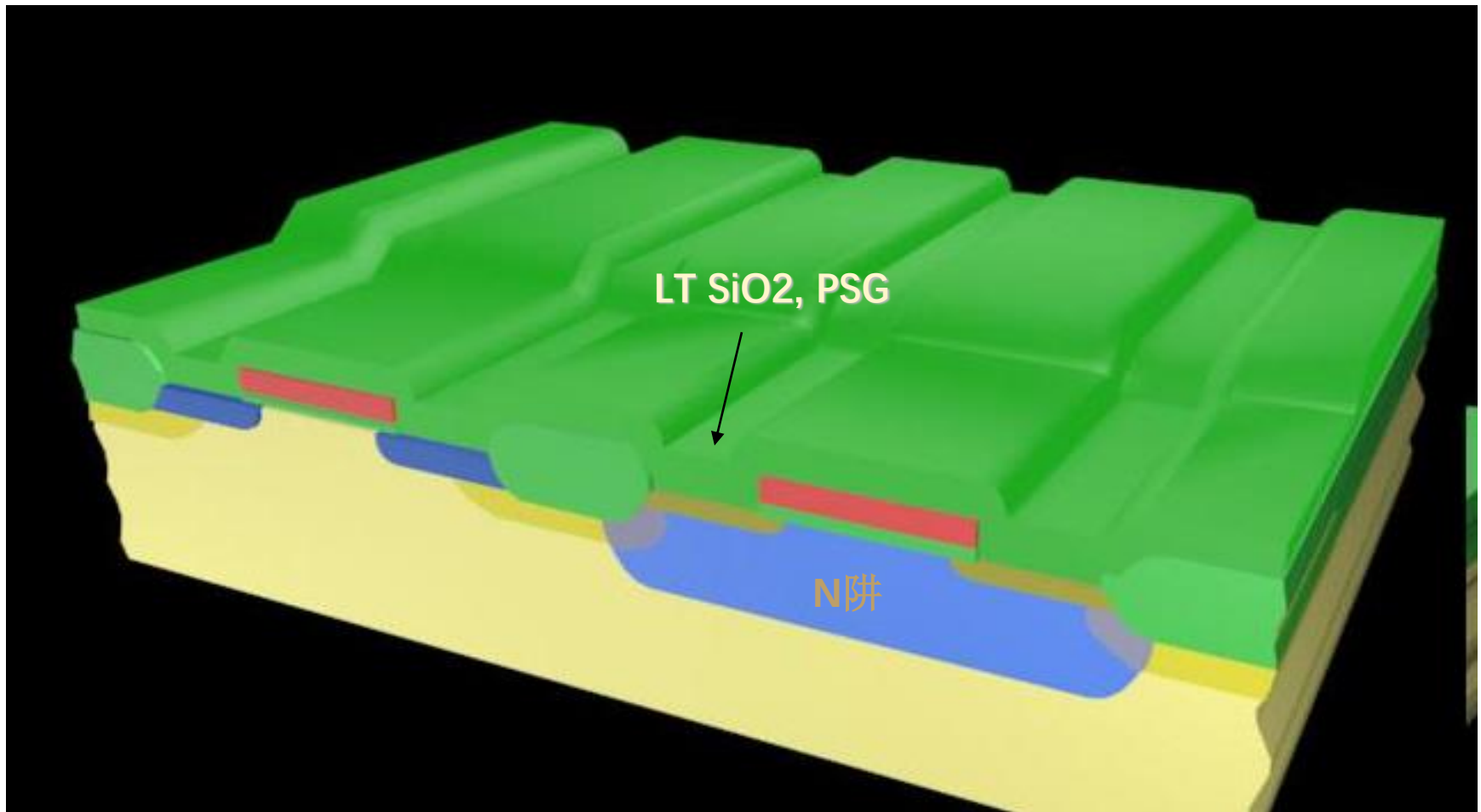
NMOS管硅栅



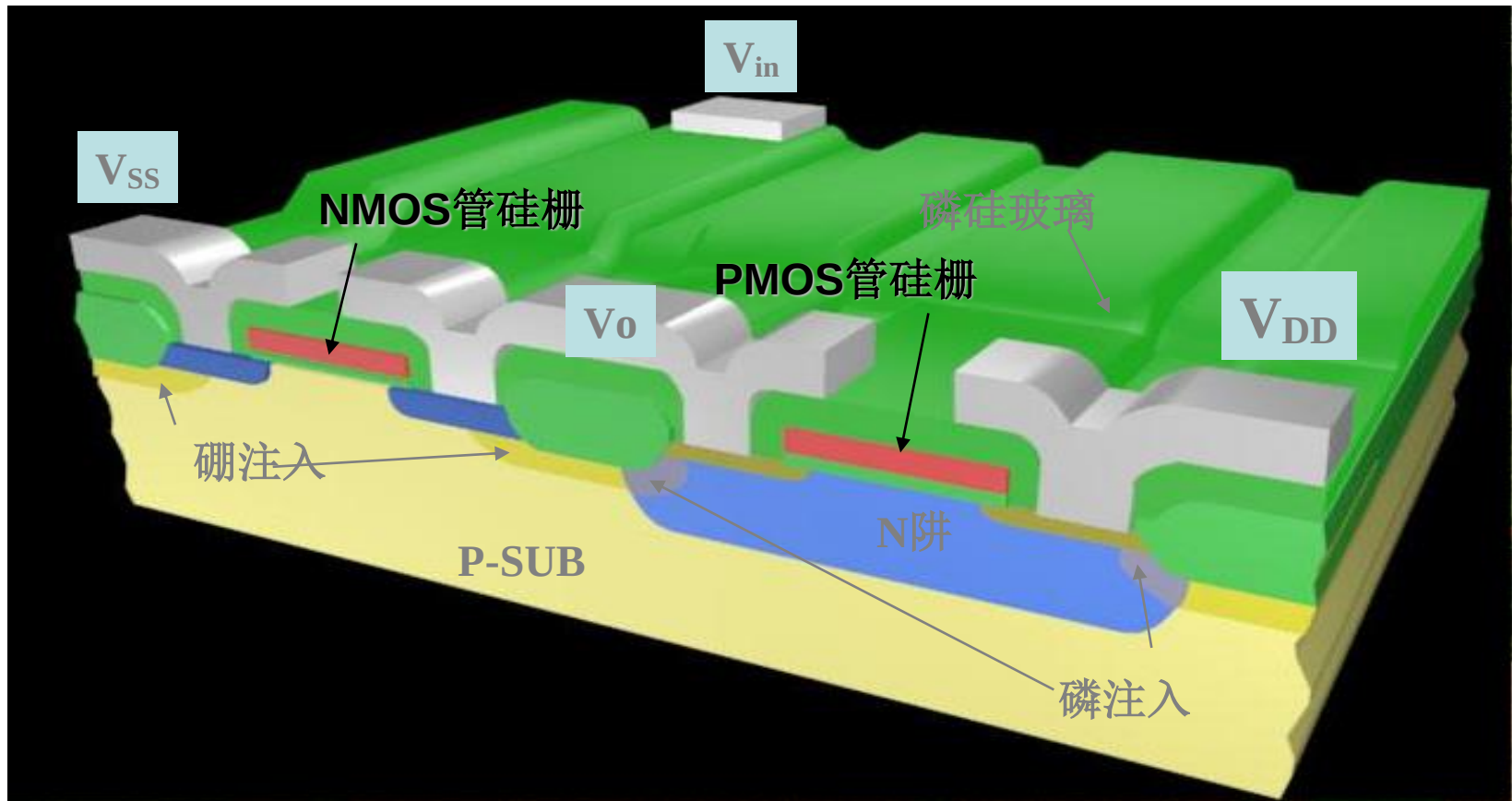
### S/D formation of PMOS



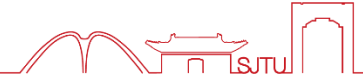
### LT-SiO<sub>2</sub> growth



### Metal deposition and etch



## 10.2.7 P-well CMOS process



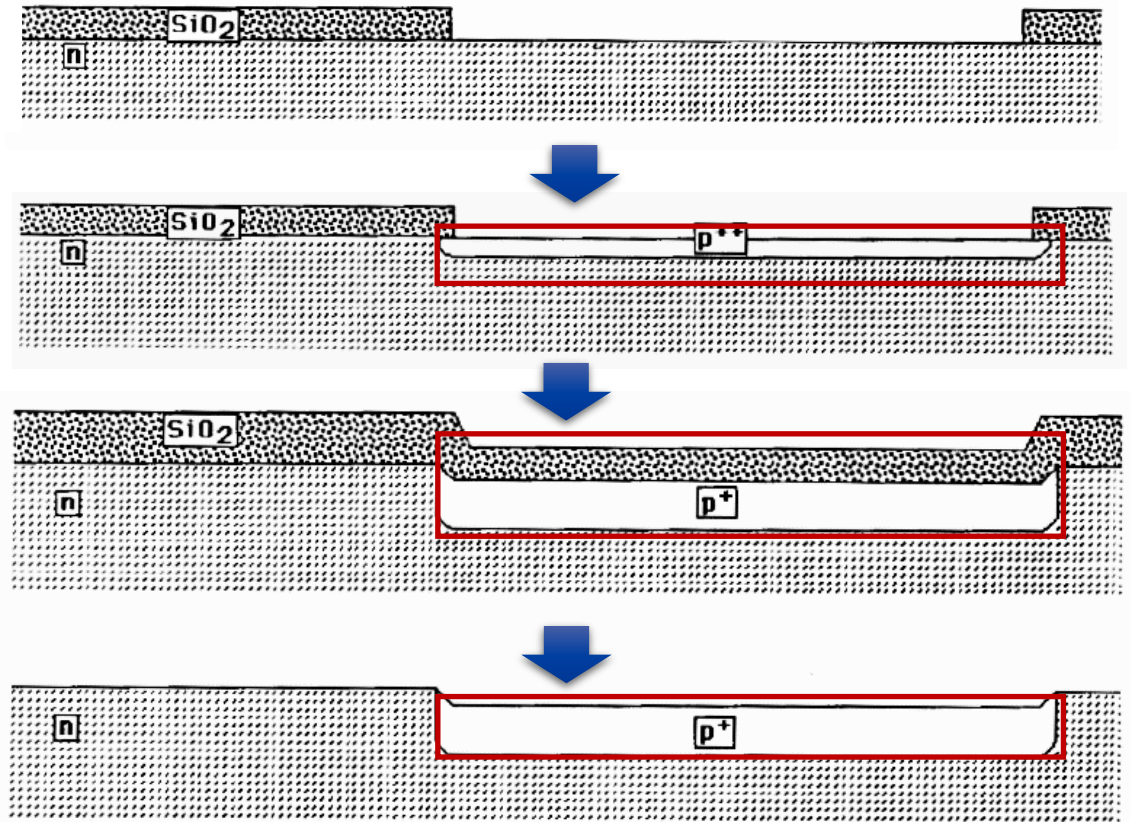
47

Pattern mask opening for P-well implantation

Shallow implantation of boron

Diffusion drive-in to form P-well in oxidizing ambient

Remove masking oxide



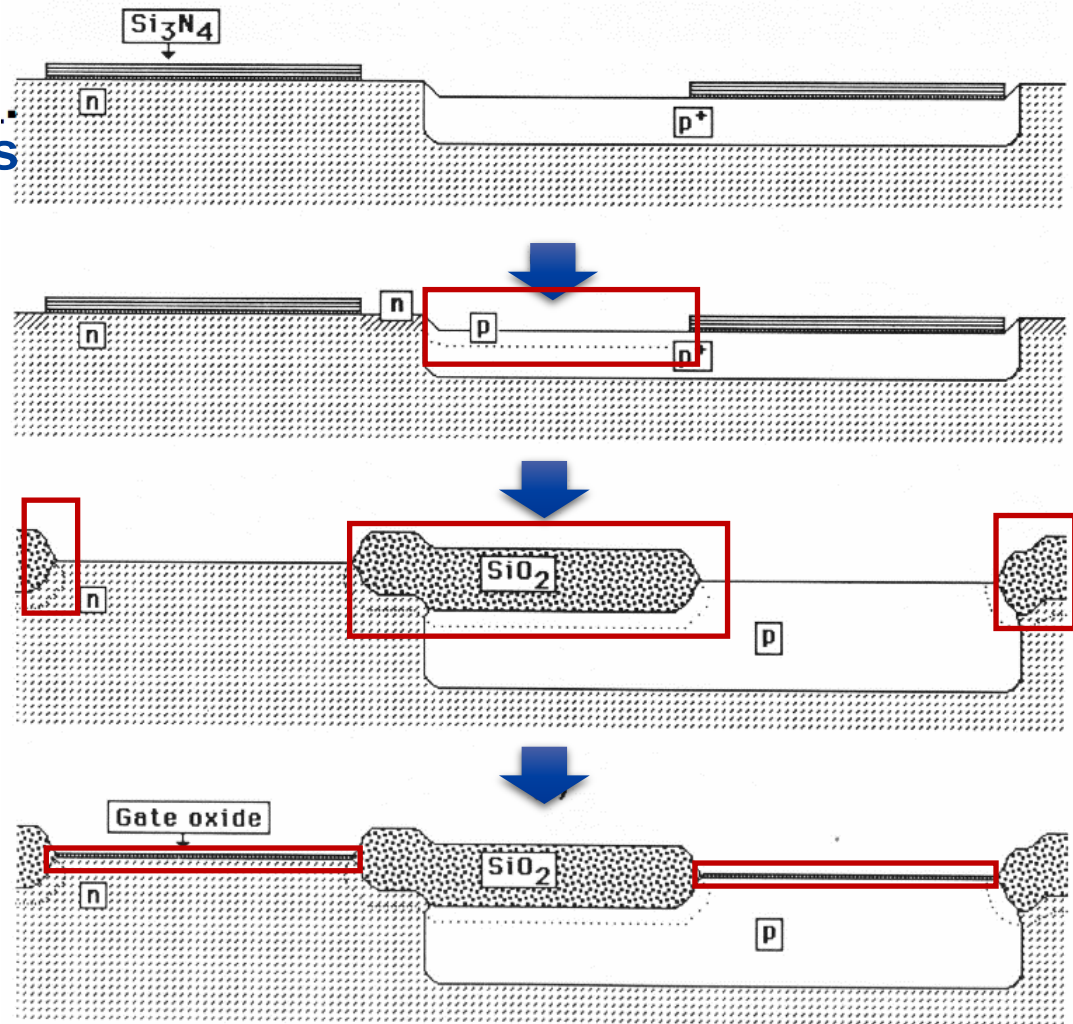
## 10.2.7 P-well CMOS process

Pad oxide growth and CVD  $\text{Si}_3\text{N}_4$ . Pattern field oxide regions

Blanket implant of Boron for n channel stop inside P--well  
Protect P--well regions with photoresist.

LOCOS Oxidation

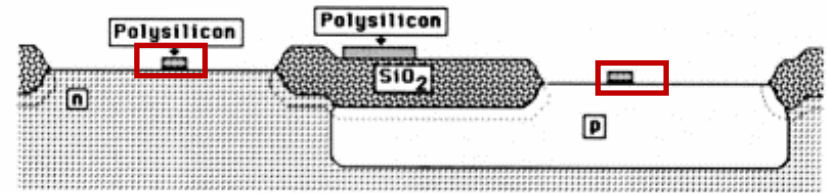
Thermal oxidation of gate  $\text{SiO}_2$



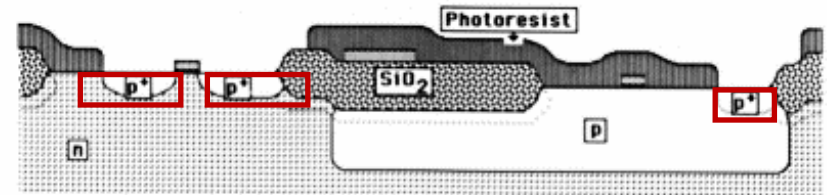
## 10.2.7 P-well CMOS process

CVD poly-Si

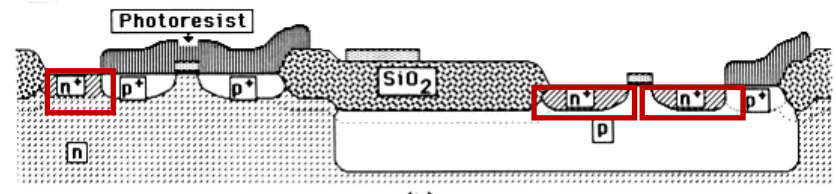
Pattern poly-Si gates



Protect All NMOSFET  
with photoresist.  
Boron implantation to form  
source/drain of PMOSFET



Form source drain of  
NMOSFET

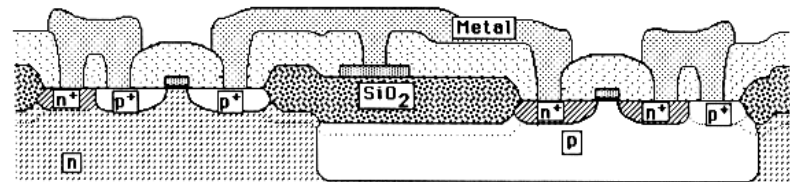
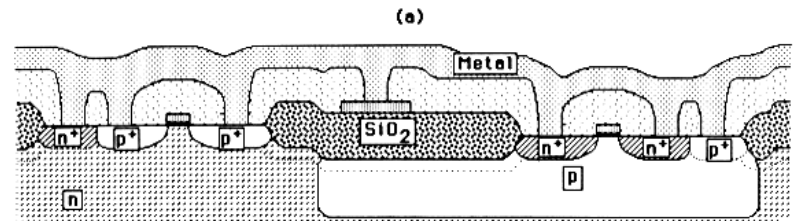
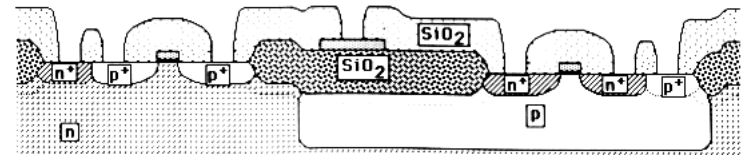
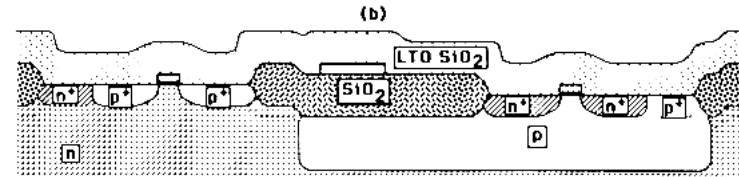


CVD SiO<sub>2</sub> (Low--  
temperature oxide)

Pattern and etch contact  
openings to source/drain, well  
contact, and substrate contact.

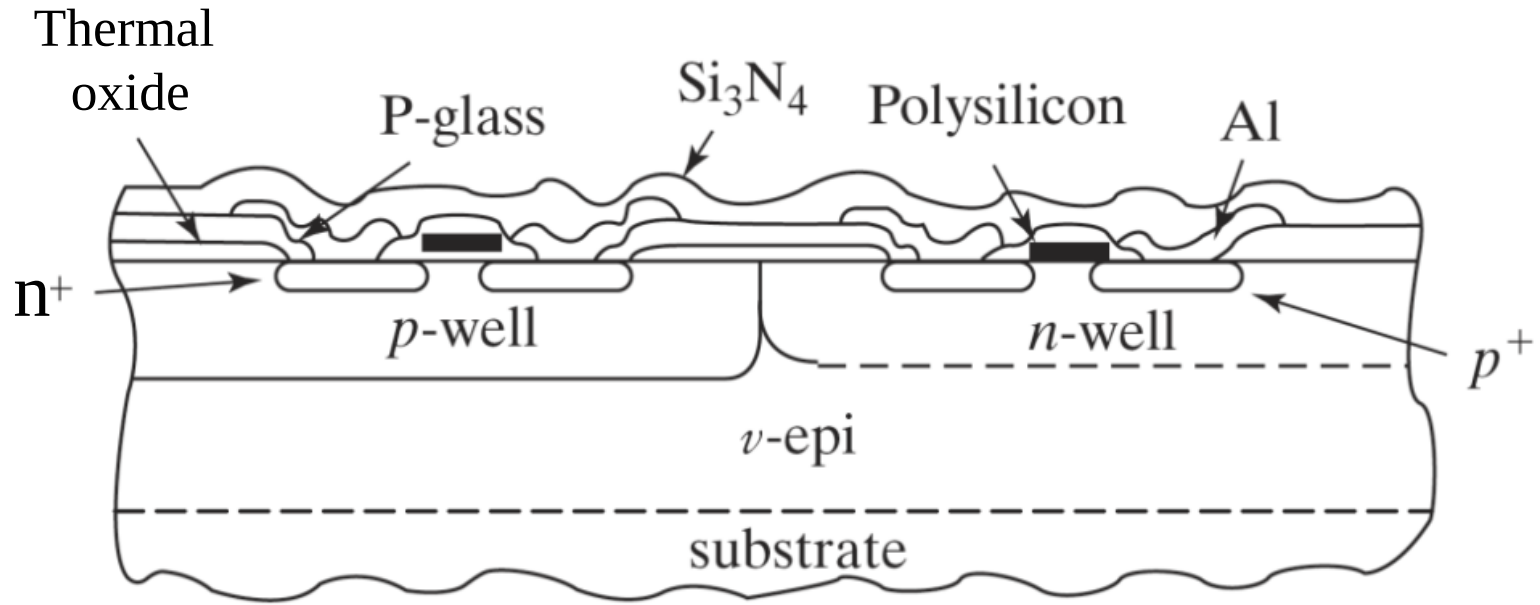
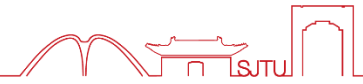
Metal deposition

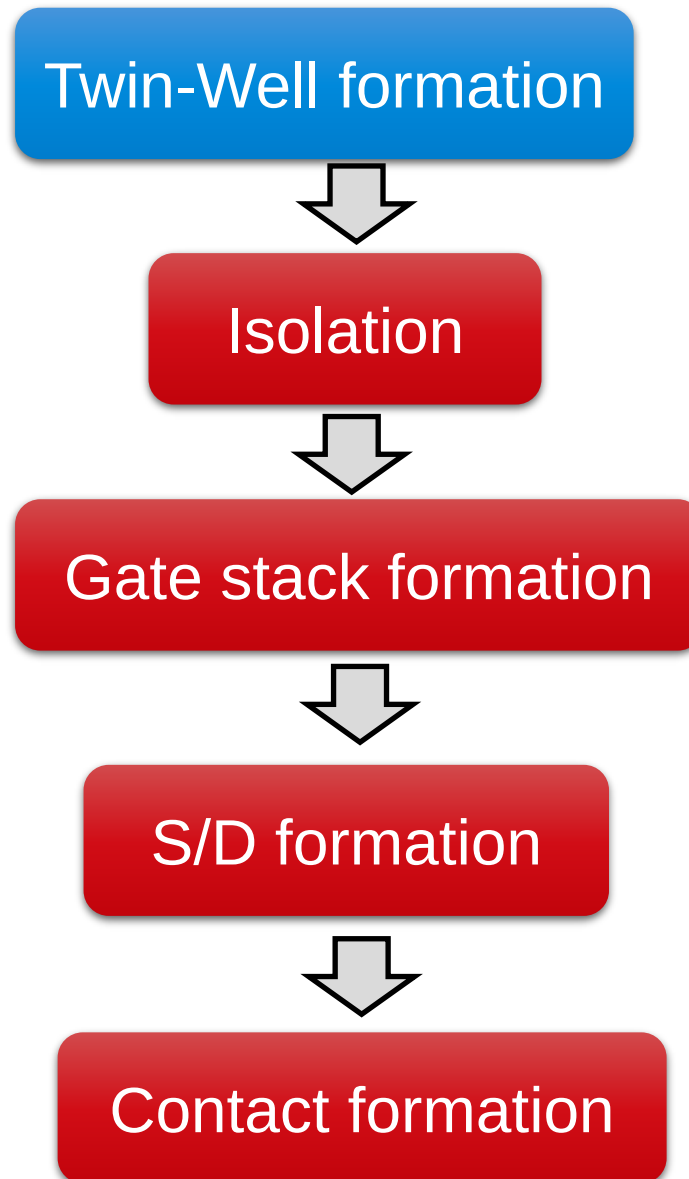
Pattern and etch  
Metal interconnects



- ✎ **10. 1 Basics of CMOS process**
- ✎ **10. 2 Single-well process**
- ✎ **10. 3 Twin-well process**
- ✎ **10. 4 FinFET process**
- ✎ **10. 5 3D NAND process**
- ✎ **10. 6 Summary and homework**

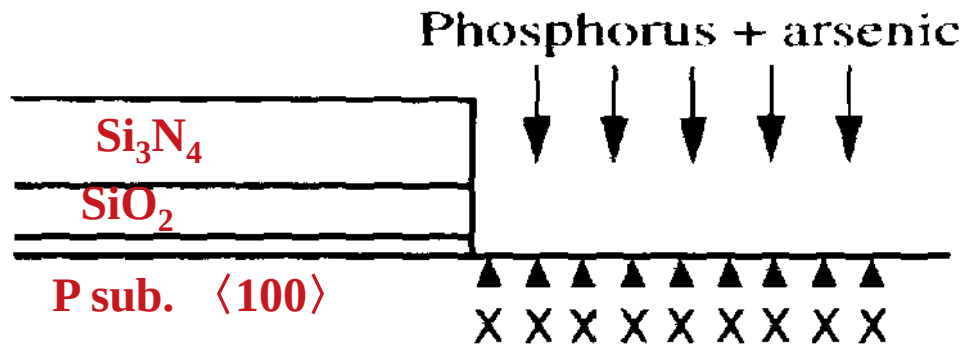
## 10.3.0 Twin-well CMOS structure





### N-well implantation

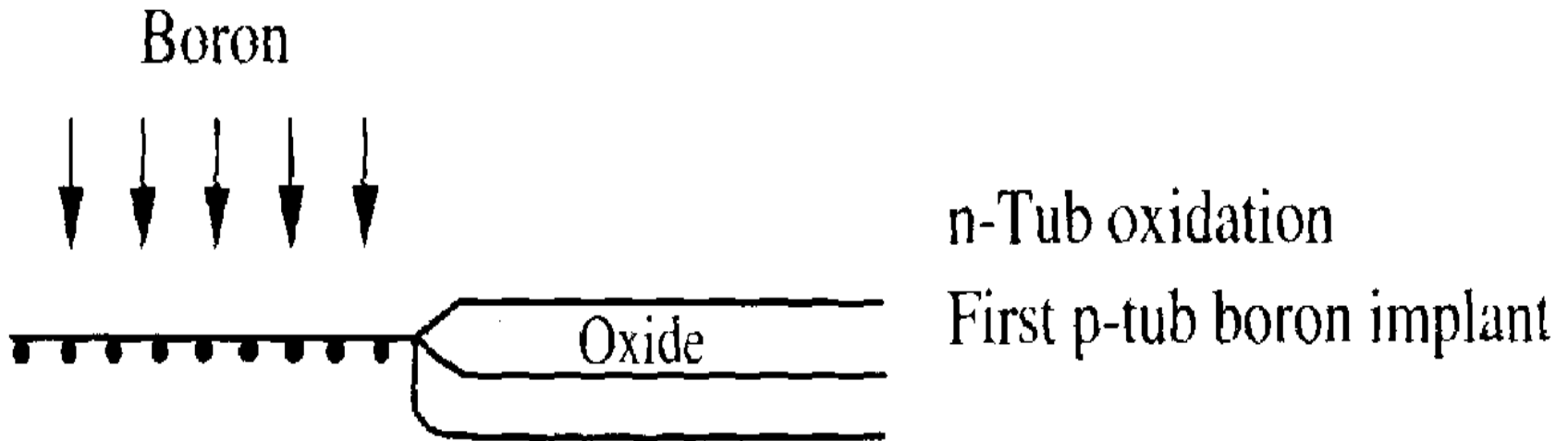
- Initial oxidation and  $\text{Si}_3\text{N}_4$  deposition
- Pattern mas opening for N-well region
- $\text{SiO}_2$  and  $\text{Si}_3\text{N}_4$  etching for N-well region
- N-well implantation



n-Tub patterning  
n-Tub phosphorus implant  
+ As implant

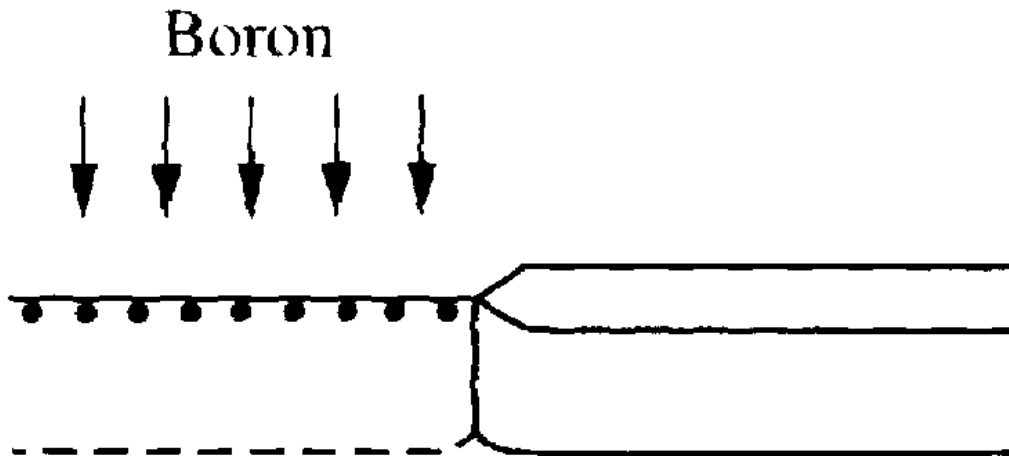
### P-well implantation

- Thick oxide growth in N-well region
- Remove  $\text{SiO}_2$  and  $\text{Si}_3\text{N}_4$  in P-well region
- P-well implantation

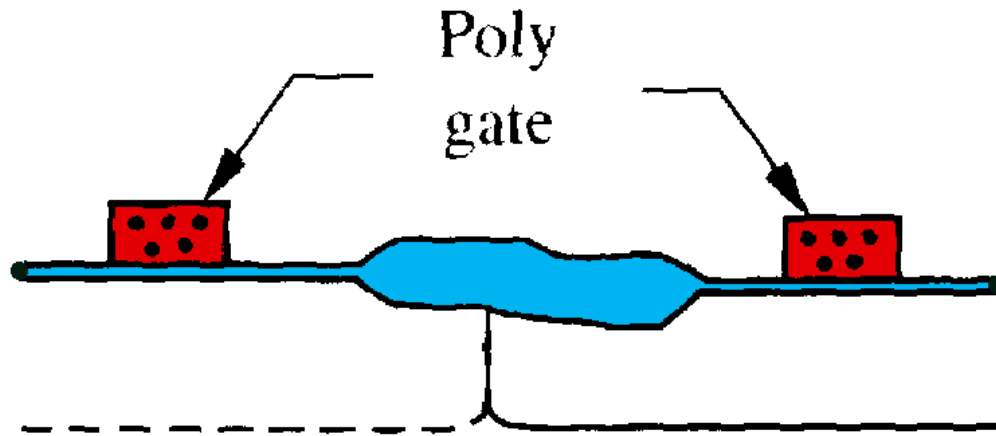


### Drive-in for twin-well formation

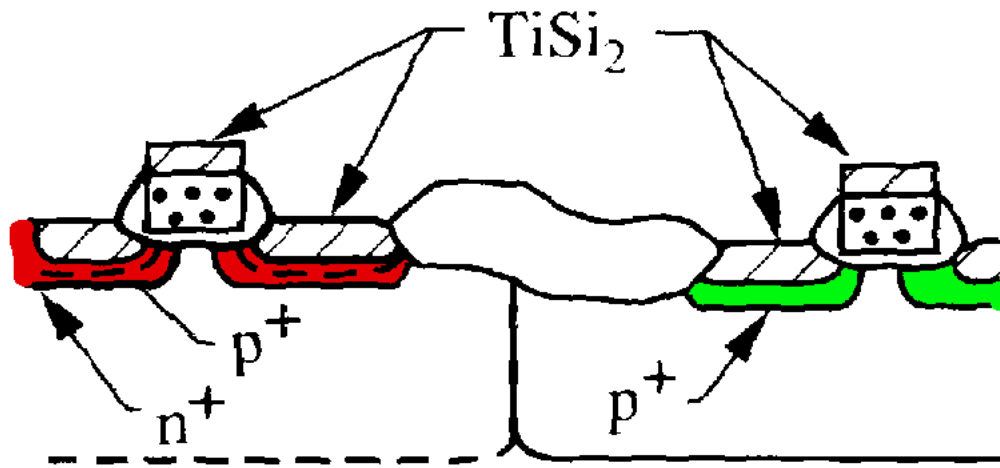
- Drive-in, junction depth is about  $1.8\mu\text{m}$
- Remove oxide on N-well region



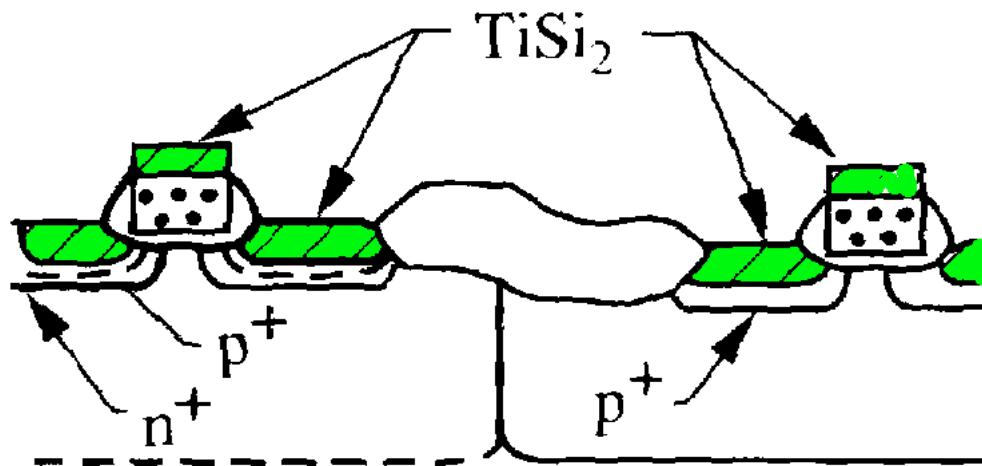
Tub drive-in  
Second and third p-tub  
boron implant  
Strip n-tub oxide



Thin-ox patterning  
HIPOX field oxidation  
Gate oxidation  
Undoped poly deposition  
Define poly gate



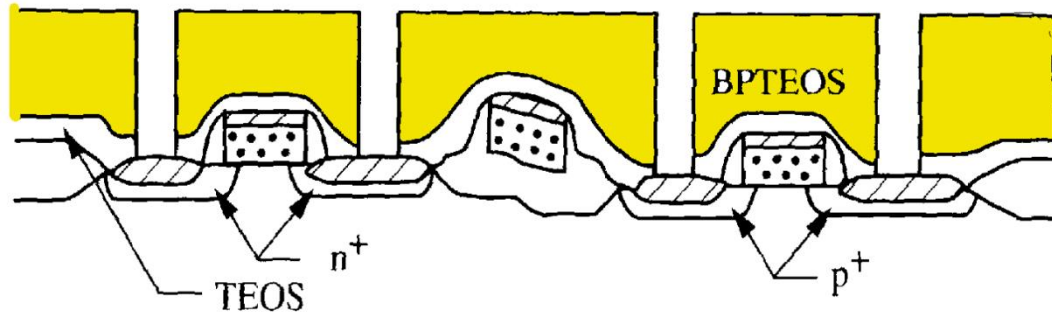
Forming sidewall spacer  
Selective  $n^+$  implant  
Selective  $p^+$  implant  
S/D dopant activation  
Self-align silicidation



Forming sidewall spacer  
Selective  $n^+$  implant  
Selective  $p^+$  implant  
S/D dopant activation  
Self-align silicidation

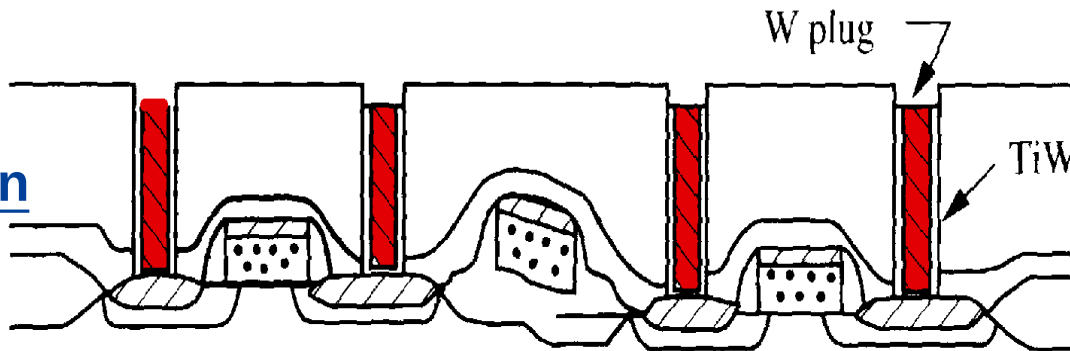
## 10.3.4 S/D and contact formation

### Via hole opening



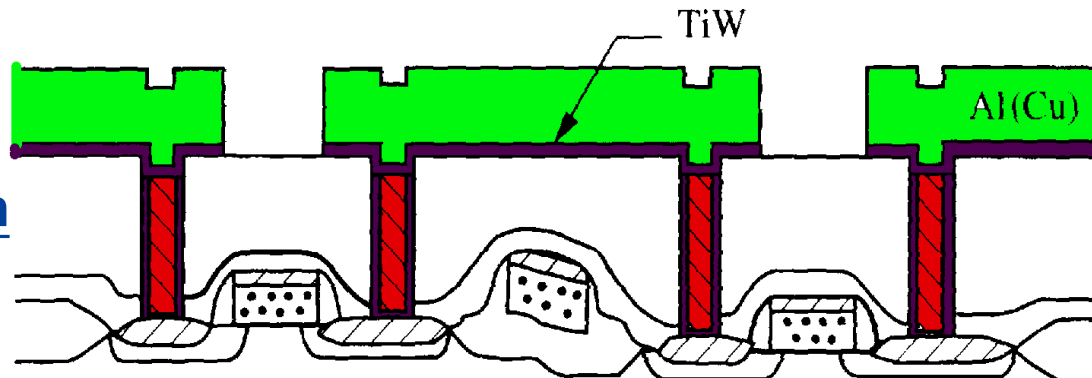
Dielectric 1 planarization:  
Undoped TEOS deposition  
BPTEOS deposition  
Flow BPTEOS  
Photoresist  
Resist etchback of BPTEOS  
Window 1 pattern transfer

### Metal deposition



Window 1 planarization:  
Inert sputter TiW  
LPVCD layer tungsten  
Blanket etchback tungsten  
to oxide

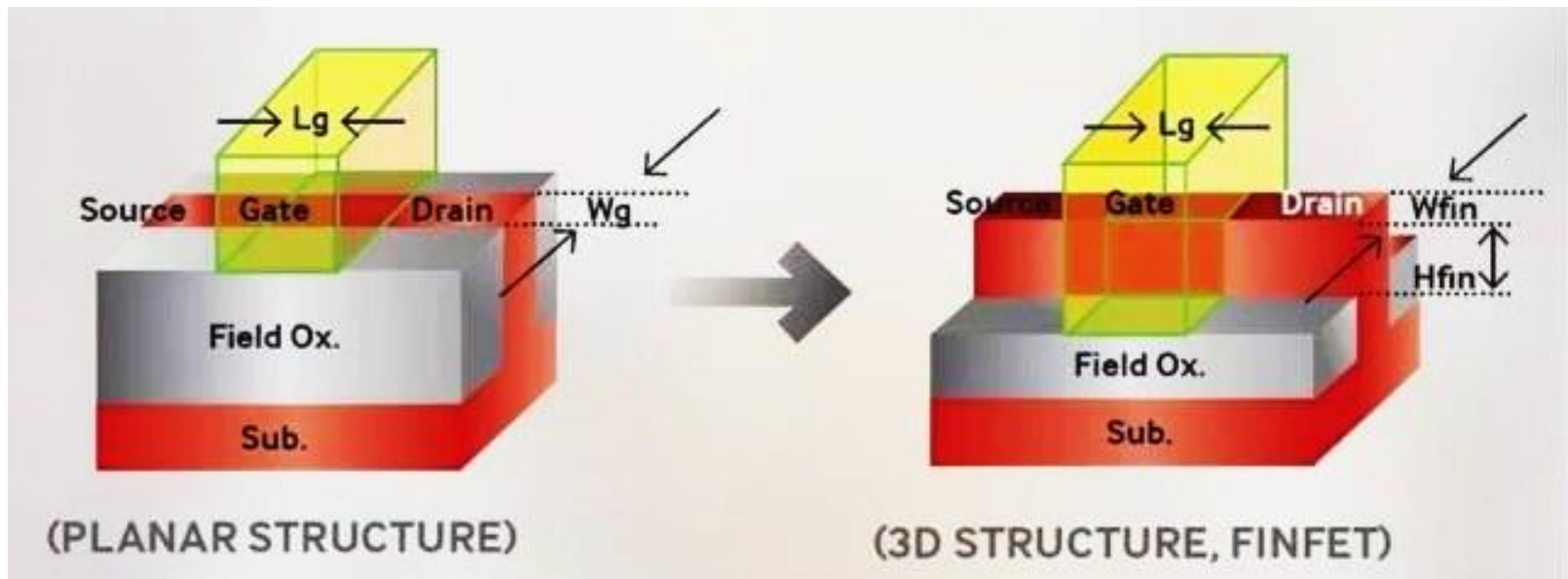
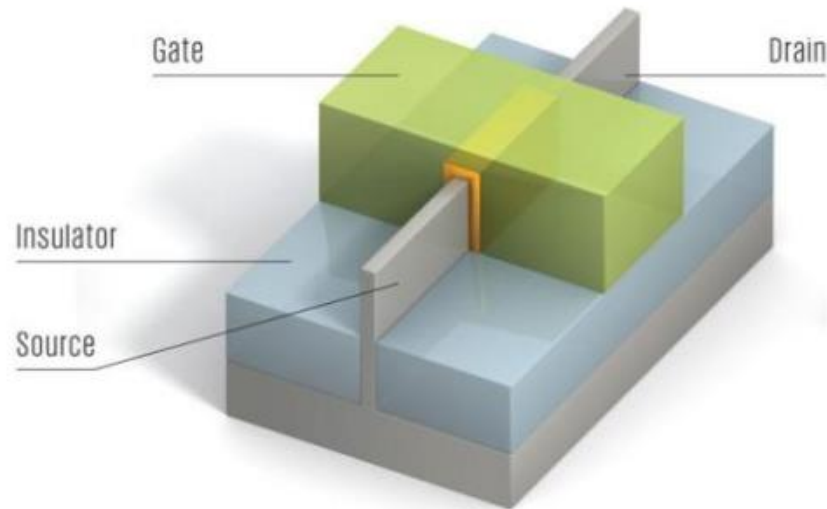
### Metal connection



Sputter TiW  
Al (0.5% Cu) deposition  
Metal 1 patterning

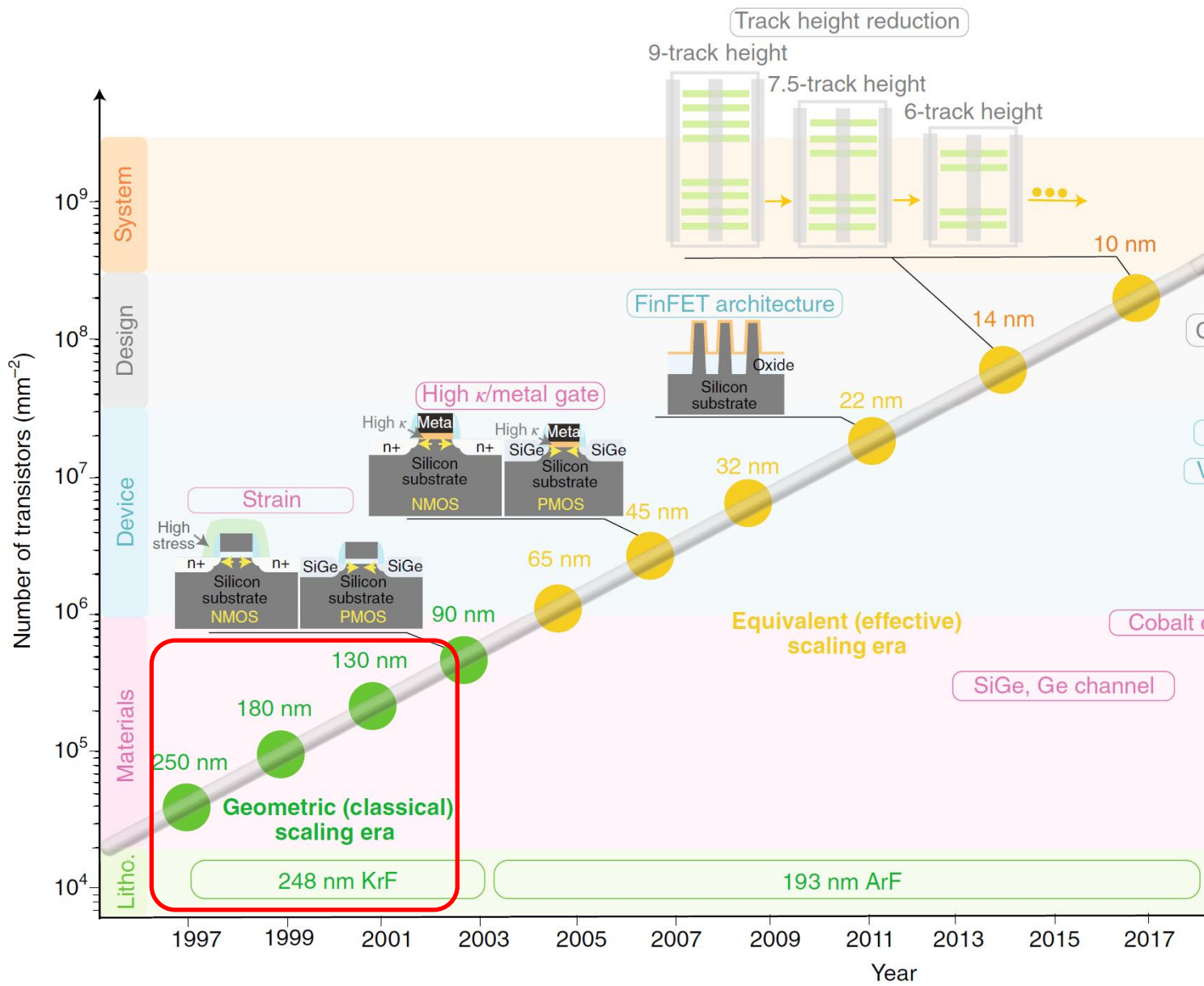
- ✎ **10. 1 Basics of CMOS process**
- ✎ **10. 2 Single-well process**
- ✎ **10. 3 Twin-well process**
- ✎ **10. 4 FinFET process**
- ✎ **10. 5 3D NAND process**
- ✎ **10. 6 Summary and homework**

## 10.4.0 structure of FinFET





# 10.4.1 scaling down of MOSFET



### Short channel effects

- 1)  $V_T$  roll-off
- 2) Channel length modulation effect
- 3) Drain-induced barrier lowering
- 4) Degradation of subthreshold characteristics
- 5) Hot carrier effects

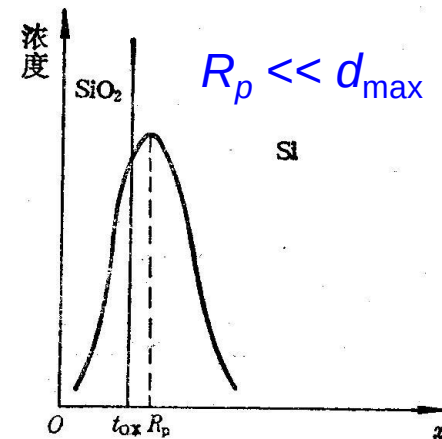
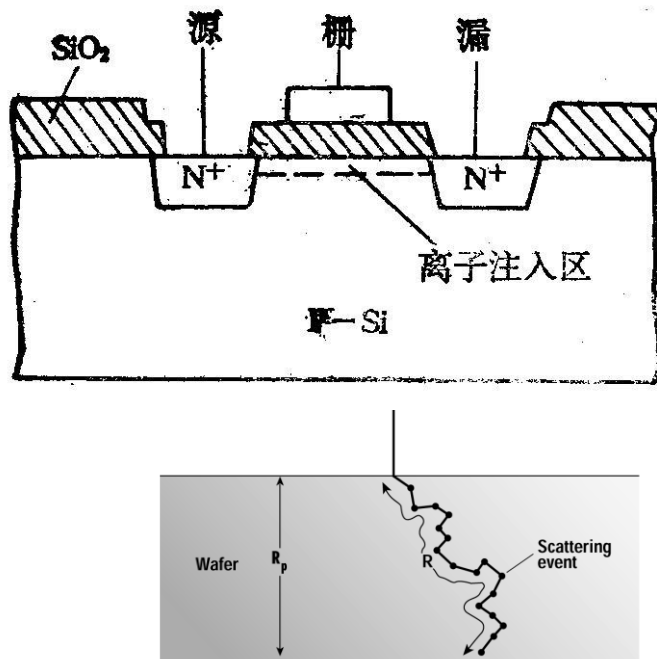
### Narrow channel effect

- 1)  $V_T$  roll-up

### Thin oxide effect

- 1) Tunneling current

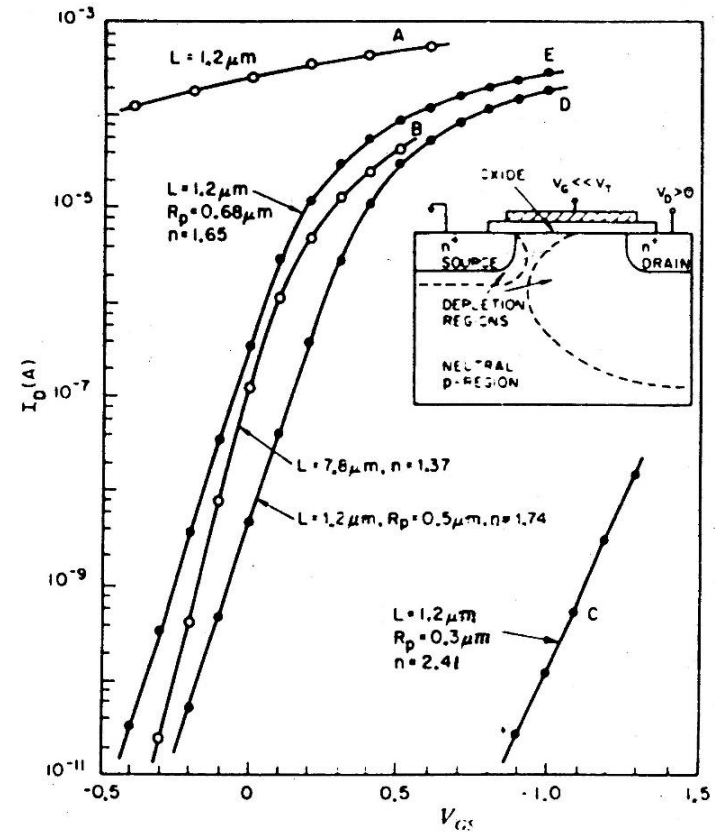
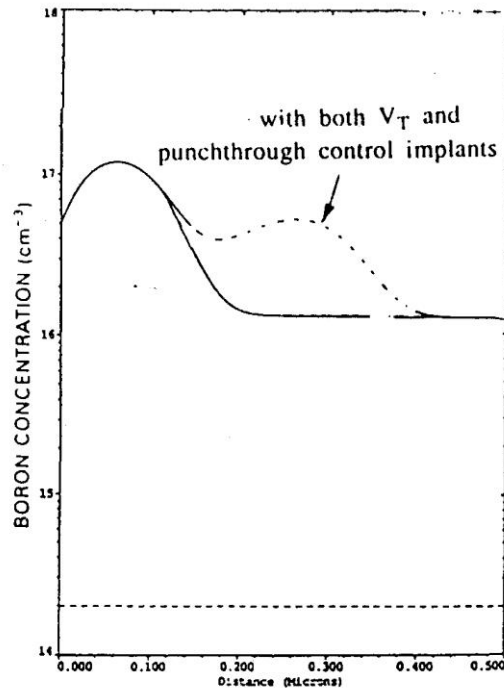
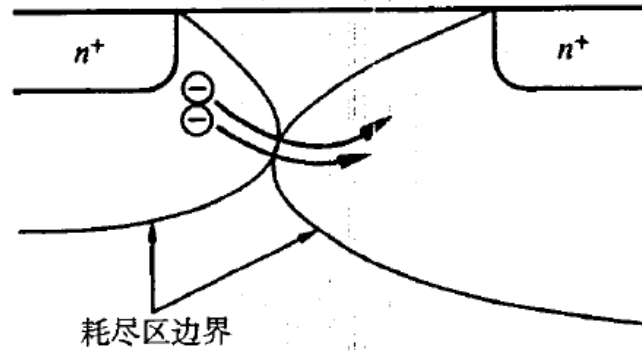
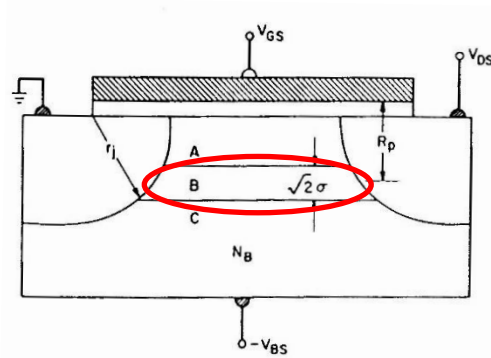
# 10.4.1 $V_T$ Adjustment implantation



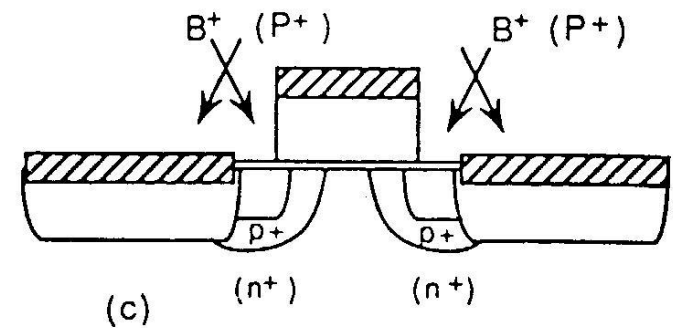
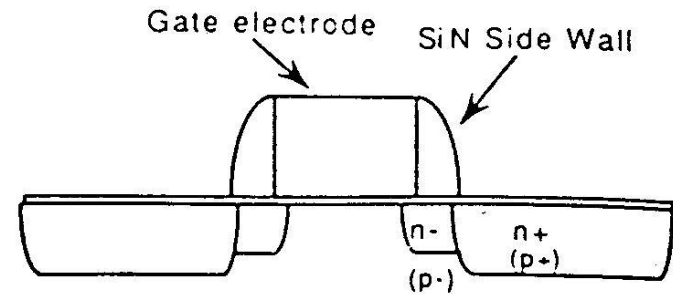
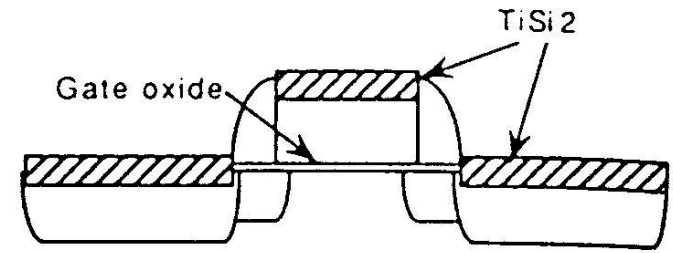
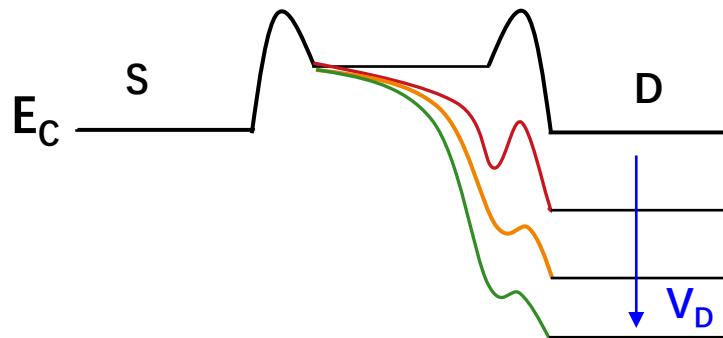
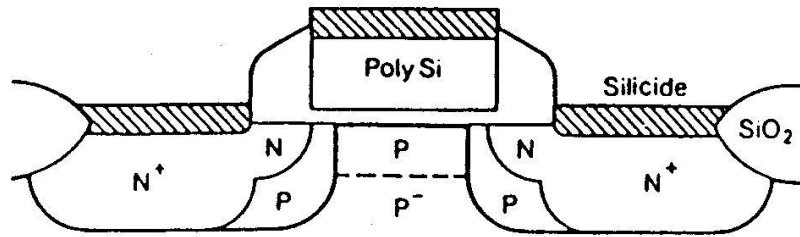
$$Q_B^{total}(d_{max}) = \int_0^{d_{max}} q[N_A + N'_A(x)]dx = Q_B(d_{max}) + \Delta Q_B(d_{max})$$

$$\Delta Q_B(d_{max}) = \int_0^{d_{max}} qN'_A(x)dx = qN_{Im} \quad \Delta V_T = \frac{\Delta Q_B}{C_{ox}} \approx \frac{qN_{Im}}{C_{ox}}$$

# 10.4.1 Anti-punchthrough implantation (PTI)



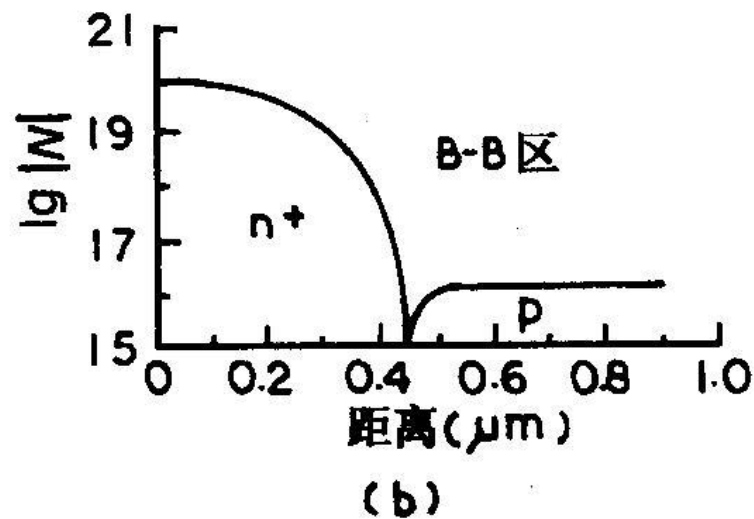
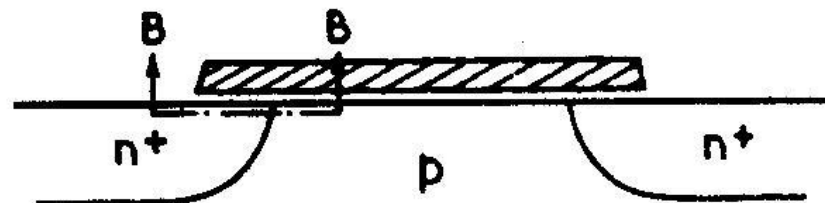
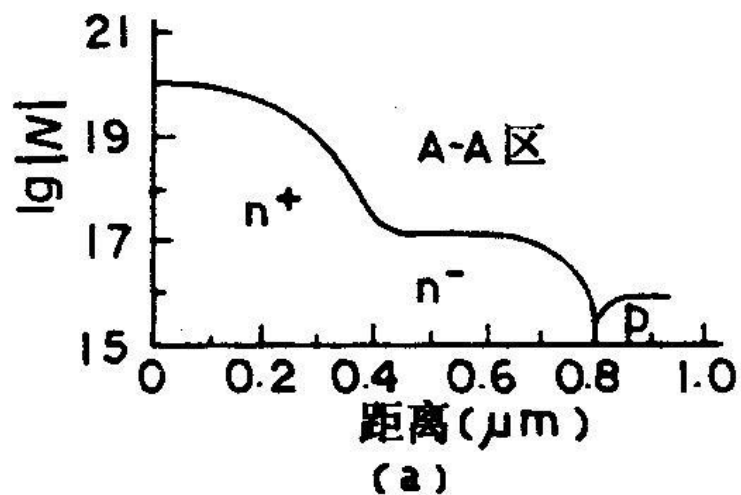
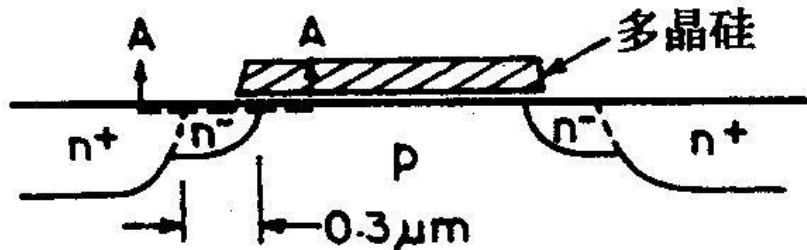
# 10.4.1 Halo implantation



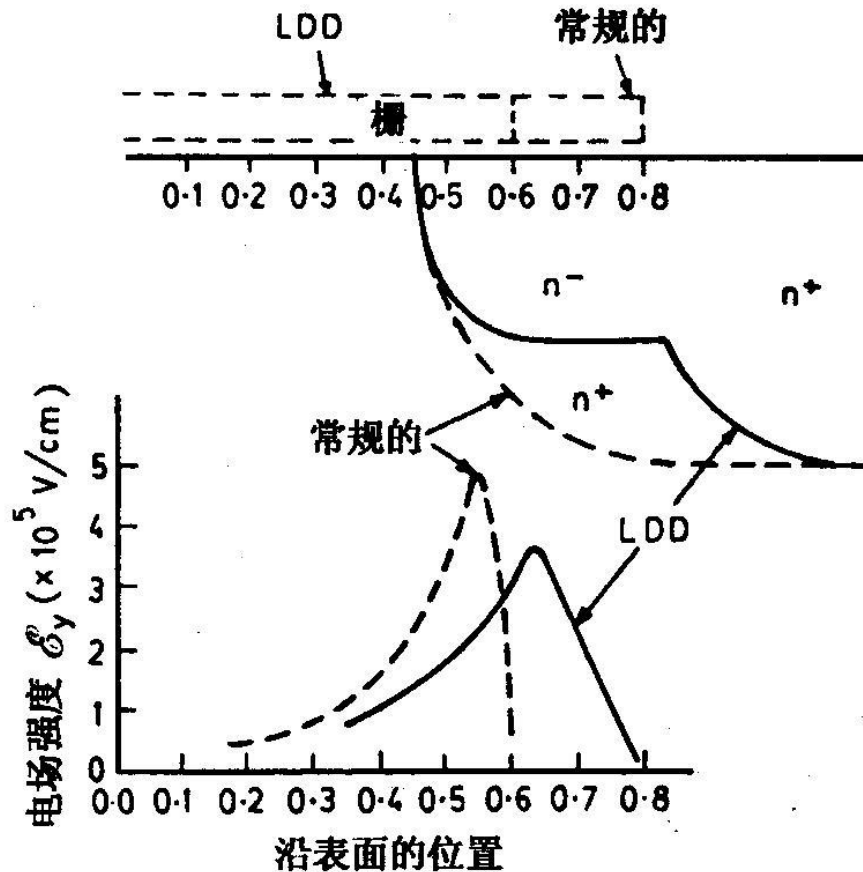
## 10.4.1 Light Drain Doping



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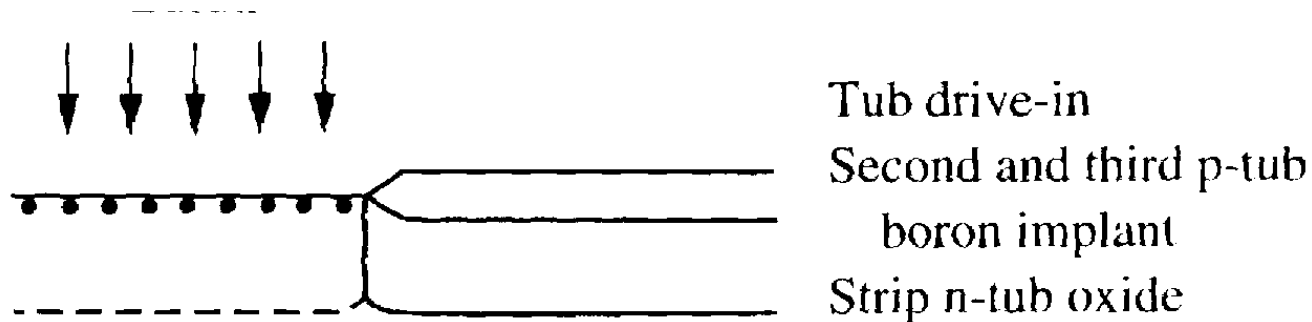
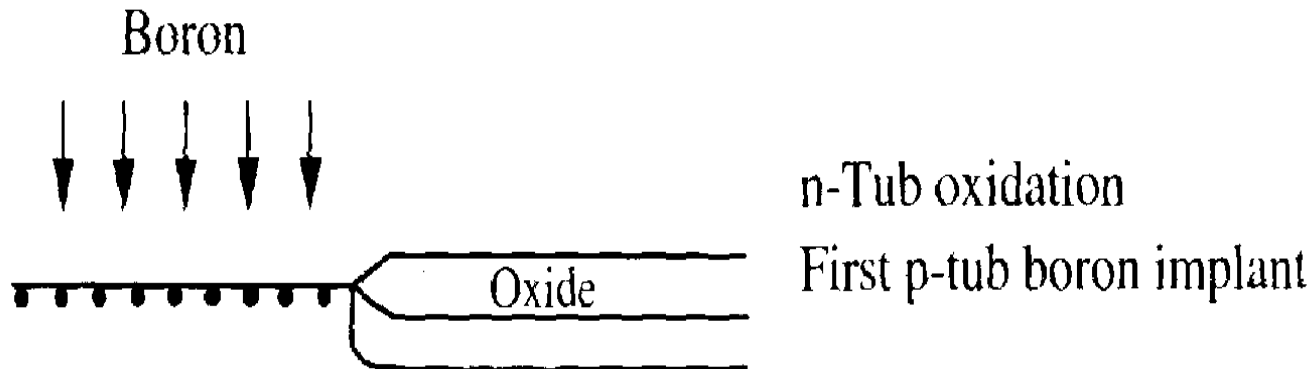
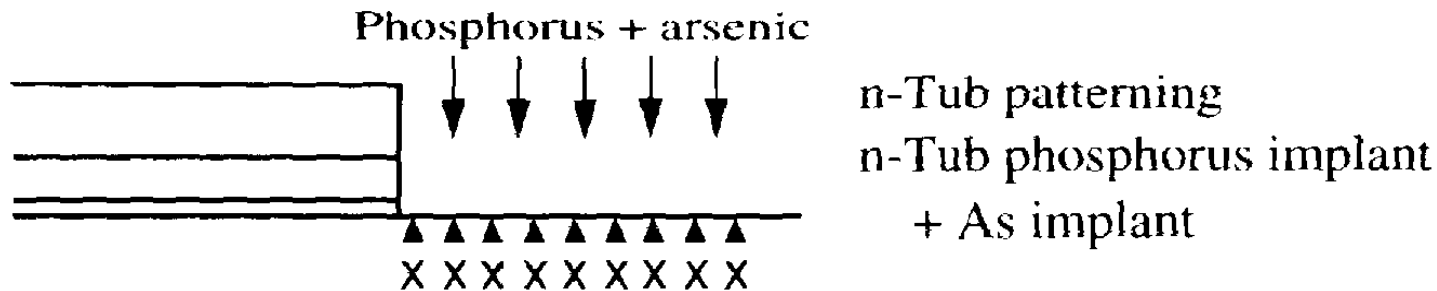
### Suppression of hot carrier effect



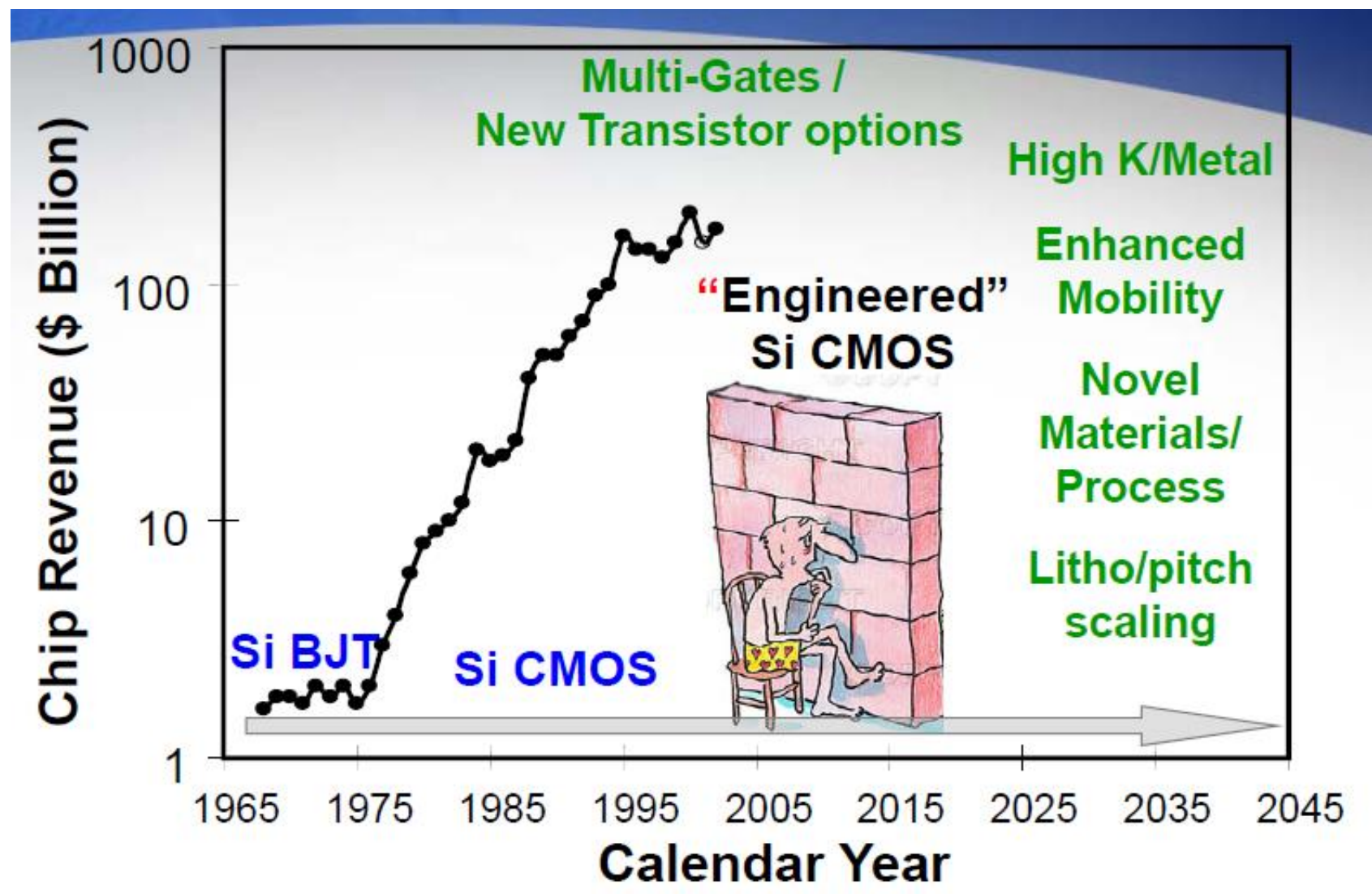
## 10.4.1 multiple step implantation

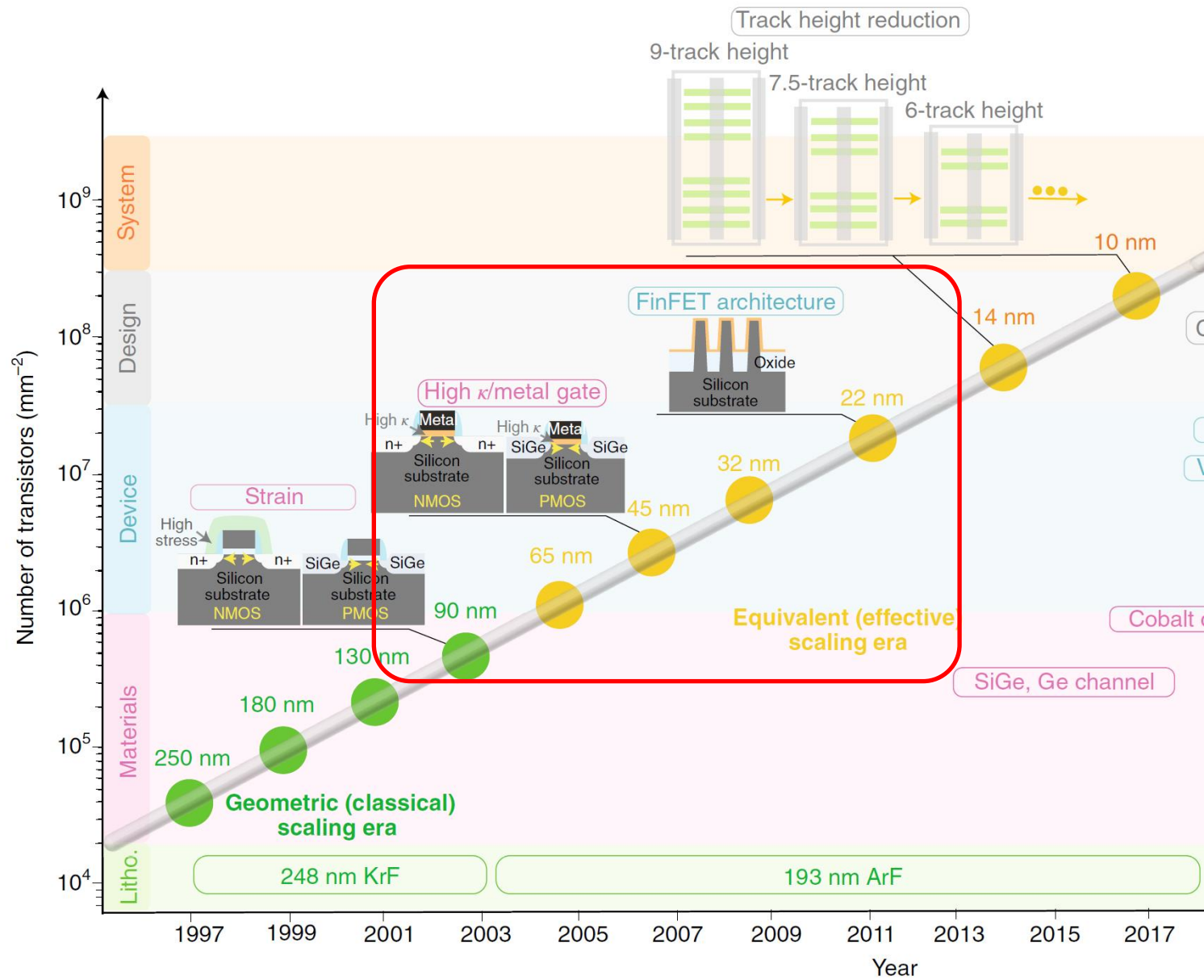


70



# 10.4.2 MOSFET development Roadmap

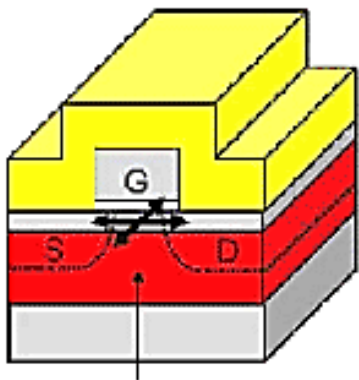




Type of Stress Needed for Enhanced Mobility

| Direction    | NMOS                | PMOS                |
|--------------|---------------------|---------------------|
| Longitudinal | Tension<br>+++      | Compression<br>++++ |
| Transverse   | Tension<br>++       | Tension<br>+++      |
| Out-of-plane | Compression<br>++++ | Tension<br>+        |

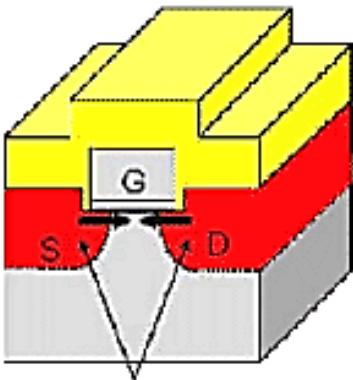
Traditional Approach



Graded SiGe Layer

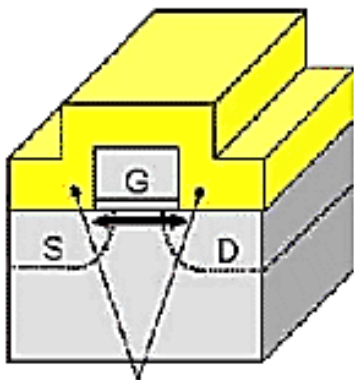
Biaxial  
Tensile Strain

Intel's 90nm Technology



Selective SiGe S-D

Uniaxial  
Compressive Strain  
for PMOS



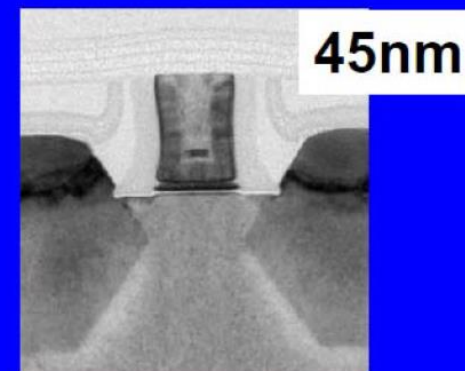
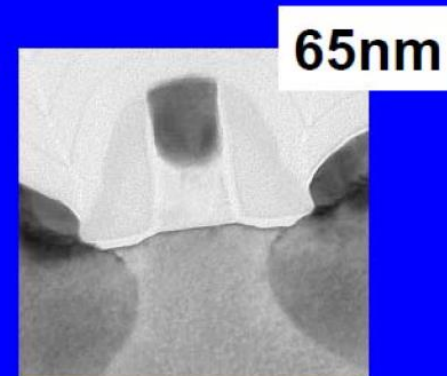
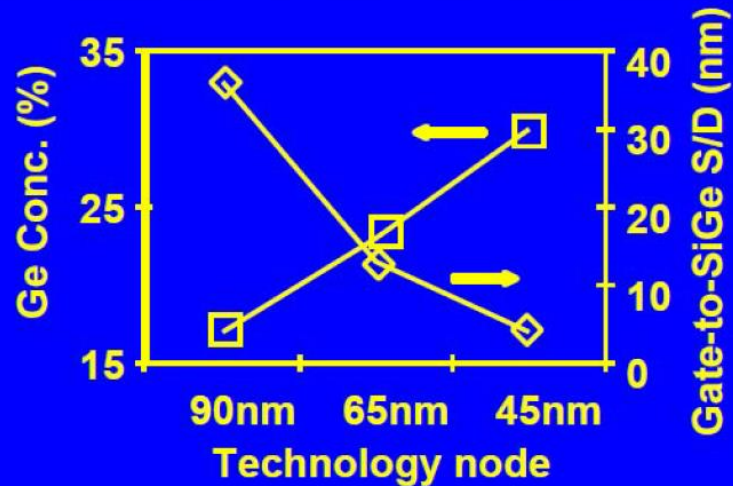
Tensile Si<sub>3</sub>N<sub>4</sub> Cap

Uniaxial  
Tensile Strain  
for NMOS

### Embedded SiGe S/D

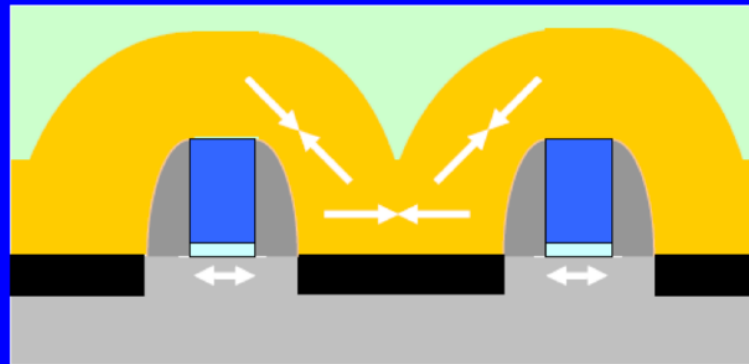
- Continued Scaling for SiGe S/D

- Ge concentration inc. to 30%
- Gate  $\rightarrow$  S/D distance reduced

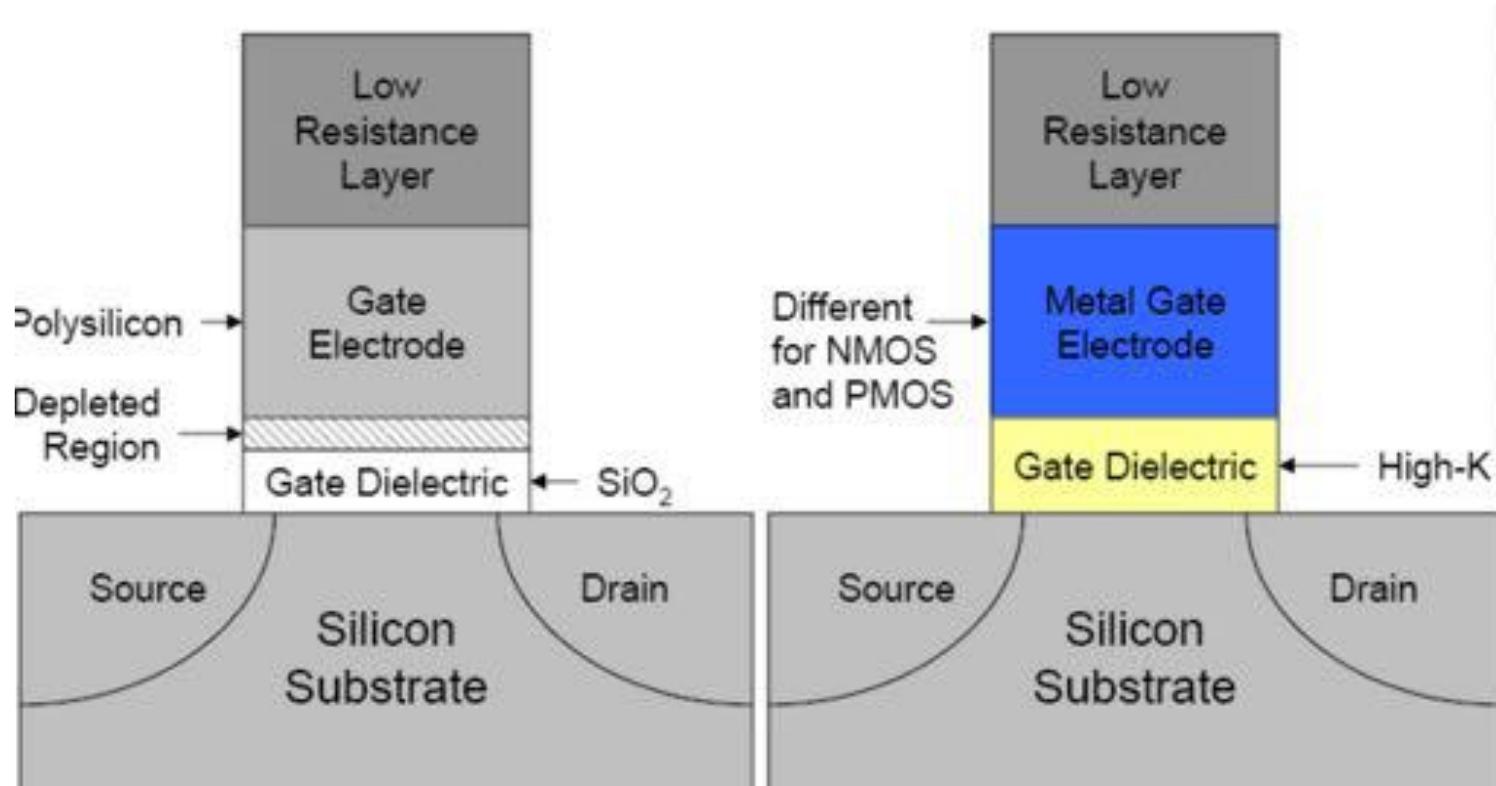


### Methods of NMOS Strain at 65nm

- Tensile Nitride Cap
- Stress Memorization  
→ Gate + S/D



## 10.4.3 high-k metal gate



### Metal Gate Flow Options

#### Gate-First

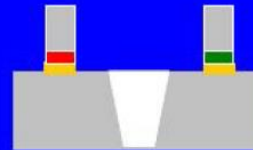
Dep Hi-k &  
Met 1



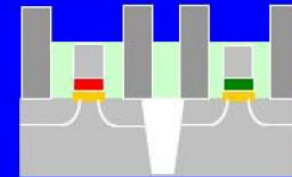
Patt Met 1 &  
Dep Met 2



Patt Met 2 &  
Etch Gates



S/D formation &  
Contacts

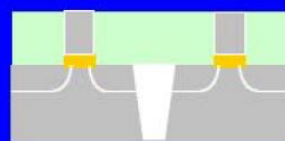


#### Hik-First, Gate-Last

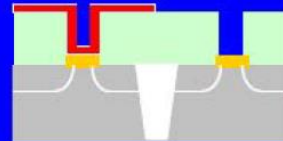
Dep & Patt  
Hik+Gate



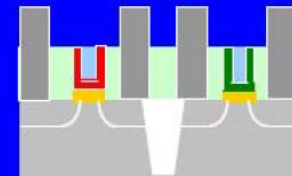
S/D formation &  
ILD dep/polish

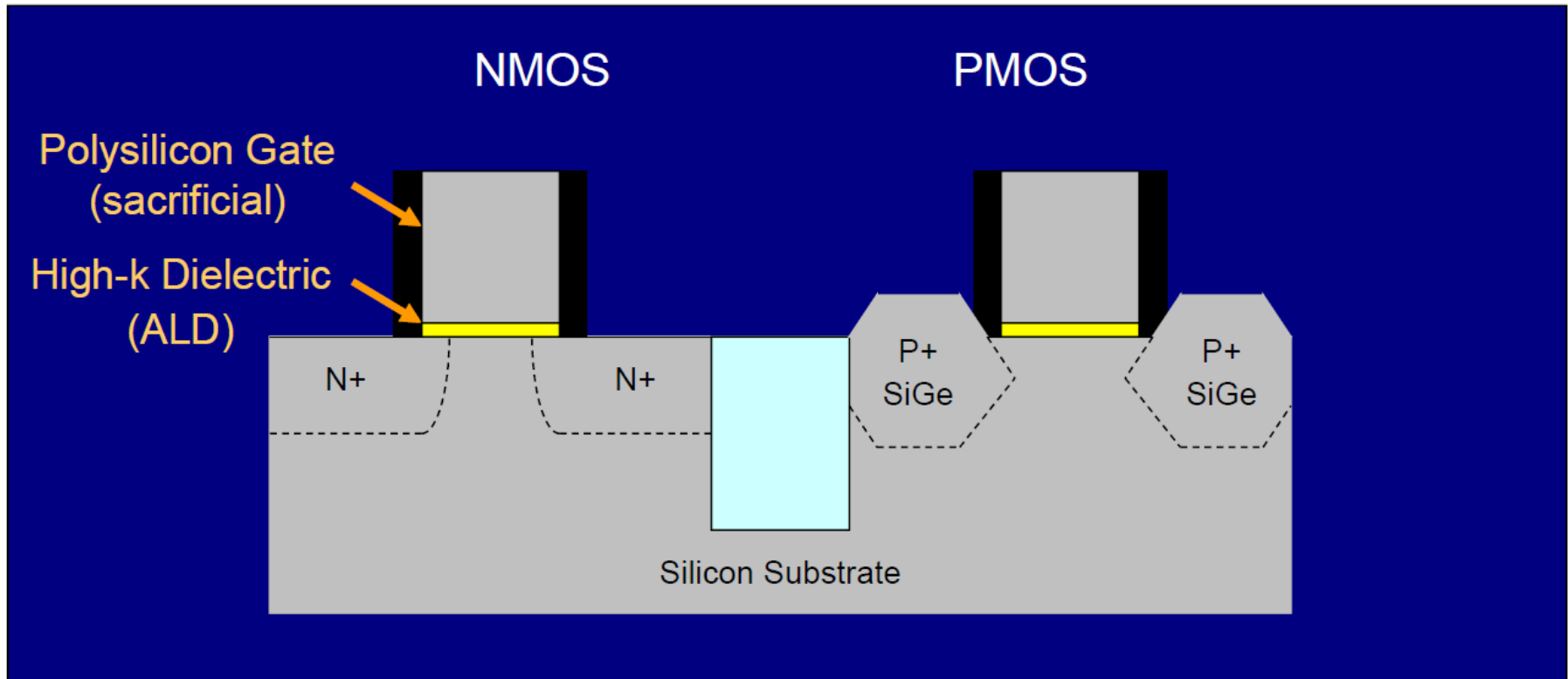


Rem Gate &  
Patt Met 1

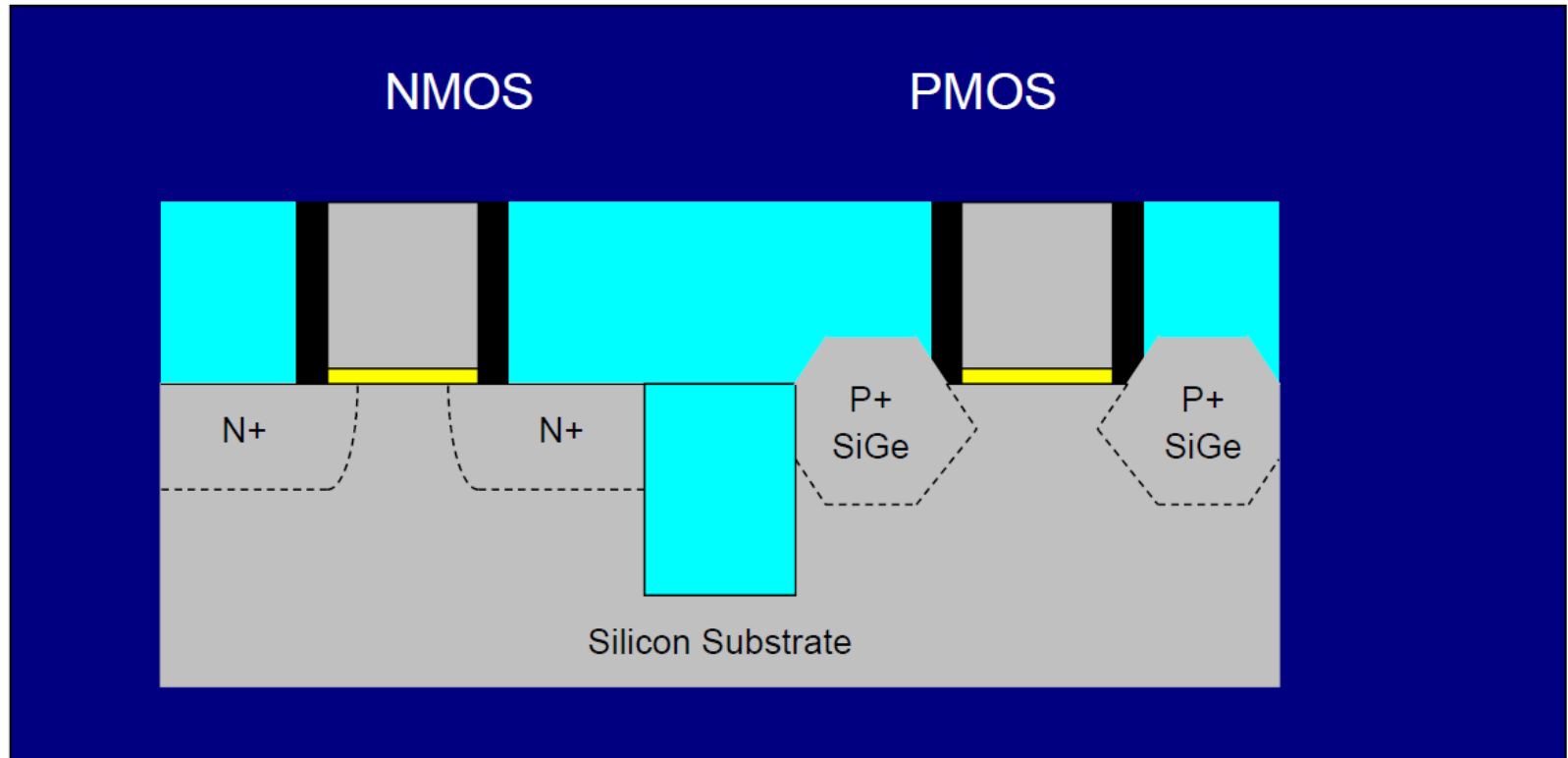


Dep Met 2+Fill &  
Polish





**Standard transistor process through source-drain formation,  
but include atomic layer deposition high-k dielectric**

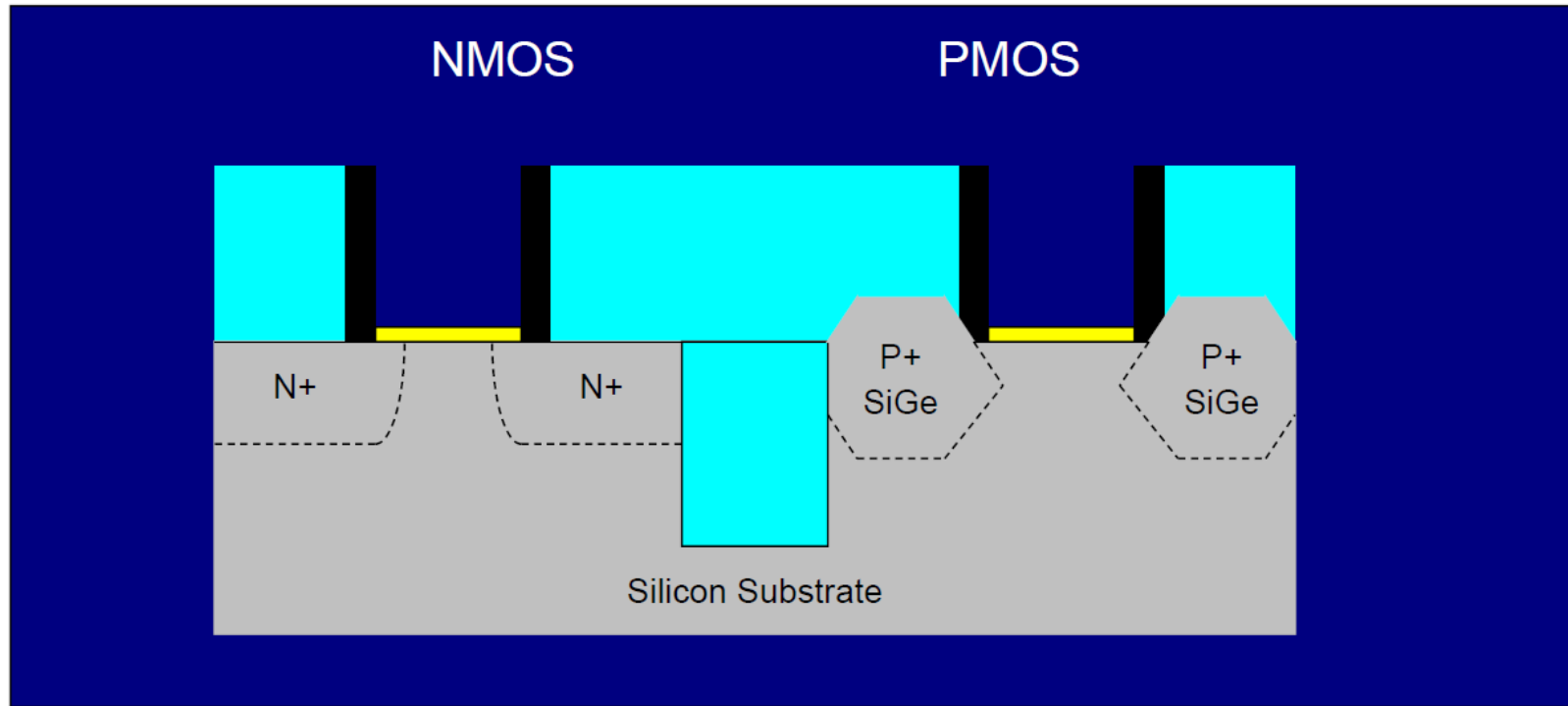


**Deposit and planarize oxide layer**

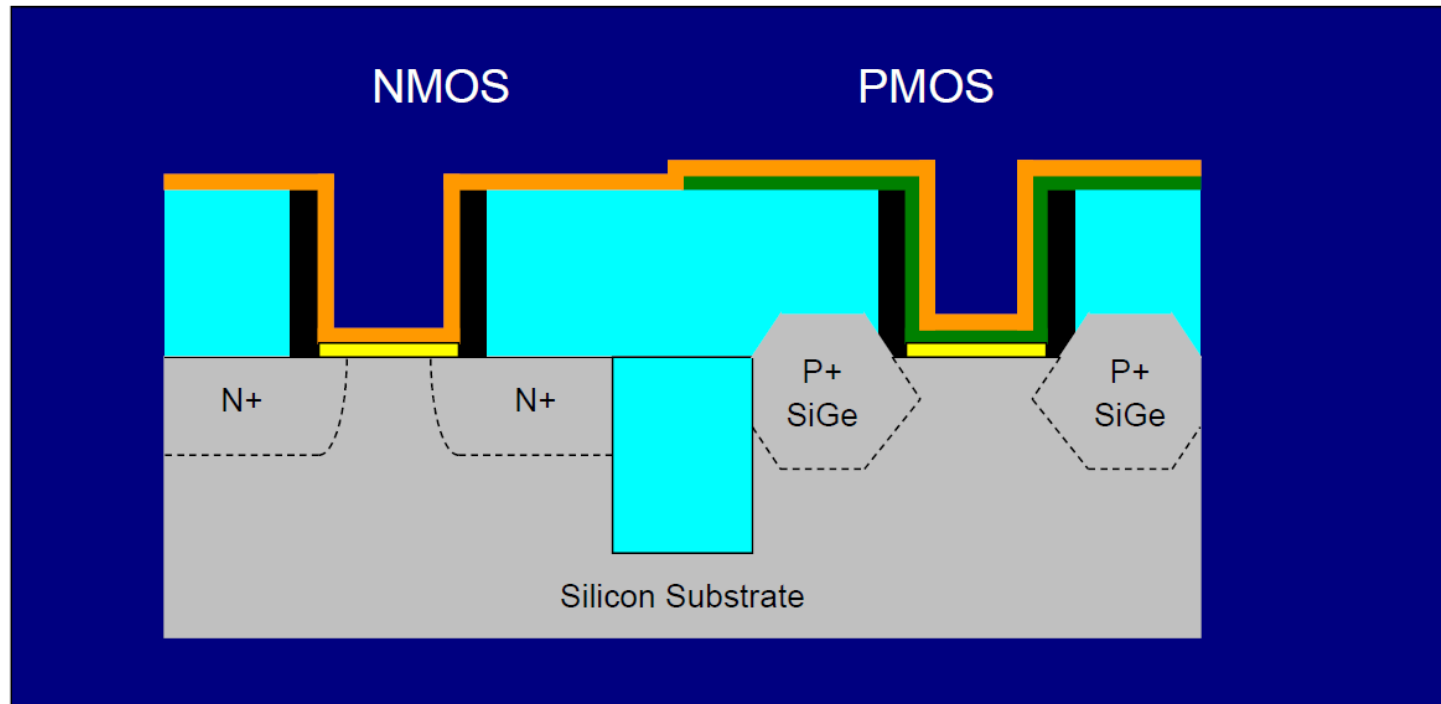
## 10.4.3 replacement gate flow



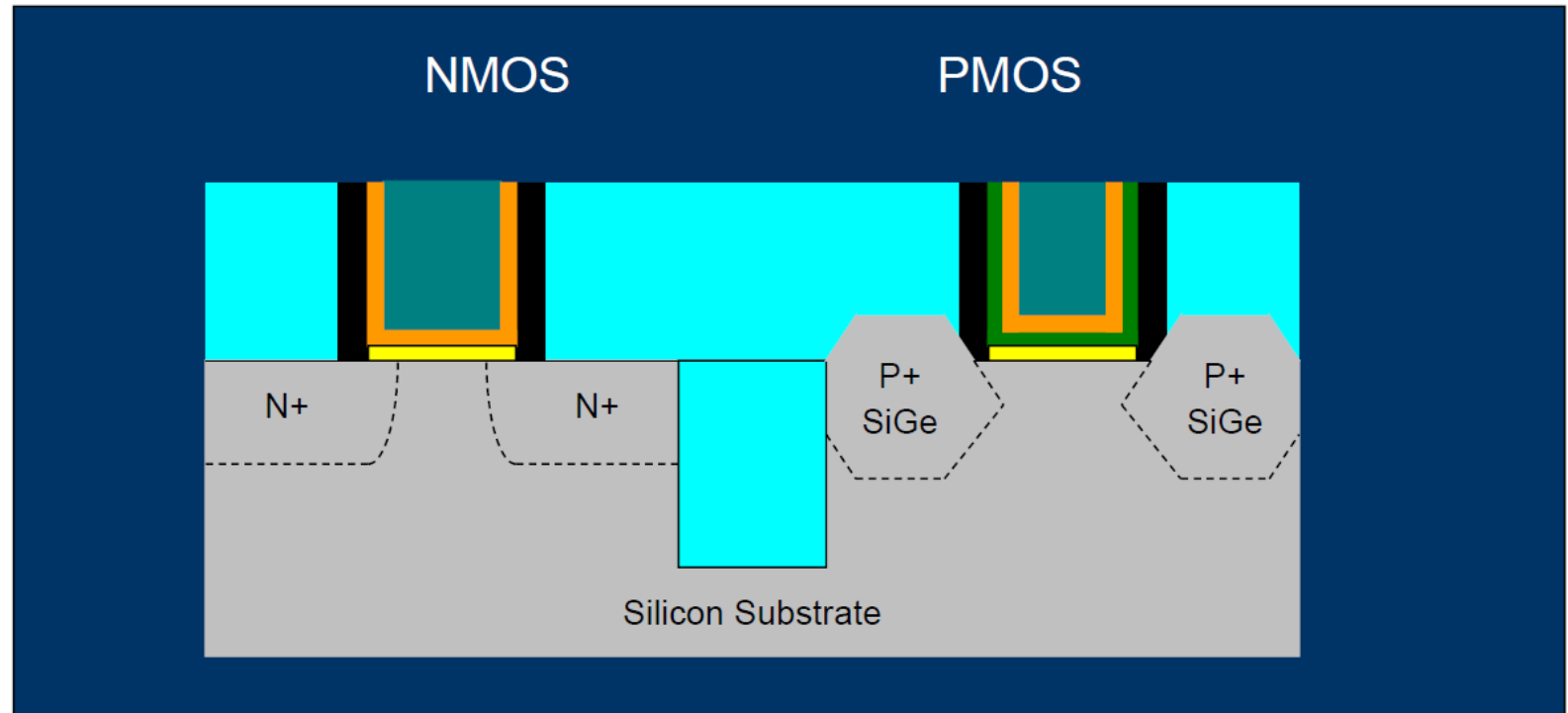
80



**Etch out sacrificial polysilicon gate**

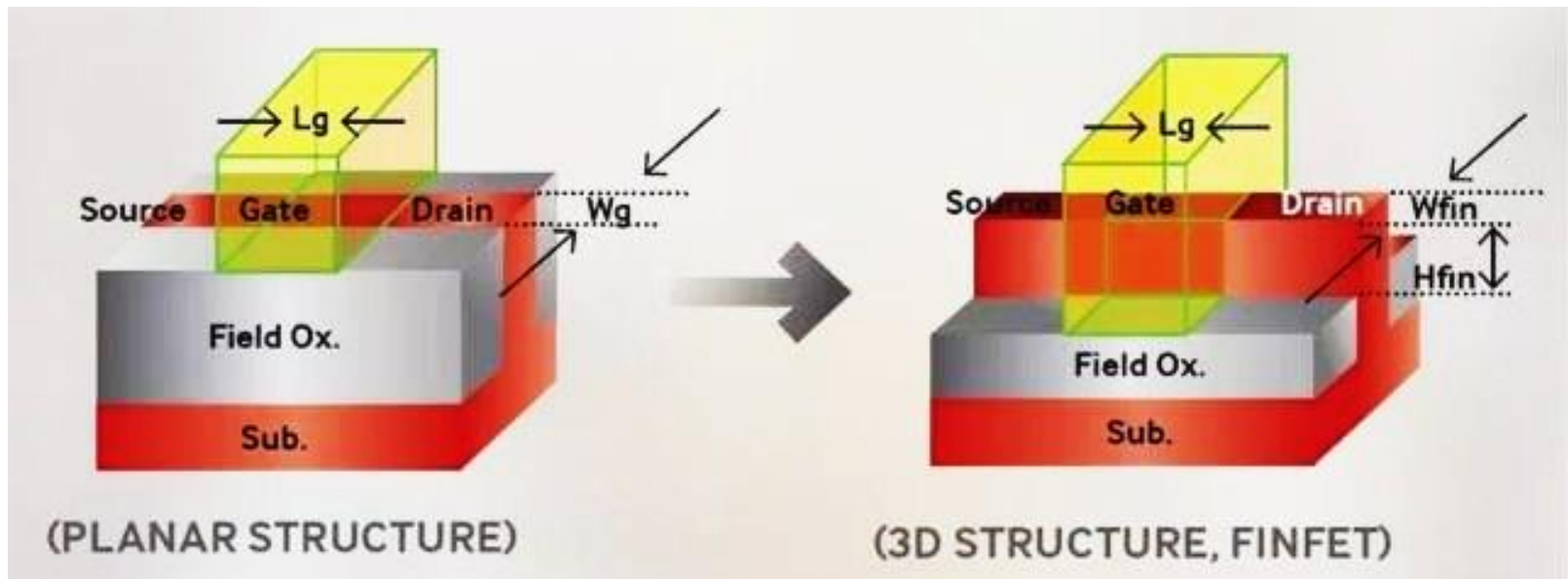
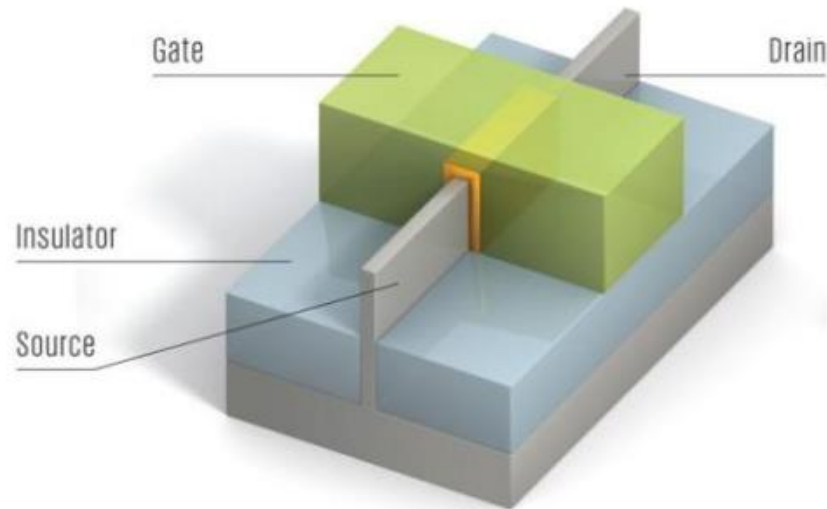


**Deposit separate NMOS and PMOS WF metal layers**



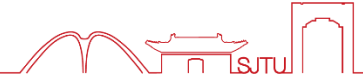
**Deposit Al fill metal, planarize surface**

## 10.4.4 FinFET: structure

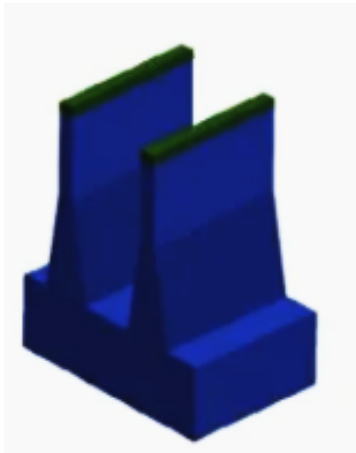
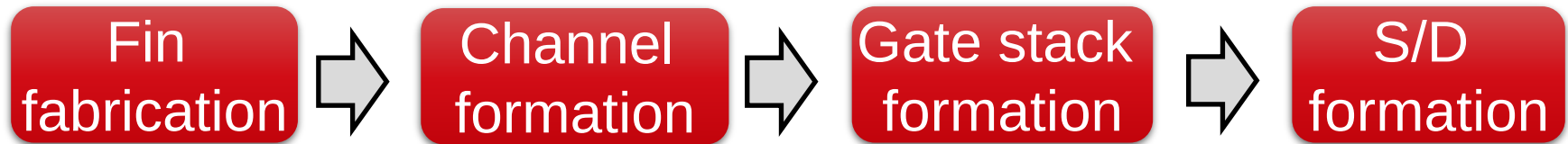




## 10.4.4 Process flow of FinFET



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**The key of FinFET**



**High-k metal gate  
(replacement gate)**



**Stained  
Silicon**



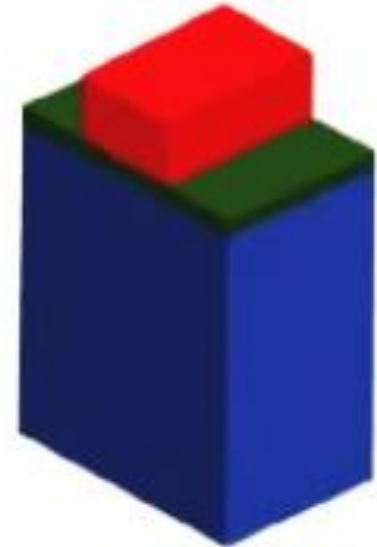
**Substrate**



**Oxide growth and  
Si<sub>3</sub>N<sub>4</sub> deposition**



**Poly-Si deposition**



**Poly-Si etching**



**SiO<sub>2</sub> deposition**  
Width of fin is  
controlled by  
SiO<sub>2</sub> thickness



**SiO<sub>2</sub> etching**  
Stop on Si<sub>3</sub>N<sub>4</sub>

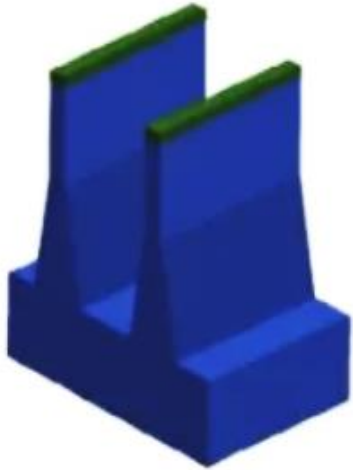
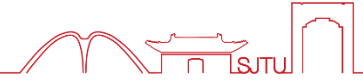


**Poly-Si etching**



**Substrate  
etching**

## 10.4.4 Channel formation



$\text{SiO}_2$  etching  
Stop on  $\text{Si}_3\text{N}_4$



$\text{SiO}_2$  deposition  
Flatting

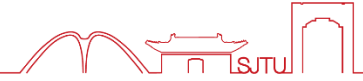


$\text{SiO}_2$  etching



$\text{Si}_3\text{N}_4$  etching

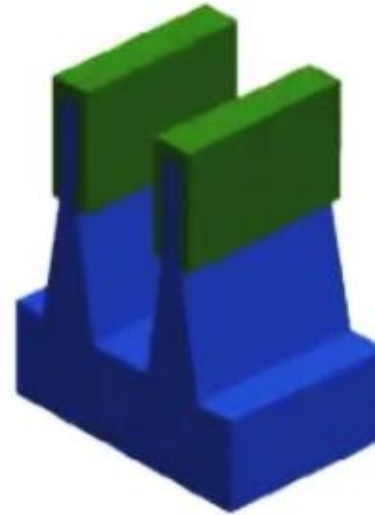
## 10.4.4 Channel formation



$\text{SiO}_2$  and  $\text{Si}_3\text{N}_4$   
deposition as  
hard mask



$\text{Si}_3\text{N}_4$  etching



$\text{SiO}_2$  etching



N-well and P-  
well formation  
via implantation



$\text{SiO}_2$  and  $\text{Si}_3\text{N}_4$   
etching



$\text{SiO}_2$  deposition



$\text{SiO}_2$  etching  
The height of  
Fin is controlled  
by etching time



Channel  
implantation

## 10.4.4 Gate stack formation



**SiO<sub>2</sub> deposition  
as buffer layer**



**Poly-Si  
deposition**



**Etch to form gate  
S/D implantation  
and activation**

**Poly-Si etching  
High-k metal gate  
formation**



**Si<sub>3</sub>N<sub>4</sub>  
deposition**



$\text{SiO}_2$  and  $\text{Si}_3\text{N}_4$  deposition and etching to form side wall



$\text{SiO}_2$  deposition as 外延阻挡层



Etching to expose S/D region of PMOS



Epitaxial growth of SiGe on S/D of PMOS



Etching  $\text{SiO}_2$  on S/D of NMOS



$\text{SiO}_2$  deposition as NMOS S/D barrier layer  
变材料阻挡层



Etching to expose S/D of NMOS



Epitaxial growth of SiC on S/D of NMOS

✎ **10. 1 Basics of CMOS process**

✎ **10. 2 Single-well process**

✎ **10. 3 Twin-well process**

✎ **10. 4 Modern CMOS process**

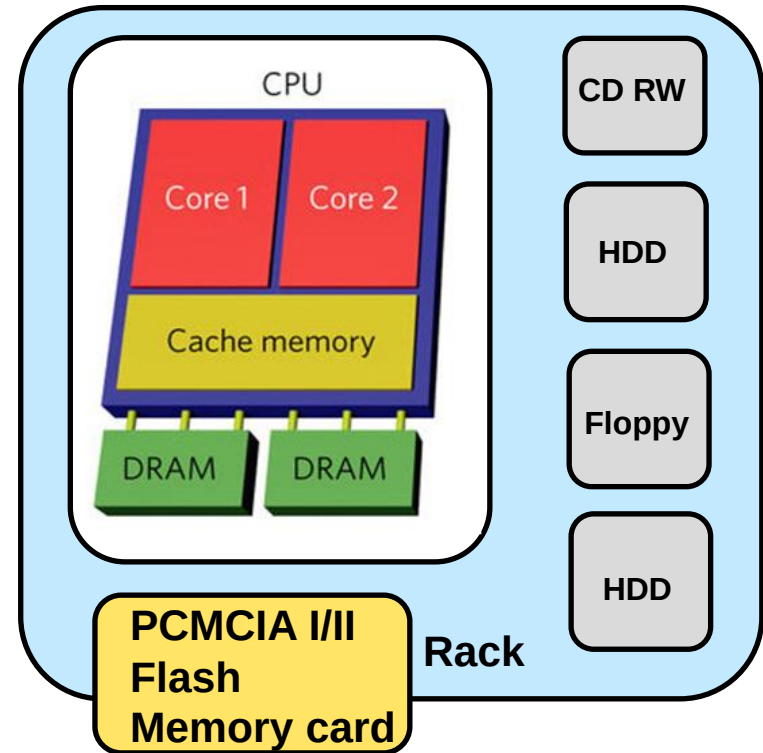
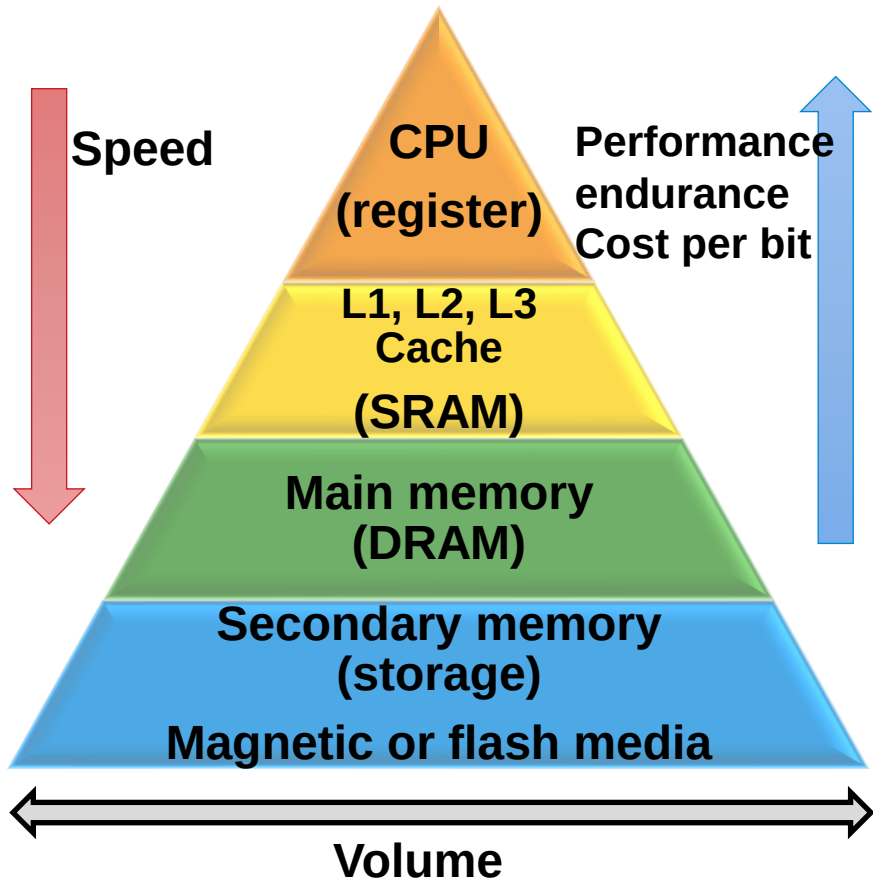
✎ **10. 5 3D NAND process**

✎ **10. 6 Summary and homework**

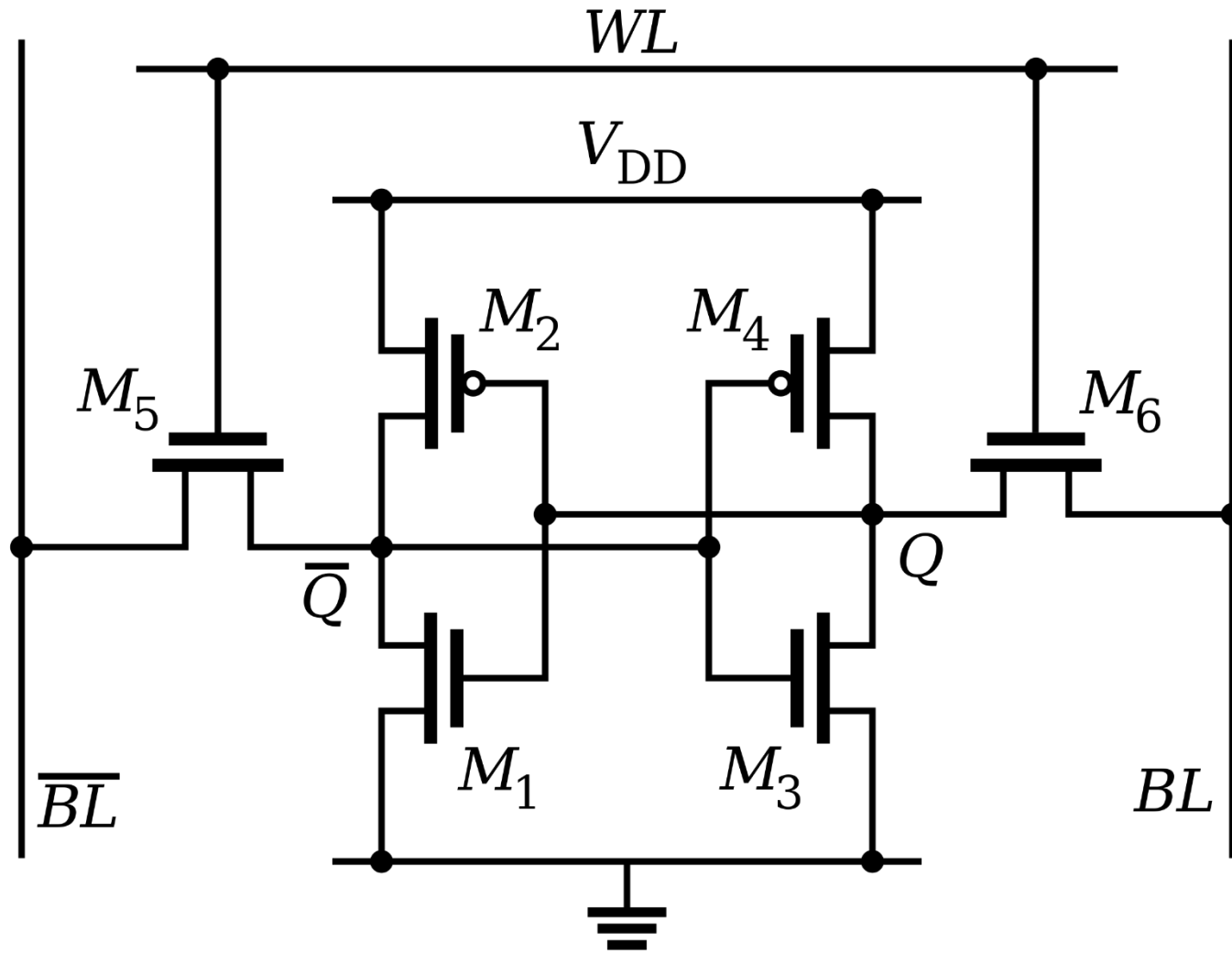
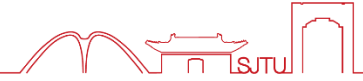
# 10.5.1 Memory devices

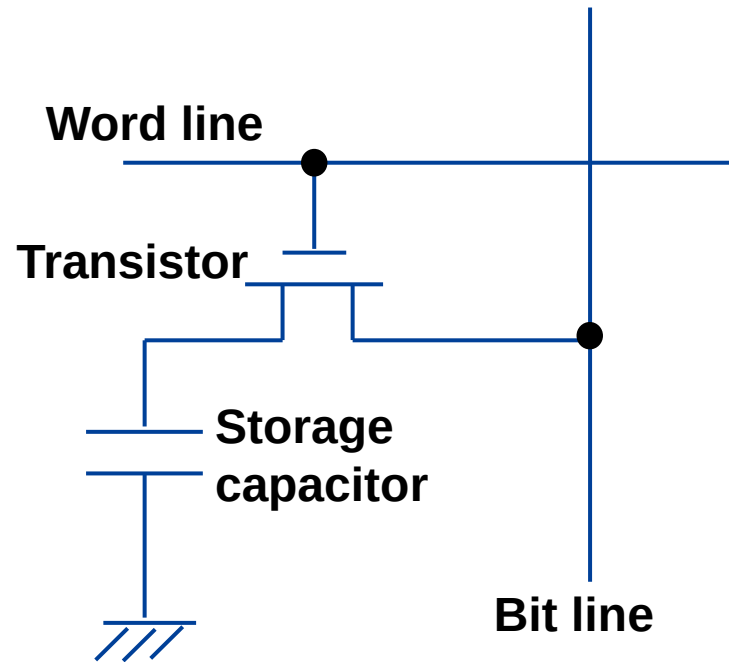


95

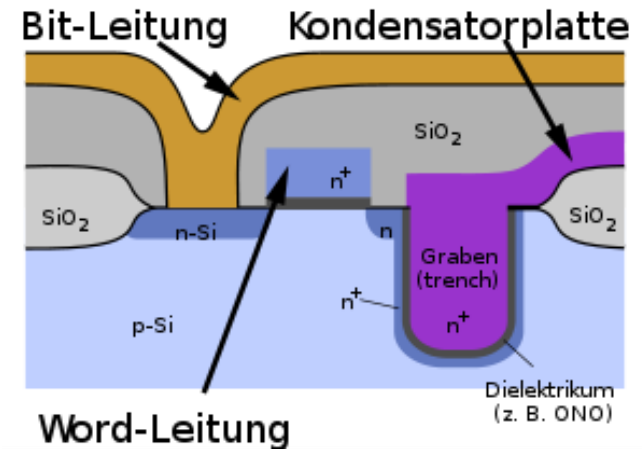


## 10.5.1 SRAM

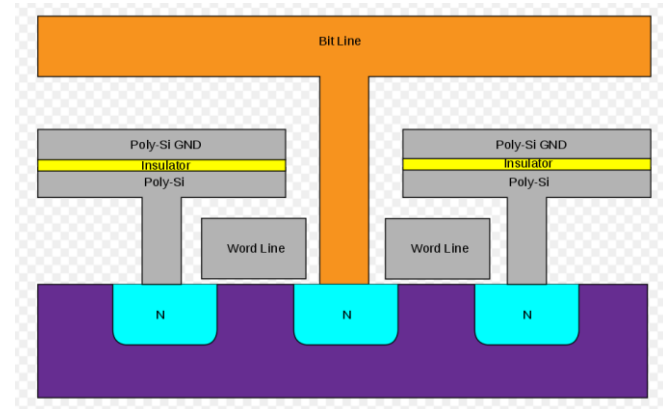




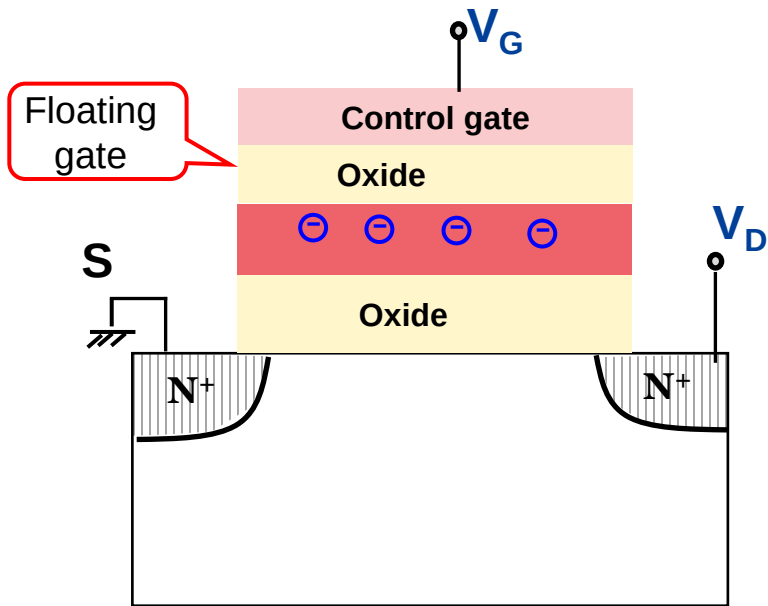
## Trench structure



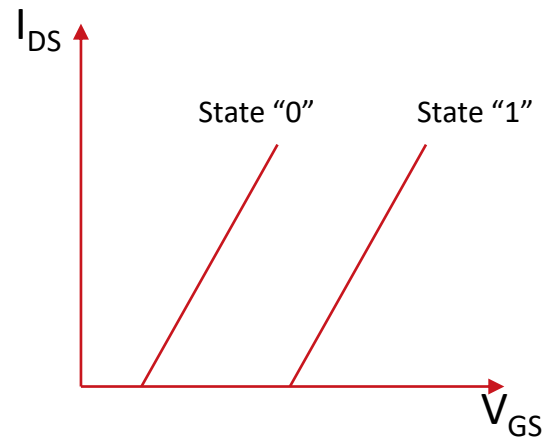
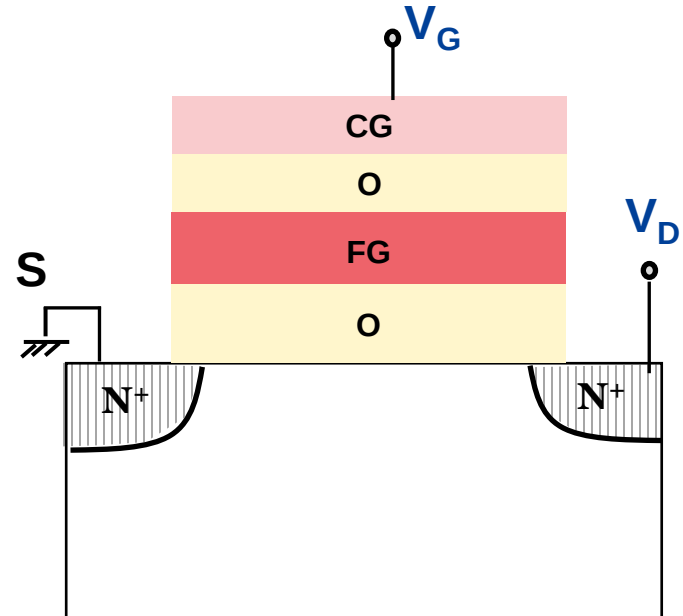
## Stacked structure



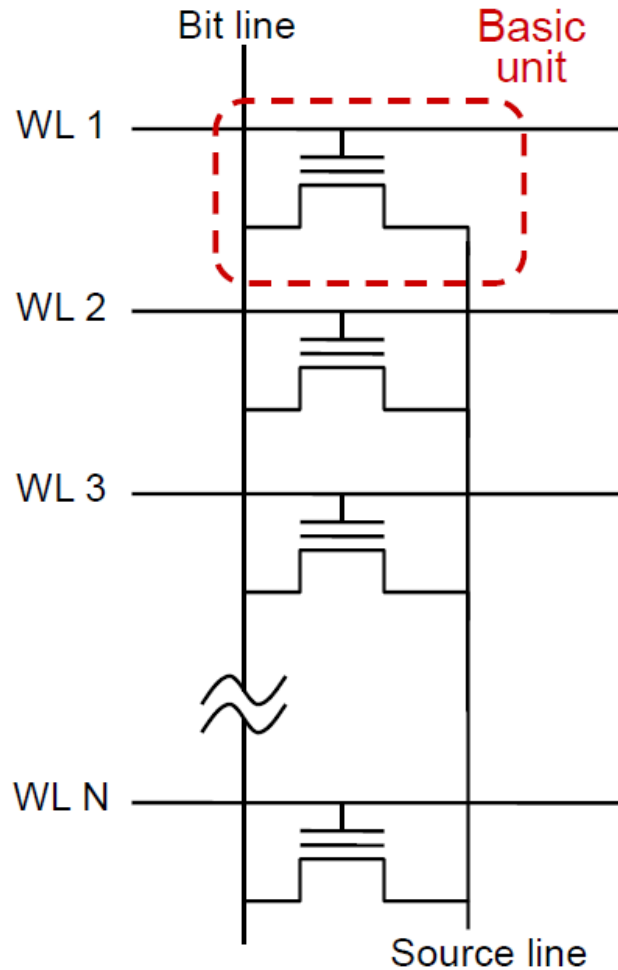
## Programmed state



## Erased state

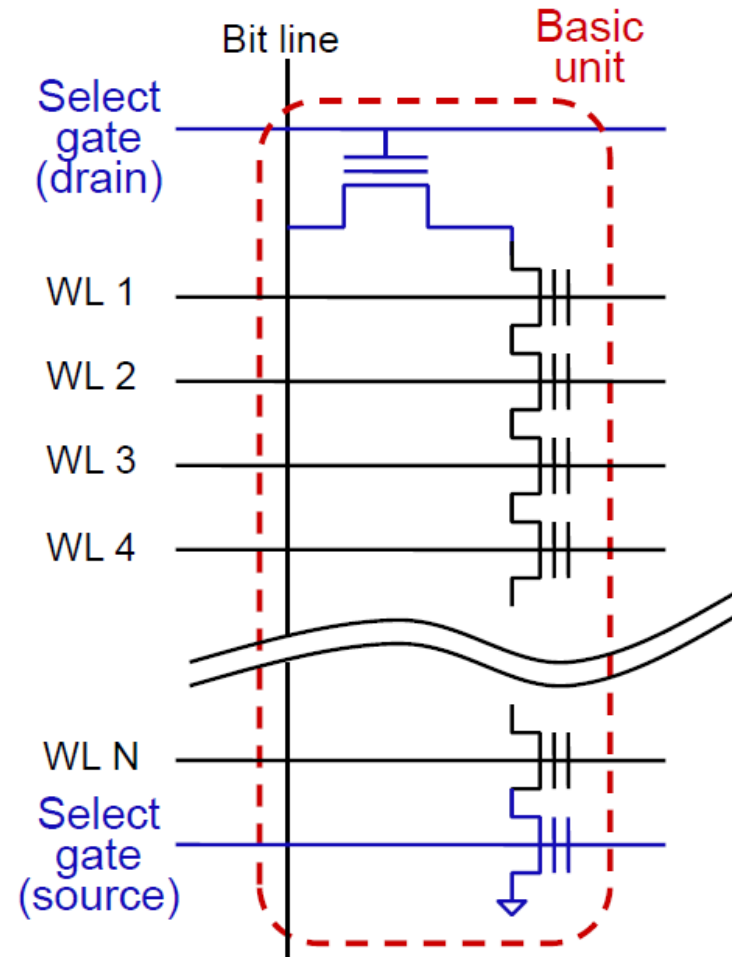


## 10.5.1 Flash-NAND



**NOR structure**

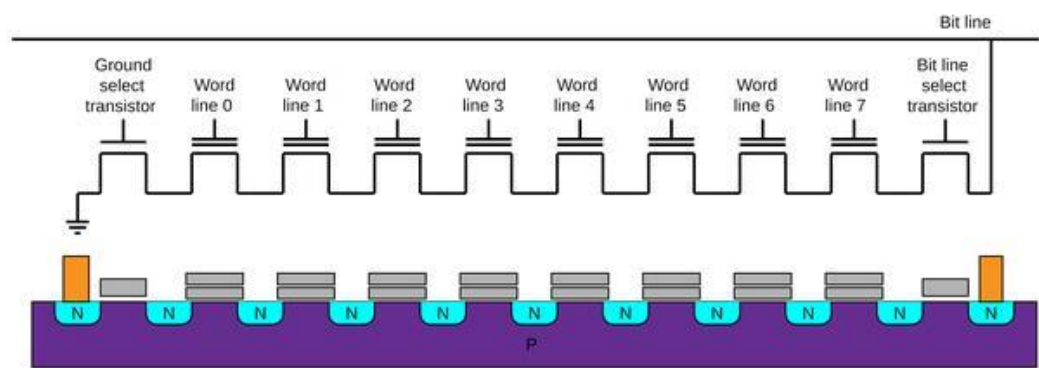
Cells are connected in parallel to the bit lines, allowing the cells to be read and programmed individually.



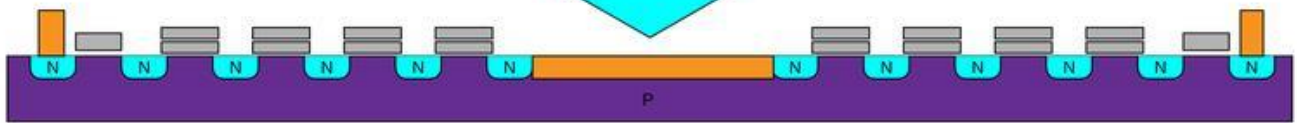
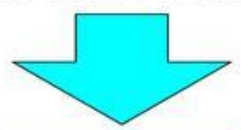
**NAND structure**

Cells are connected in series to the bit line, consuming less space and reducing the cost.

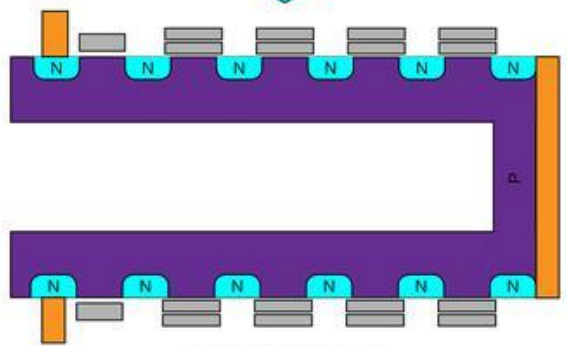
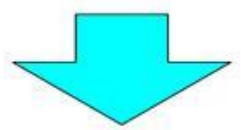
# 10.5.2 From planar NAND to vertical NAND



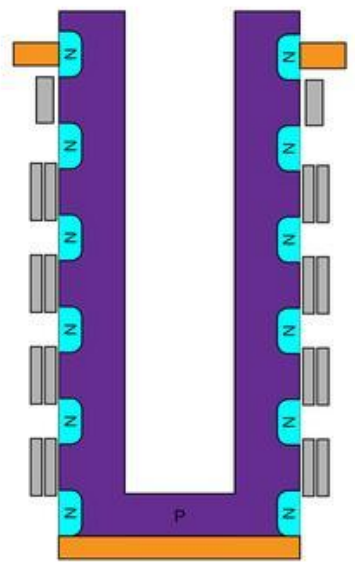
Typical 2D NAND FLASH String



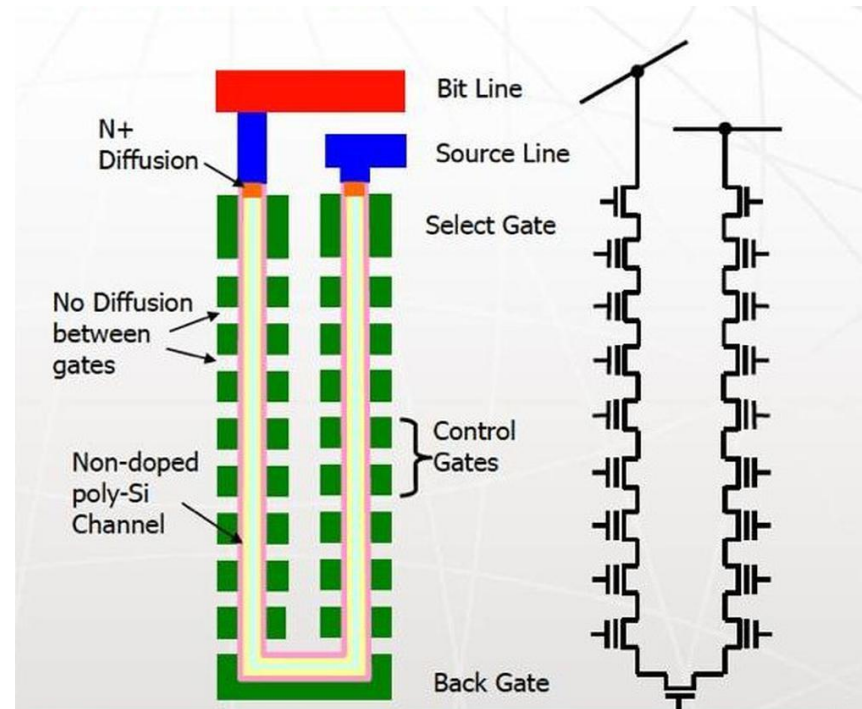
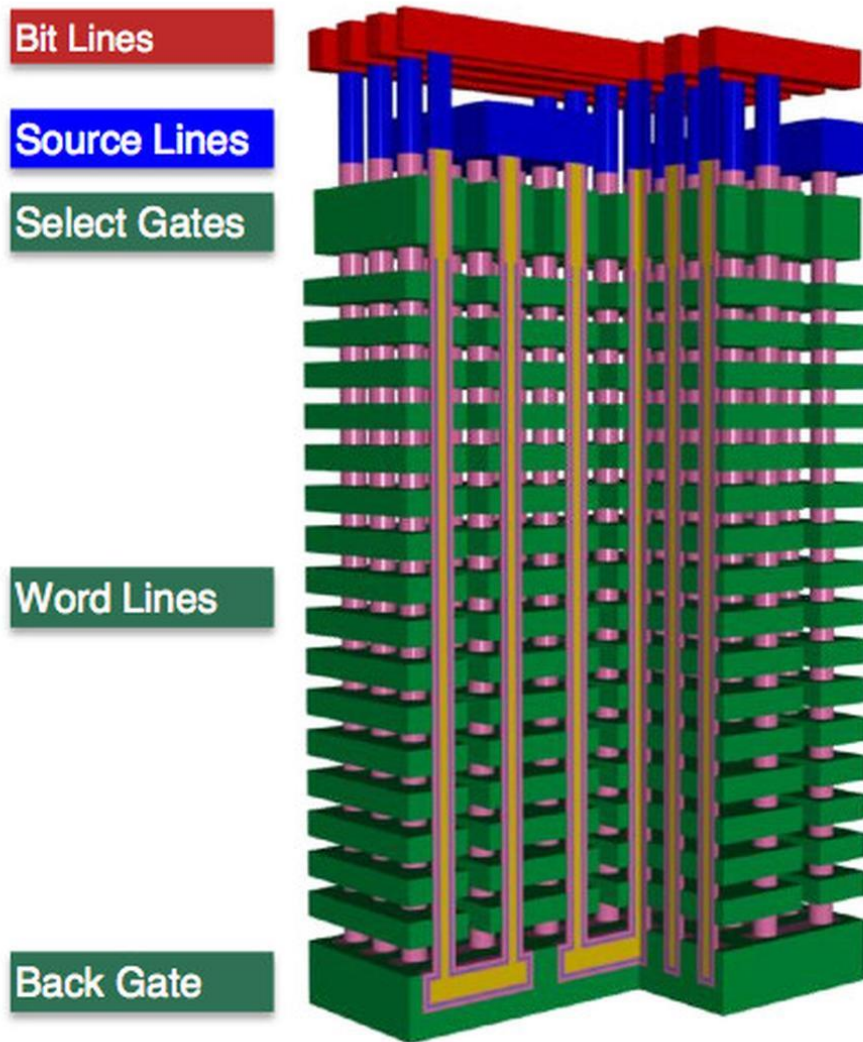
Stretch It Out In The Middle



Fold It Over



Stand It Vertically





- ✎ **10. 1 Basics of CMOS process**
- ✎ **10. 2 Single-well process**
- ✎ **10. 3 Twin-well process**
- ✎ **10. 4 Modern CMOS process**
- ✎ **10. 5 3D NAND process**
- ✎ **10. 6 Summary and homework**

- **Basics of CMOS process**

**Substrate selection, well designing, isolation**

- **Single-well process**

**Well formation, isolation, gate stack formation, S/D formation, contact formation and connection**

- **Twin-well process**

- **Modern CMOS process**

**Method to suppress short channel effect,  
strained-Si, high-k metal gate, FinFET**

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# 谢 谢

